

Speech Technology



Nguyen Thi Thu Trang

About me

- Engineer & Master in CS, HUST, Vietnam
- PhD in CS, Université Paris-Sud 11, France
- Senior Lecturer and Researcher, SoICT, HUST
 - Software Engineering, Speech Language Processing, Natural Language Processing
 - <https://dblp.org/pid/140/2812.html>
 - <https://scholar.google.com/citations?user=nTljOQYAAAAJ>
- National Golden Globe Award in ICT (2009)
- Founder of Vbee (vbee.ai, vbee.vn)
 - Highest Award at Vietnam Talent for Text-To-Speech (2018)
 - Winner at Grab Venture Ignite for Conversational AI (2020)
 - Winner at Qualcomm Innovation Challenge for on-device Conversational AI (2024)
 - and many more...



0. Course Introduction



Nguyen Thi Thu Trang

Course Introduction

1. Acoustics & Phonetics
2. Speech Dataset
3. Speech Applications

Course Introduction

1. Acoustics & Phonetics

- Sound Representation. Prosody, Intonation, and Accents
- Phonetics Basics. Vietnamese phonology and phonetics
- Signal Representation, Feature Extraction
- Noise Reduction and Silence Removal
- Speech Augmentation
- Normalization and Feature Scaling

2. Speech Dataset

3. Speech Applications

Course Introduction

1. Acoustics & Phonetics

2. **Speech Dataset**

- Dataset Acquisition
- Label Alignment
- Dataset Preprocessing
- Dataset Augmentation
- Dataset Evaluation
- Large Audio-Text Datasets

3. Speech Applications

Course Introduction

1. Acoustics & Phonetics

2. Speech Dataset

3. Speech Applications

- Model Evaluation Metrics
- Model Debugging and Optimization
- Text-to-Speech (TTS)
- Voice Cloning
- Automatic Speech Recognition
- Speech Profiling

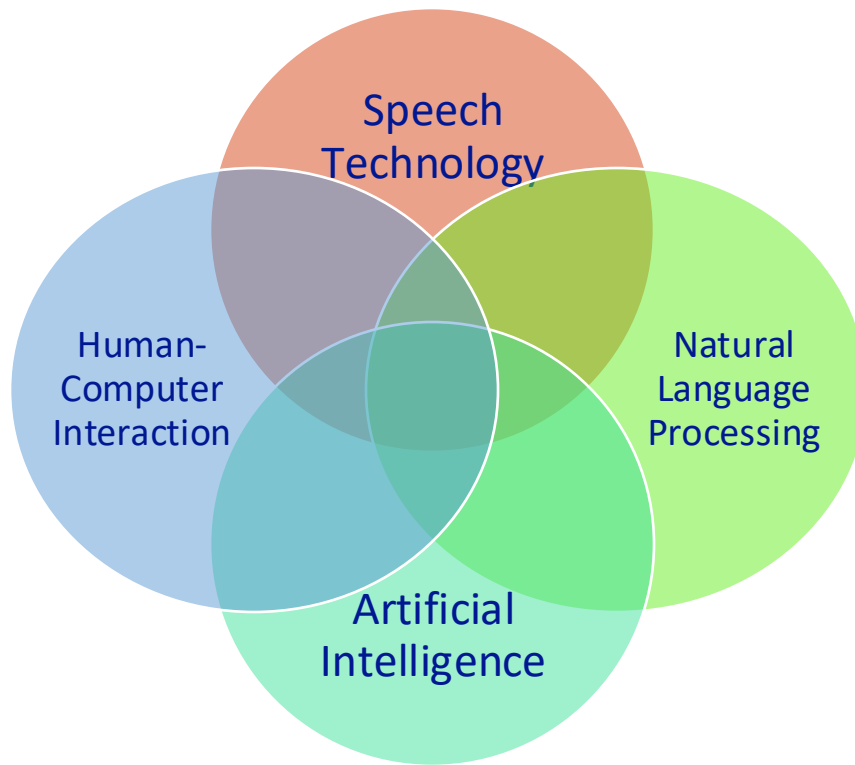
1. Introduction to Speech Technology



Nguyen Thi Thu Trang

Speech Technology

- Recognition, understanding and generation of human speech
- Why it Matters:
 - Communication efficiency in human-computer interaction
 - Accessibility for people with disabilities.



Applications of Speech Technology

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Applications of Speech Technology

- Virtual Assistant (e.g. Siri, Alexa)
- Speech Translation
- User Authentication
- Automated customer service systems
- Accessibility for people with disabilities, e.g. Screen Readers



Applications of Speech Technology

- **Healthcare:** Diagnosing Alzheimer's through speech patterns.
- **Education:** Language learning tools powered by TTS.
- **Accessibility:** Smart homes controlled via voice.
- **Entertainment:** Synthetic voices for personalized audiobooks.



Tasks of Speech Technology, e.g. TTS

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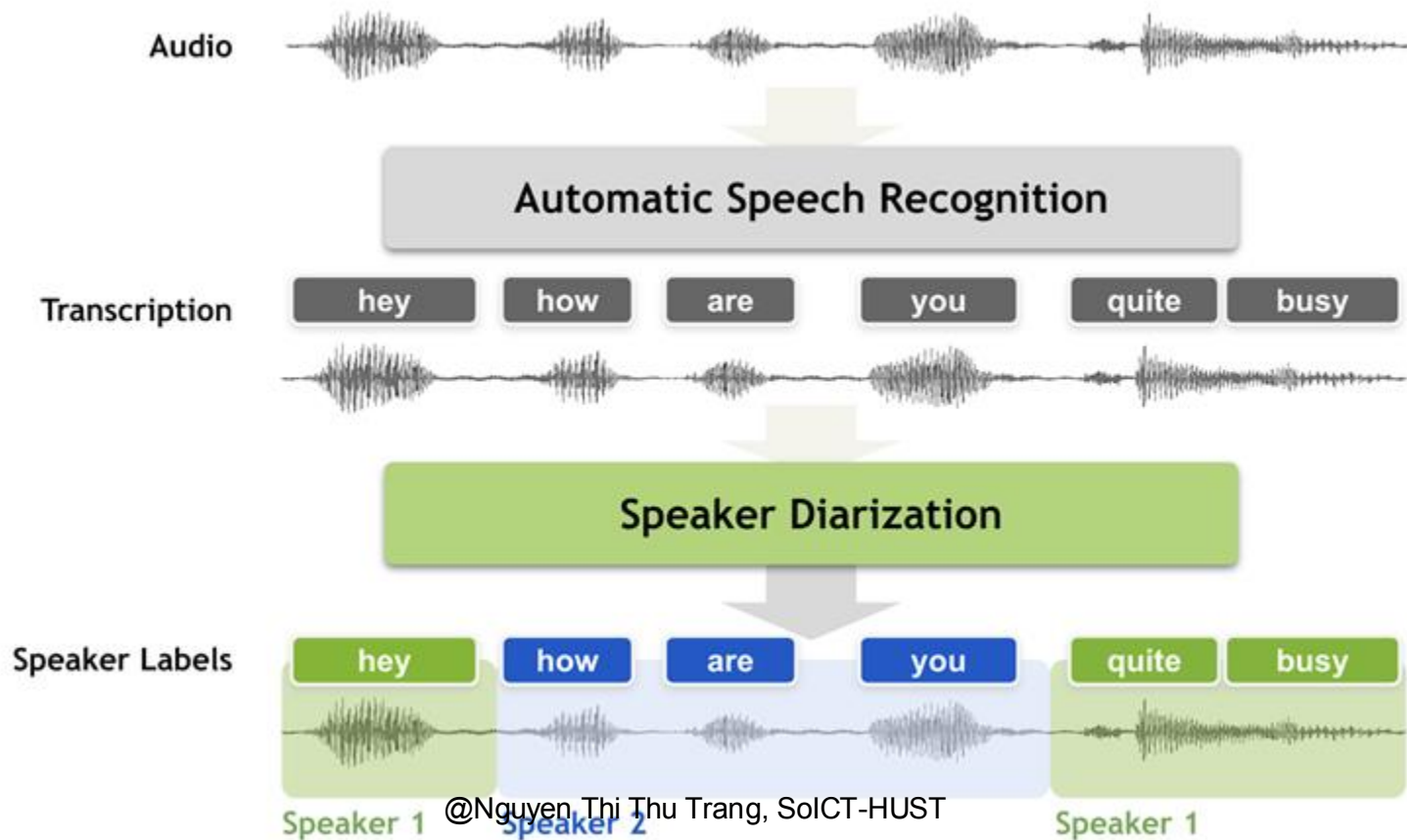
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Speech Technology Tasks

- Speaker Recognition
- Speaker Diarization
- Speech Emotion Recognition (SER)
- Automatic Speech Recognition (ASR)
- Spoken Language Understanding (SLU)
- Speech Synthesis / Text-To-Speech (TTS)
- Mispronunciation Detection and Diagnosis (MD&D)

Speech Technology Tasks



E.g. Speech Transcription



The screenshot shows a web browser displaying a news article on the VnExpress website. The article is titled "G8 bùng phiên họp chất vấn Chiều 8/11/2019" (G8 erupts in a questioning session on the evening of 8/11/2019). The main image shows a man in a suit speaking at a podium. The text of the article discusses the G8 summit and the questioning session. The article is written in Vietnamese and is dated 8/11/2019. The browser's address bar shows the URL "http://vnexpress.vn/". The page also features a sidebar with related news items and a bottom section with social media links.

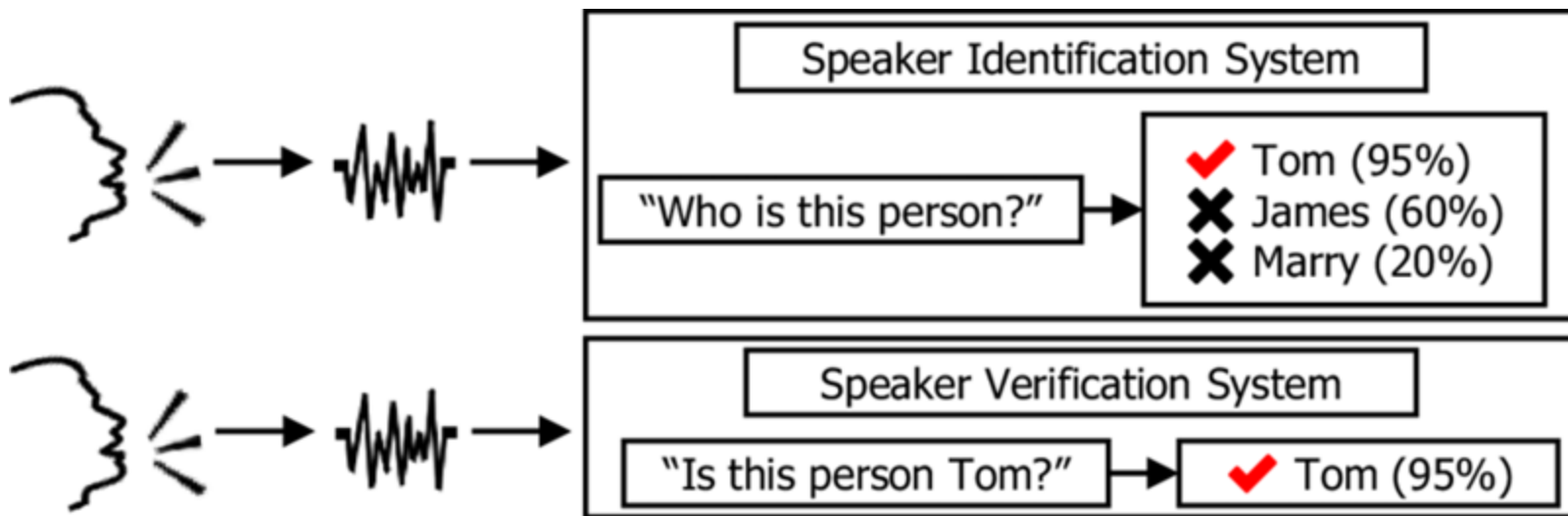
G8 bùng phiên họp chất vấn Chiều 8/11/2019

Chiều 8/11/2019, các báo và video trực tuyến đã đưa tin về phiên họp chất vấn của các thành viên G8.

Phiên họp chất vấn đã diễn ra vào chiều 8/11/2019, với sự tham gia của các thành viên G8. Phiên họp chất vấn đã diễn ra vào chiều 8/11/2019, với sự tham gia của các thành viên G8. Phiên họp chất vấn đã diễn ra vào chiều 8/11/2019, với sự tham gia của các thành viên G8.

Speech Technology Tasks

Speaker Recognition

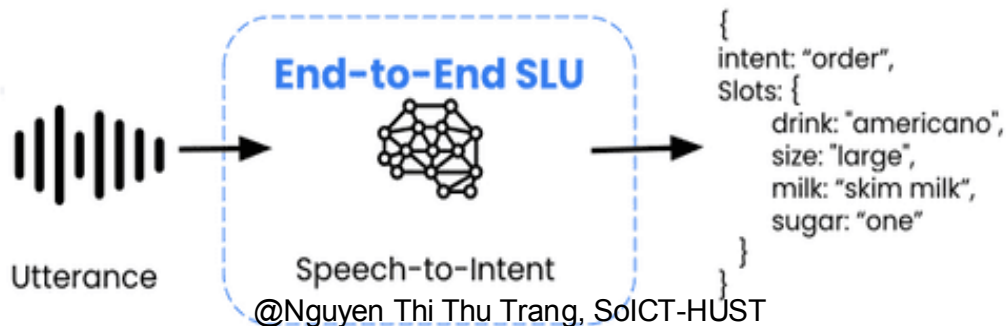
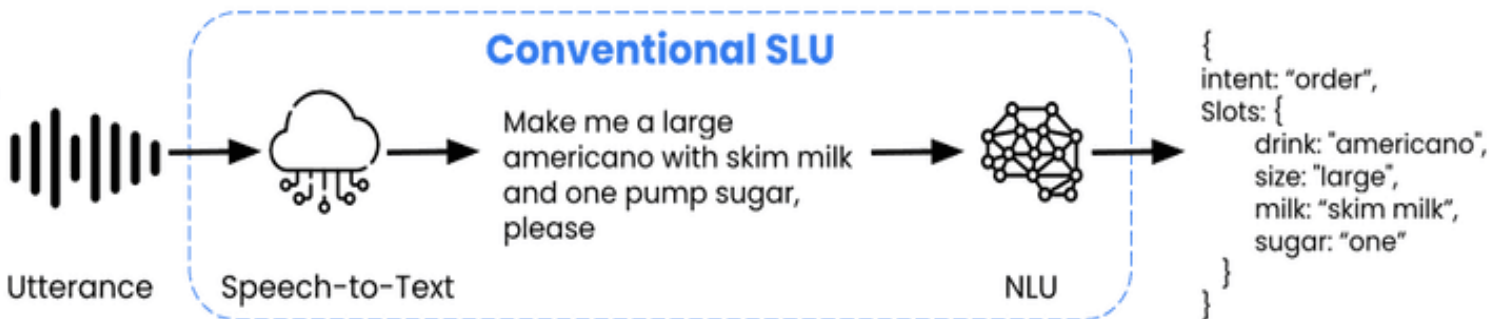


Nuance Voice Biometrics: Customer Verification/Identification (2018)



Spoken Language Processing Tasks

Spoken Language Understanding



Google Assistant: Calls Local Businesses To Make Appointments (2018)

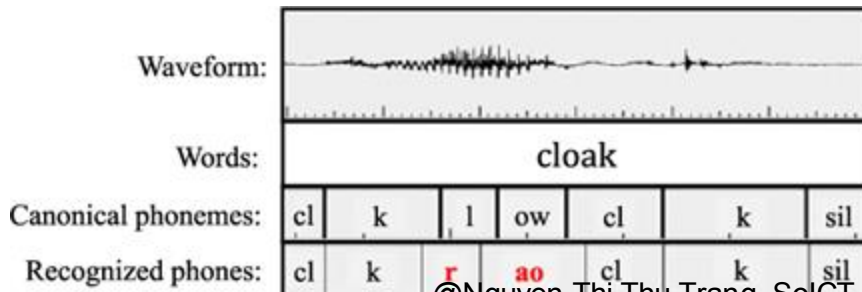


Spoken Language Processing Tasks

Mispronunciation Detection and Diagnosis (MD&D)



ELSA Speak



interesting /'ɪn.trɪ.stɪŋ/

73%

SOUND

YOU SAID

/r/

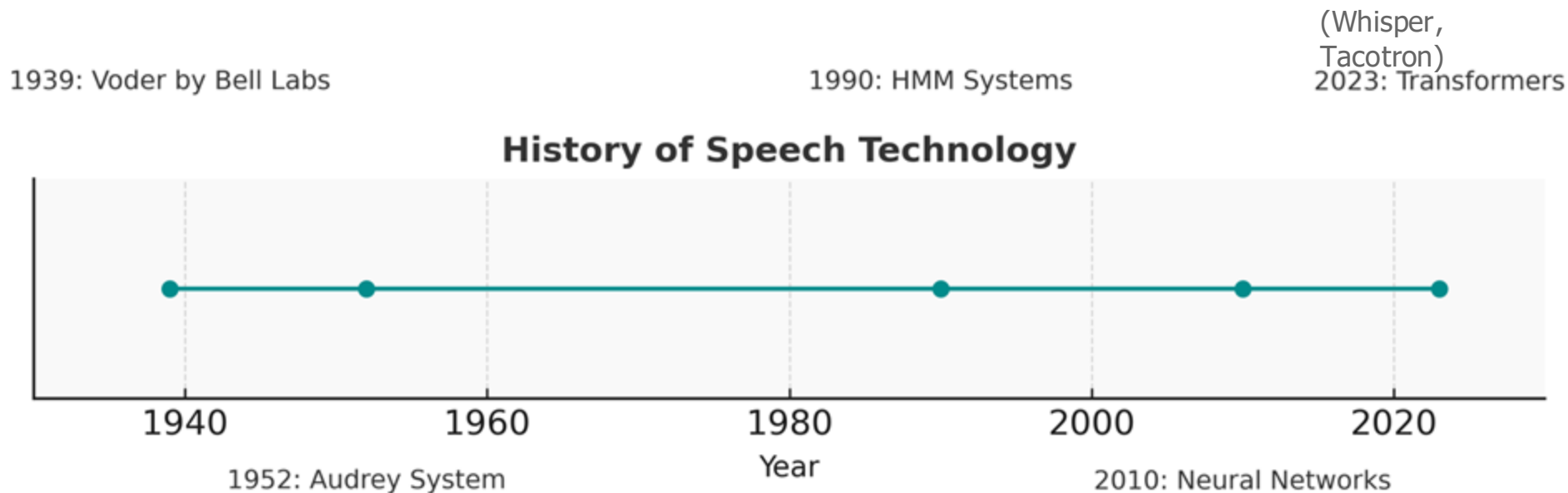
Incorrect

To pronounce the /r/ sound, round your lips and curl your tongue. Don't make contact with your teeth.

/ə/

Excellent

History of Speech Technology



(First automatic digit recognition)

**Comparison of accuracy from
early systems to modern ASR?**

What is the breakthrough difference
in accuracy
between early ASR systems and modern models?

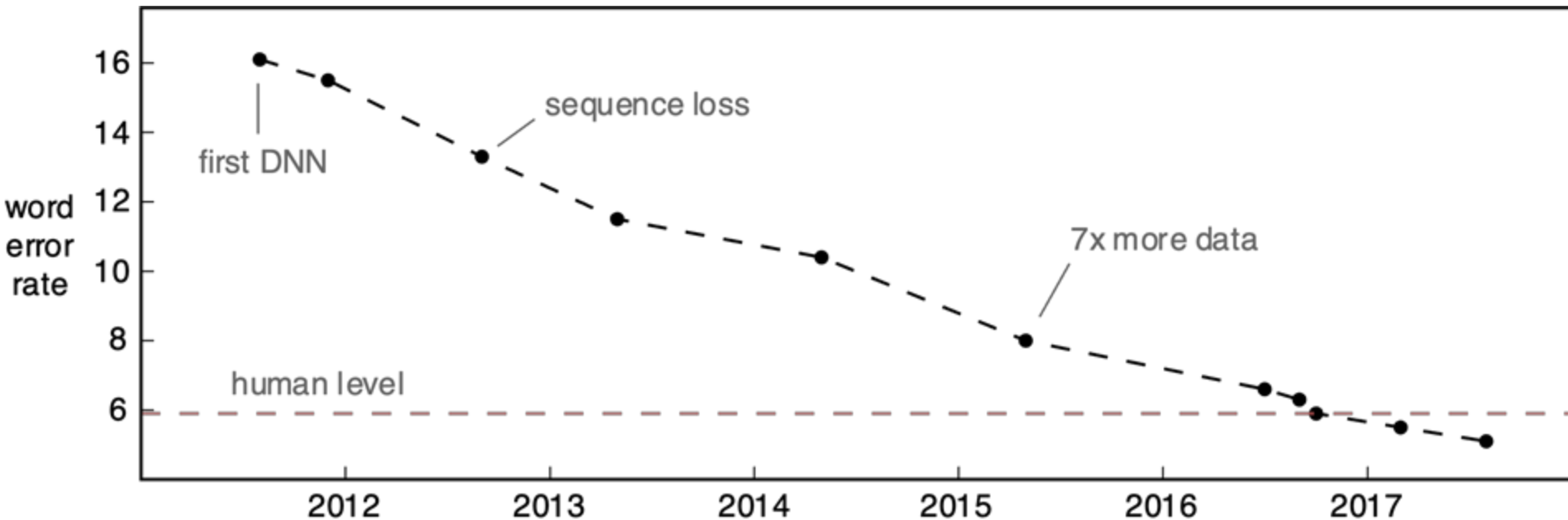
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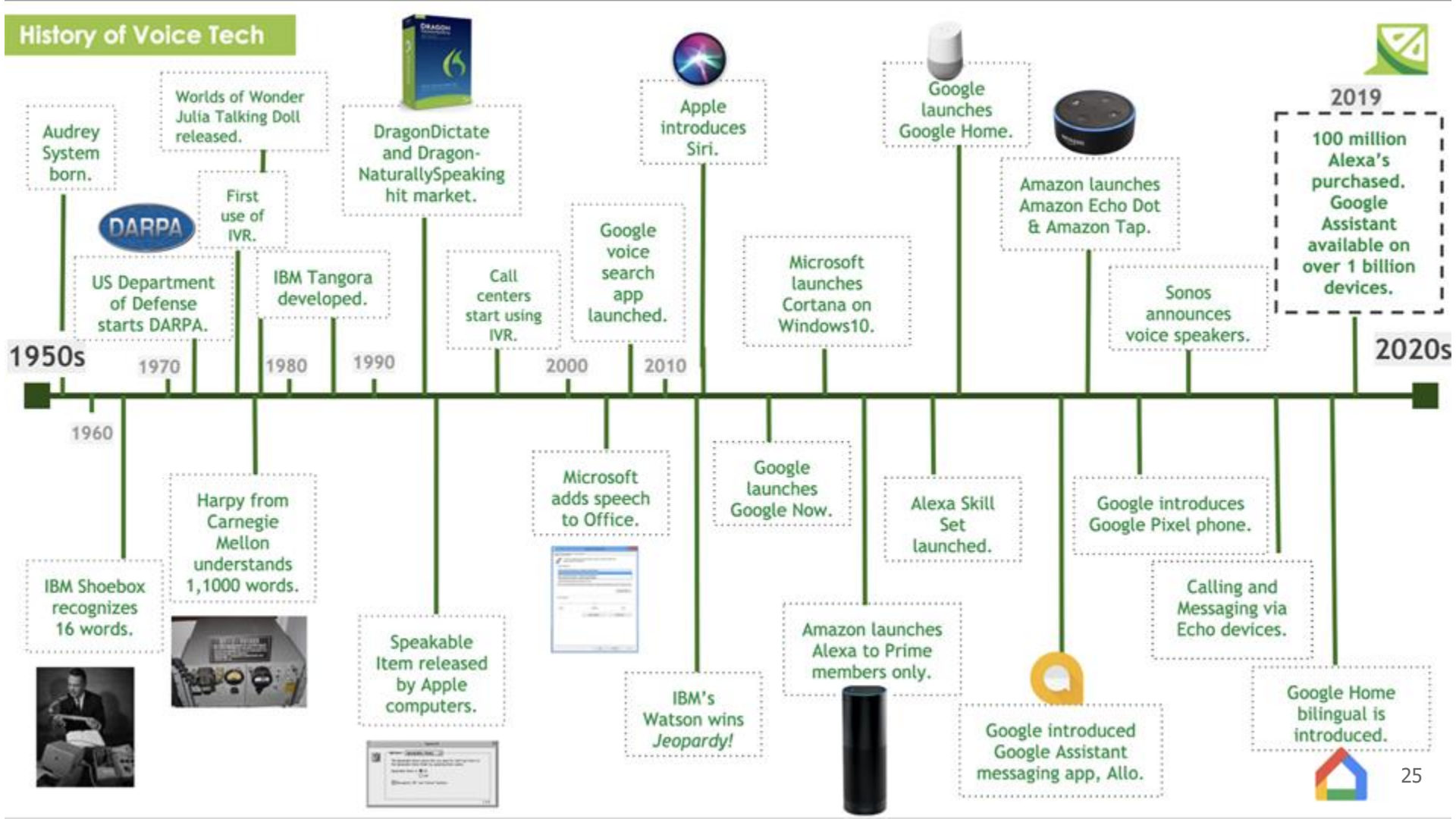


Challenges in ASR

Switchboard Conversational Speech Recognition (WER)



History of Voice Tech



Current Trends in Speech Technology

- **Deep Learning Dominance:**
 - End-to-end ASR systems outperforming traditional methods.
- **Real-Time Applications:**
 - On-device processing for privacy (e.g., offline voice commands).
- **Multimodal Systems:**
 - Combining speech with vision for enhanced understanding (e.g., lip-reading systems).
- **Ethical AI:**
 - Bias mitigation, responsible voice cloning policies.

Challenges in Speech Technology

- **Noise and Variability:**
 - Real-world environments vs. controlled datasets.
 - **Solution:** Data augmentation techniques.
- **Accents and Dialects:**
 - Underrepresentation in training data.
 - **Solution:** Collecting diverse datasets (e.g., Common Voice).
- **Data Scarcity:**
 - Low-resource languages facing neglect.
 - **Solution:** Transfer learning and synthetic data.
- **Ethical Concerns:**
 - Deepfake risks and privacy violations.
 - **Solution:** Regulation and ethical guidelines.

Pakistan's former prime minister is using an AI voice clone to campaign from prison



/ Imran Khan's party crafted a four-minute message using a tool from the AI firm ElevenLabs.

By [Amrita Khalid](#), one of the authors of audio industry newsletter Hot Pod. Khalid has covered tech, surveillance policy, consumer gadgets, and online communities for more than a decade.

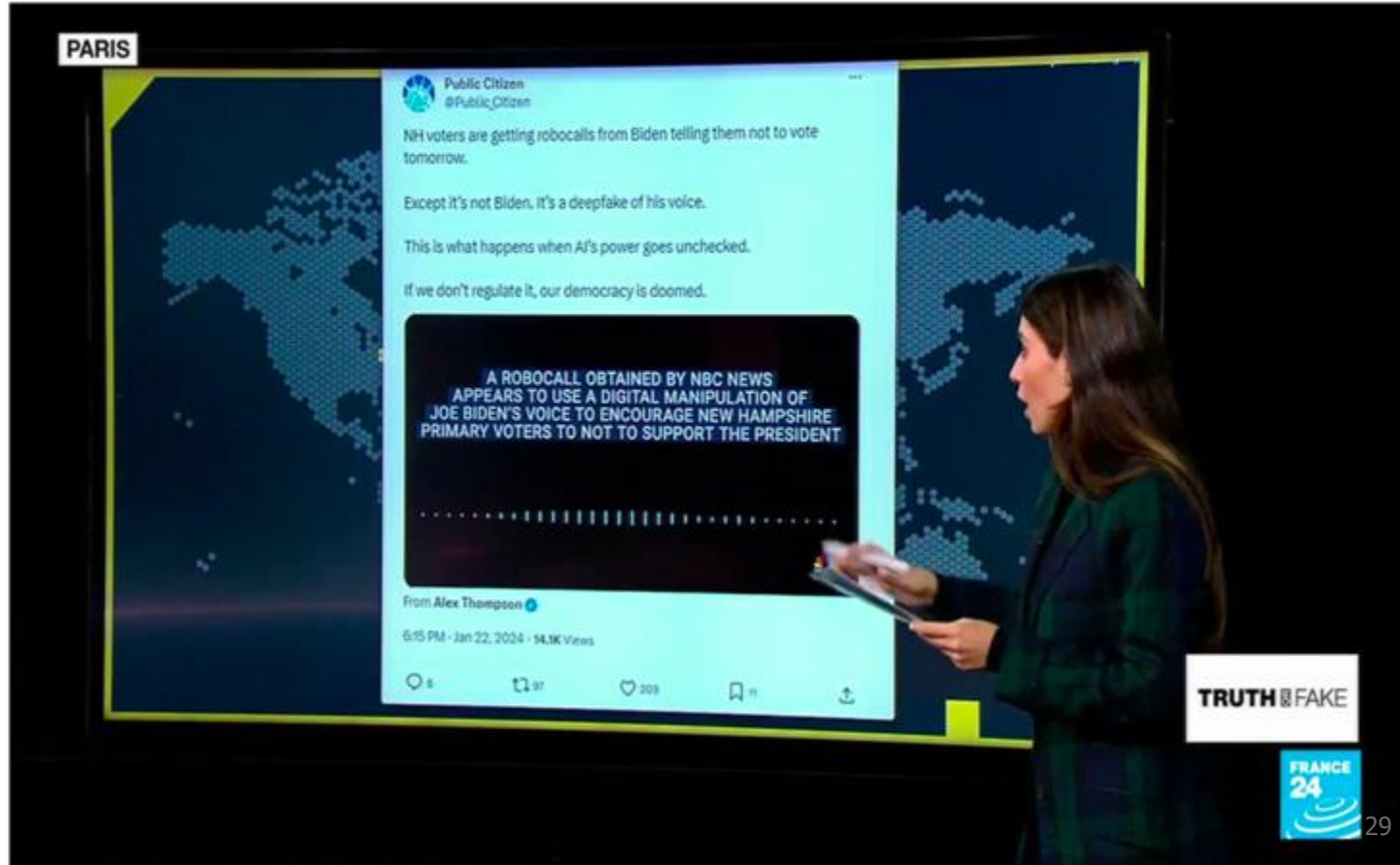
Dec 18, 2023, 2:41 PM PST

[P](#) [f](#) [t](#) | 1 Comment (1 New)

The Biden Deepfake Robocall Is Only the Beginning

An uncanny audio deepfake impersonating President Biden has sparked further fears from lawmakers and experts about generative AI's role in spreading disinformation.

A New Era of Spoken Language Applications and Impact



Future Directions

- **Universal Models:**
 - Speech systems that generalize across languages and domains.
- **Emotion Recognition:**
 - Enhanced user interaction.
- **Low-Power Models:**
 - Expanding access to speech technology in remote areas.

Vision: Democratizing speech technology to make it accessible and ethical for everyone.

Conversational AI on device (Vbee 2024)



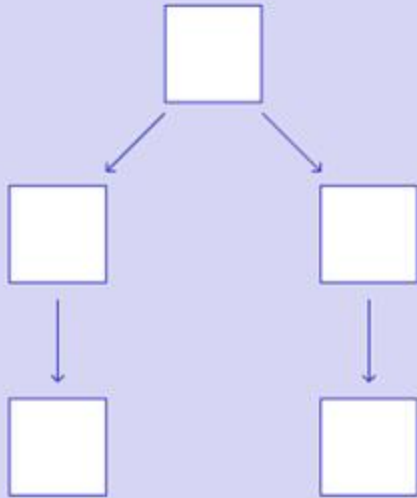
2. Linguistics and Phonetics



Nguyen Thi Thu Trang

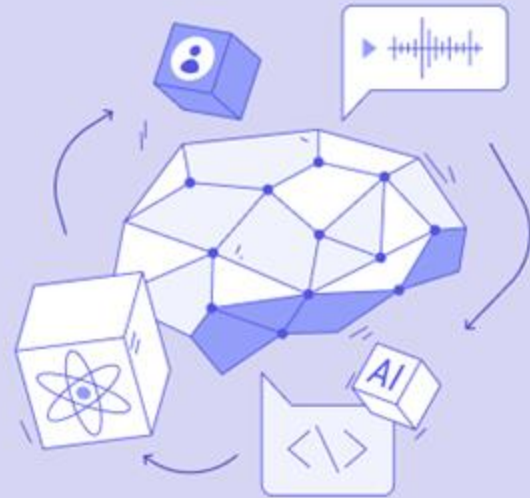
What is the main difference between
Rule-based AI and
Machine Learning?

Rule-Based Systems



*Predefined business rules and
business logic by experts*

Machine Learning Systems



*Learns patterns from large datasets
over time*

What is the main difference between
Machine Learning and
Deep Learning?

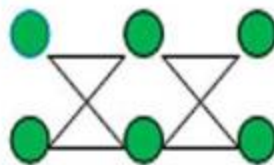
Machine Learning



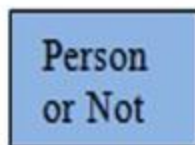
Input image



Feature Extraction



Classification

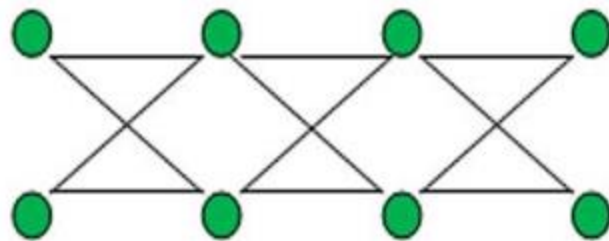


Output

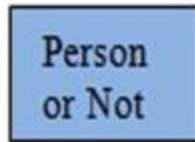
Deep Learning



Input image



Feature Extraction + Classification



Output

Phonology

vs.

Phonetics

- **Sound systems** of a language.
- How **sounds function** in a particular language or languages.
- **Abstract, mental representations of sounds.**

- **Physical production and perception** of speech sounds.
- **Actual acoustic and articulatory properties** of sounds.
- **Universal properties of speech sounds**, not language-specific

rules



Why Phonology and Phonetics?

Do we need phonology and phonetics to build systems that accurately process spoken language?

- Modern systems (based on deep learning) are far less reliant on encoding phonetic domain knowledge directly than previous approaches
- Allowing deep learning models to learn letter-sound mappings from data can perform much better than hand engineering phonetic structure into a recognition or synthesis system

Why Phonology and Phonetics?

However ...

- Basic understanding of phonology and phonetics and speech production helps with describing and debugging spoken language systems
 - E.g. how does an accent/tone change the sound of pronunciations?
- Phonetic categories are not arbitrary. They model the biology of how humans produce speech
 - Understanding the space of possible speech sounds gives a nice perspective on comparing spoken languages across the world, and how they evolve

What is Phonetics?

- **Definition:**

- The study of speech sounds, their production, transmission, and perception.

- **Three Main Branches:**

- **Articulatory Phonetics:** How speech sounds are produced.
- **Acoustic Phonetics:** Physical properties of speech sounds.
- **Auditory Phonetics:** How speech sounds are perceived by the listener

Example: Different between /b/ and /p/ sounds



Which parts of the body are involved in producing speech?

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Letter vs. Sound



Grapheme vs. Phoneme

- **Grapheme**

The smallest written unit representing a sound in speech.

Examples: "d", "th", "ngh", "a", "au".

- **Phoneme**

The smallest unit of sound in speech.

Examples: /d/, /th/, /ŋ/, /a/, /ăw/.

IPA (International Phonetic Alphabet)

BẢNG PHIÊN ÂM TIẾNG ANH IPA

VOWELS	monophthongs				diphthongs		Phonemic Chart	
	i:	ɪ	ʊ	u:	ɪə	eɪ		
	leave	live	put	pool	year	paper		
	e	ə	ɜ:	ɔ:	ʊə	ɔɪ		
	pen	a	girl	door	tourist	boy	voiced unvoiced	
	æ	ʌ	ɑ:	ɒ	eə	aɪ		
	man	cup	car	hot	chair	fine	Oʊ(əʊ)	
						house		
CONSONANTS	p	b	t	d	tʃ	dʒ	k	g
	pen	baby	table	door	chip	January	key	girl
	f	v	θ	ð	s	z	ʃ	ʒ
	fan	van	think	this	sun	zoo	shoes	vision
	m	n	ŋ	h	l	r	w	j
	mouth	nose	ring	hear	letter	rain	window	yellow

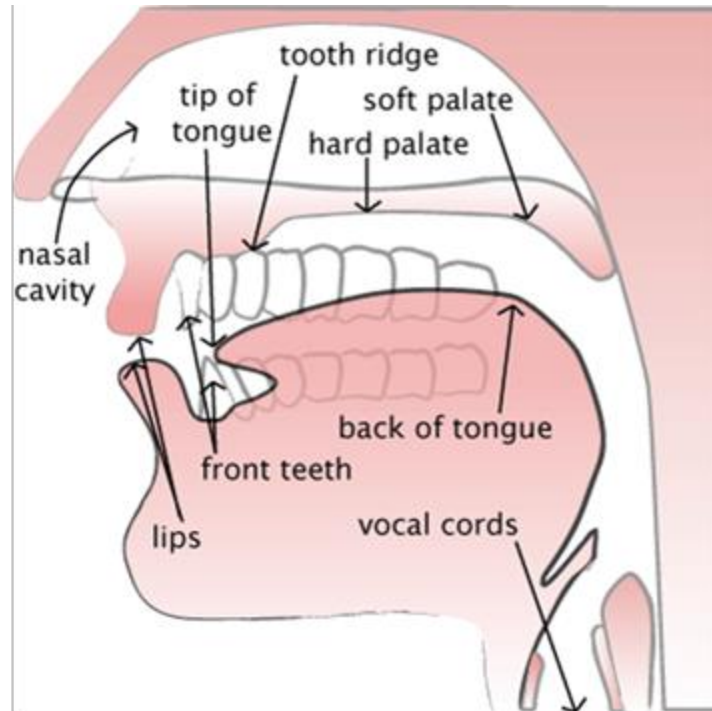
leave /li:v/
manly /'mænli/

Speech Production

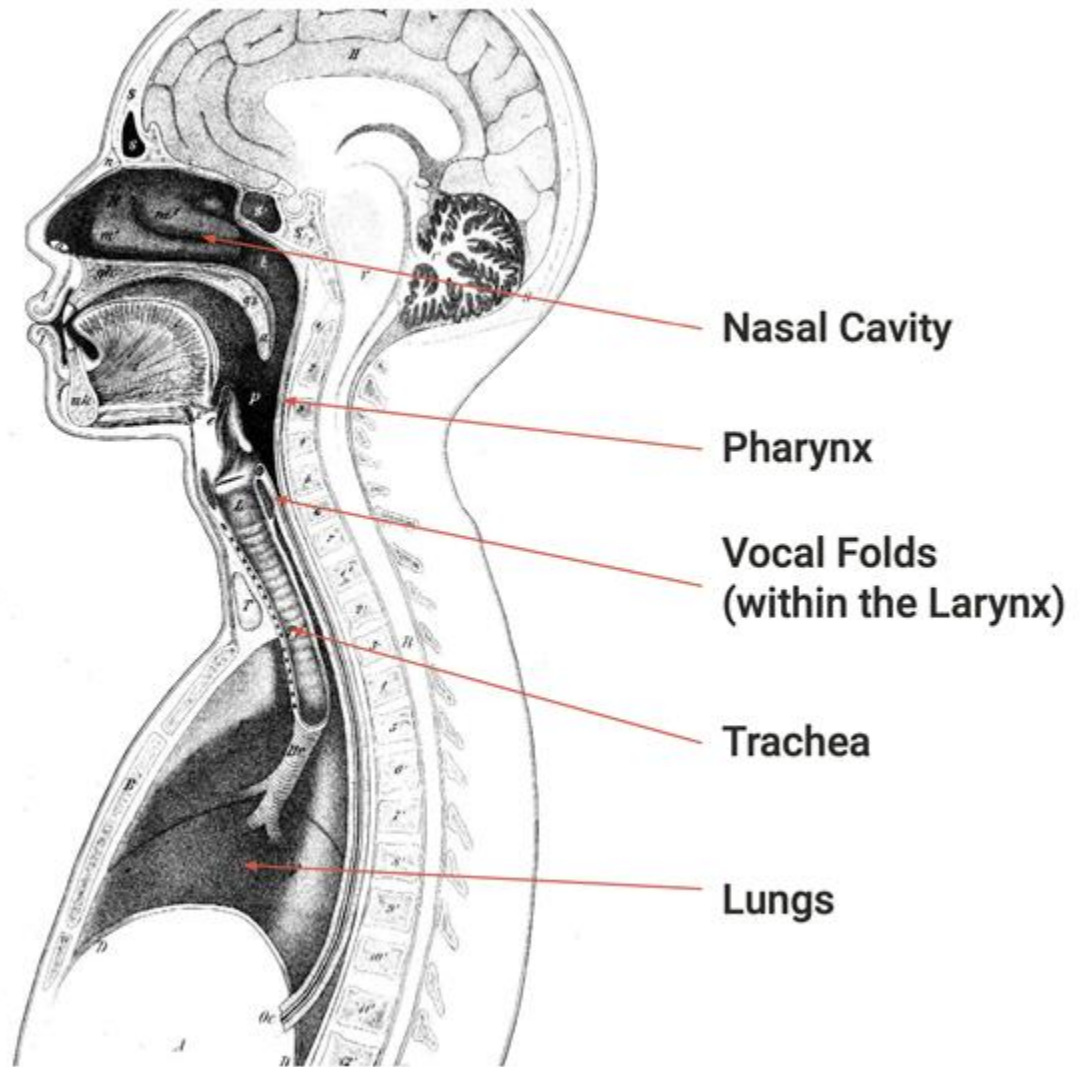
- **Flow:** we (normally) speak while breathing out. “Pulmonic egressive airstream” (Luồng khí từ phổi)
 - Airstream sets vocal folds in motion. Vibration of vocal folds (rung dây thanh âm) produces sounds.
- **Resonance** (Điều chỉnh âm thanh): shape of vocal tract causing harmonics
- **Articulation** (Phát âm): manipulation of airflow
 - Oral tract (Đường miệng): uvula (lưỡi gà), soft palate (velum - vòm miệng mềm), hard palate, tongue, lips, teeth
 - Nasal tract (Đường mũi)

Làm thế nào để tạo ra tiếng nói?

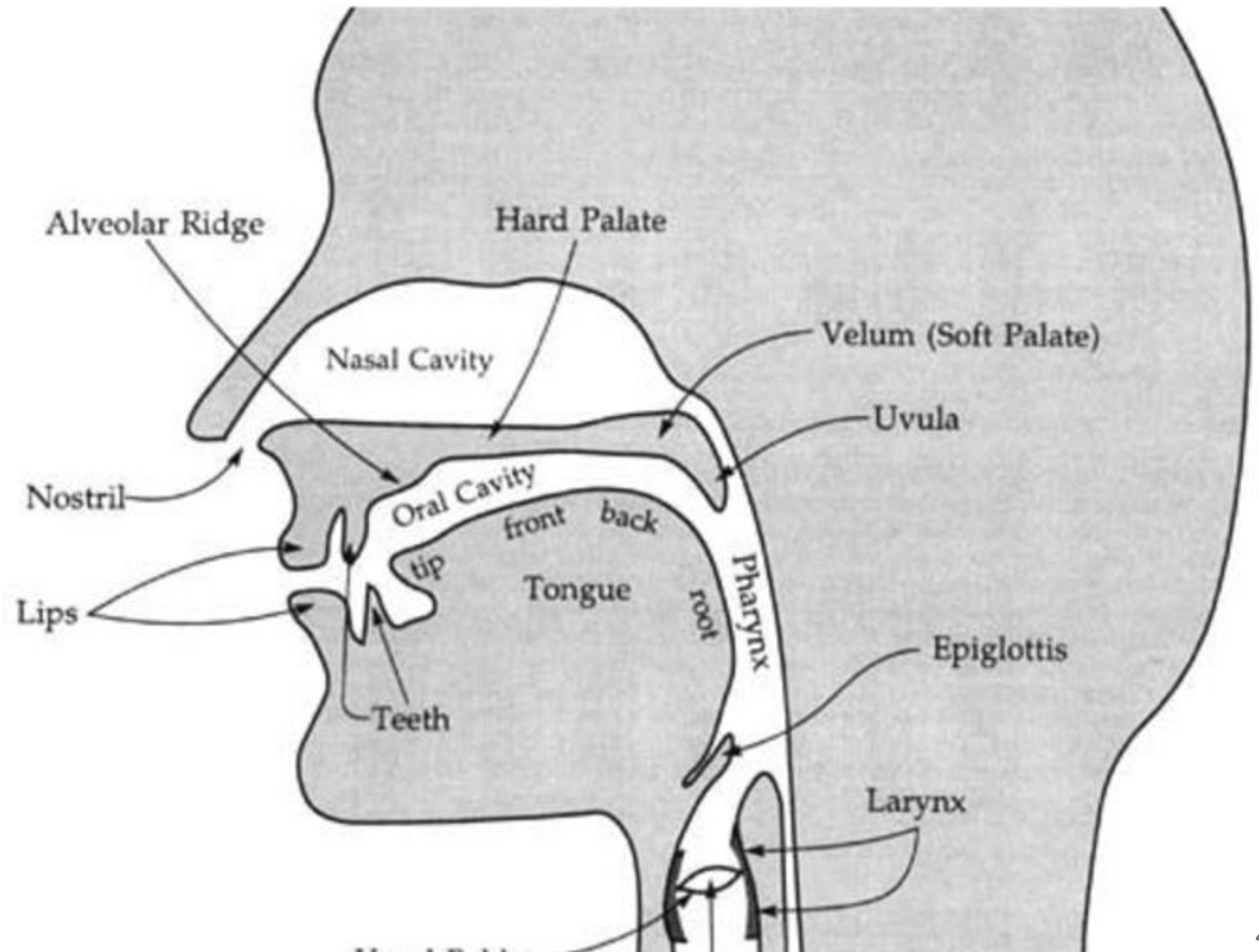
- **Hít vào và chuẩn bị không khí:** Phổi hít vào và điều chỉnh luồng khí qua cơ hoành (diaphragm).
- **Đẩy không khí qua thanh quản:** Dây thanh âm rung tạo âm thô.
- **Hình thành âm thanh:** Lưỡi, môi và khoang miệng định hình âm thanh.
- **Điều chỉnh âm rõ ràng:** Phổi hợp lưỡi, môi, răng và hàm để



Sagittal
section of
the vocal
tract

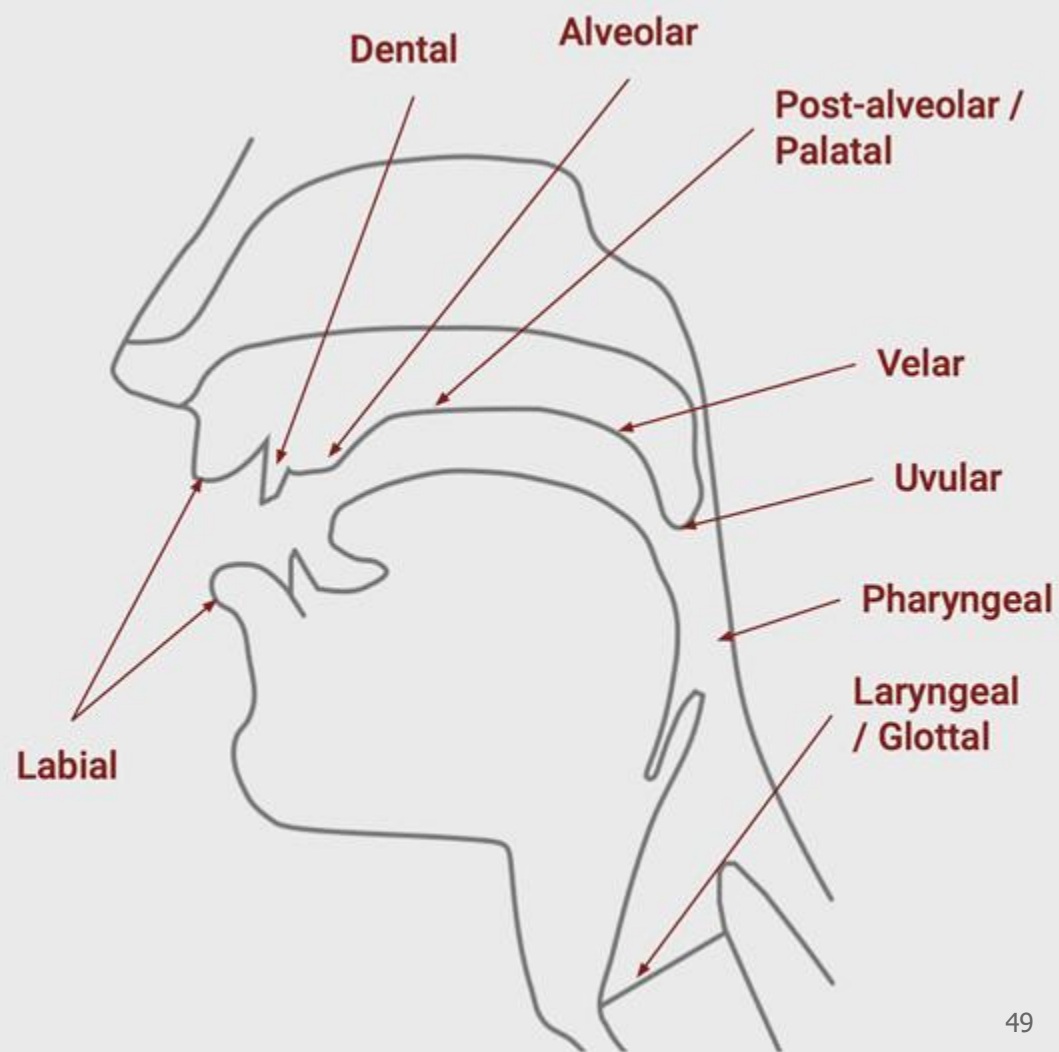


Sagittal
section of
the vocal
tract



Place of Articulation

- **Consonants** are classified according to the location where the airflow is most constricted
 - This is called **place of articulation**
- Three major kinds of place articulation:
 - **Labial** (with lips)
 - **Coronal** (using tip or blade of tongue)
 - **Dorsal** (using back of tongue)



THE INTERNATIONAL PHONETIC ALPHABET (revised to 2020)

CONSONANTS (PULMONIC)

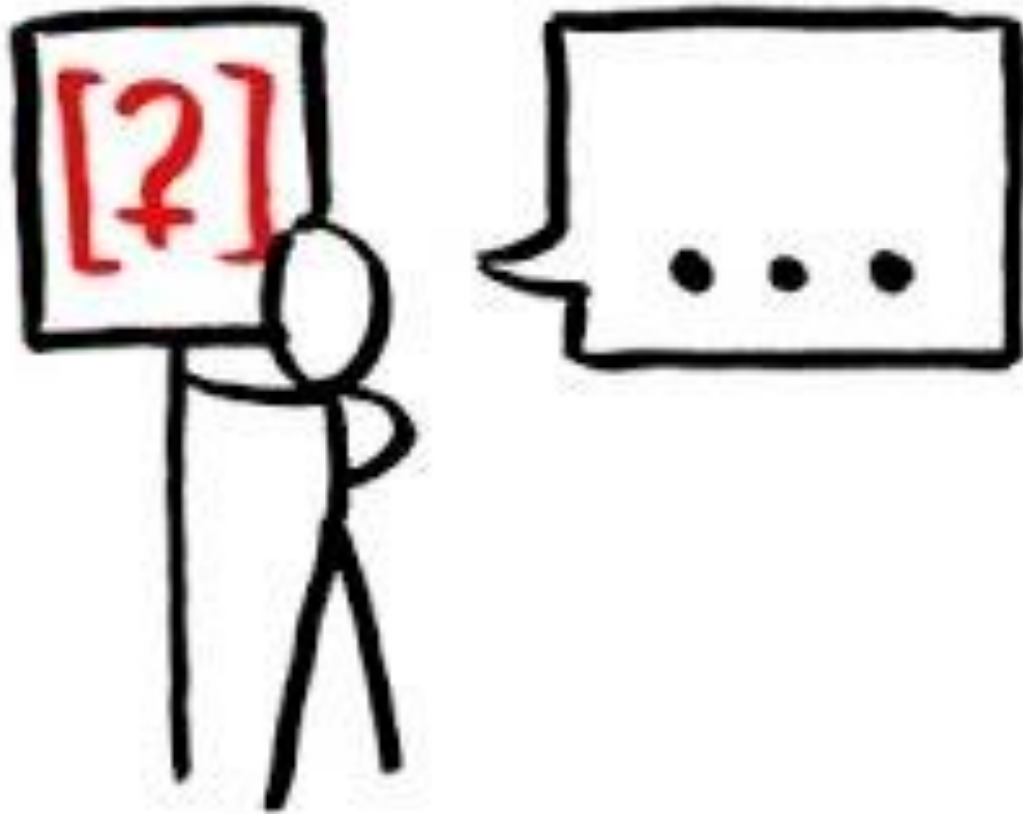
© 2020 IPA

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b			t d		ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ		n		ɳ	ɲ	ŋ	ɴ		
Trill	ʙ			r					ʀ		
Tap or Flap		ⱱ		ɾ		ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative				ɬ ɮ							
Approximant		ʋ		ɹ		ɻ	j	ɰ			
Lateral approximant				l		ɭ	ʎ	ʟ			

Symbols to the right in a cell are voiced, to the left are voiceless. Shaded areas denote articulations judged impossible.

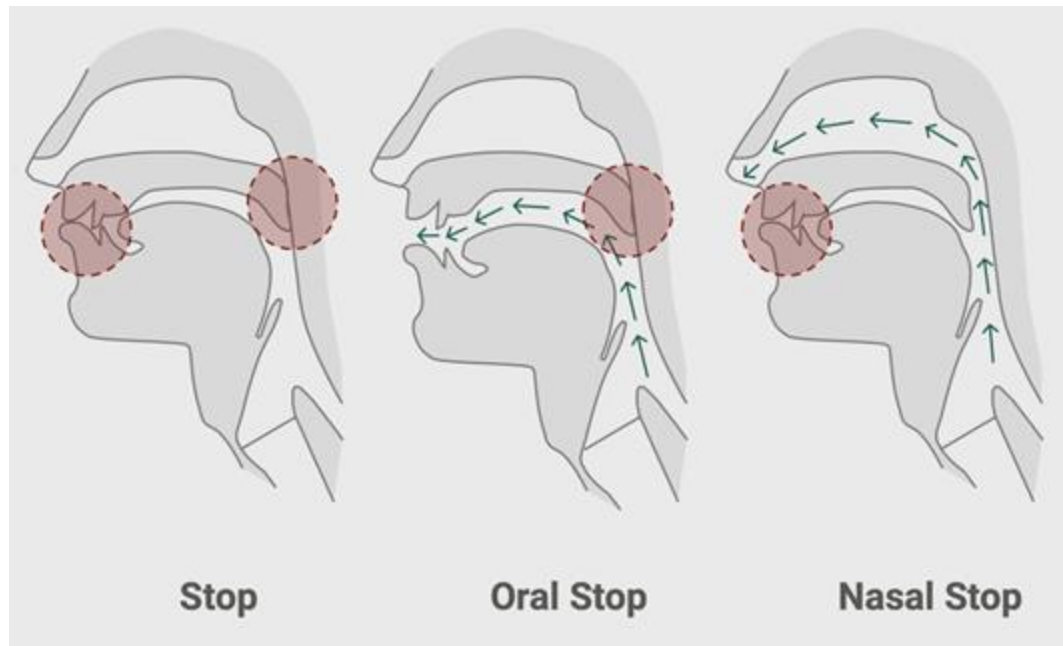
Place of Articulation

Place of Articulation

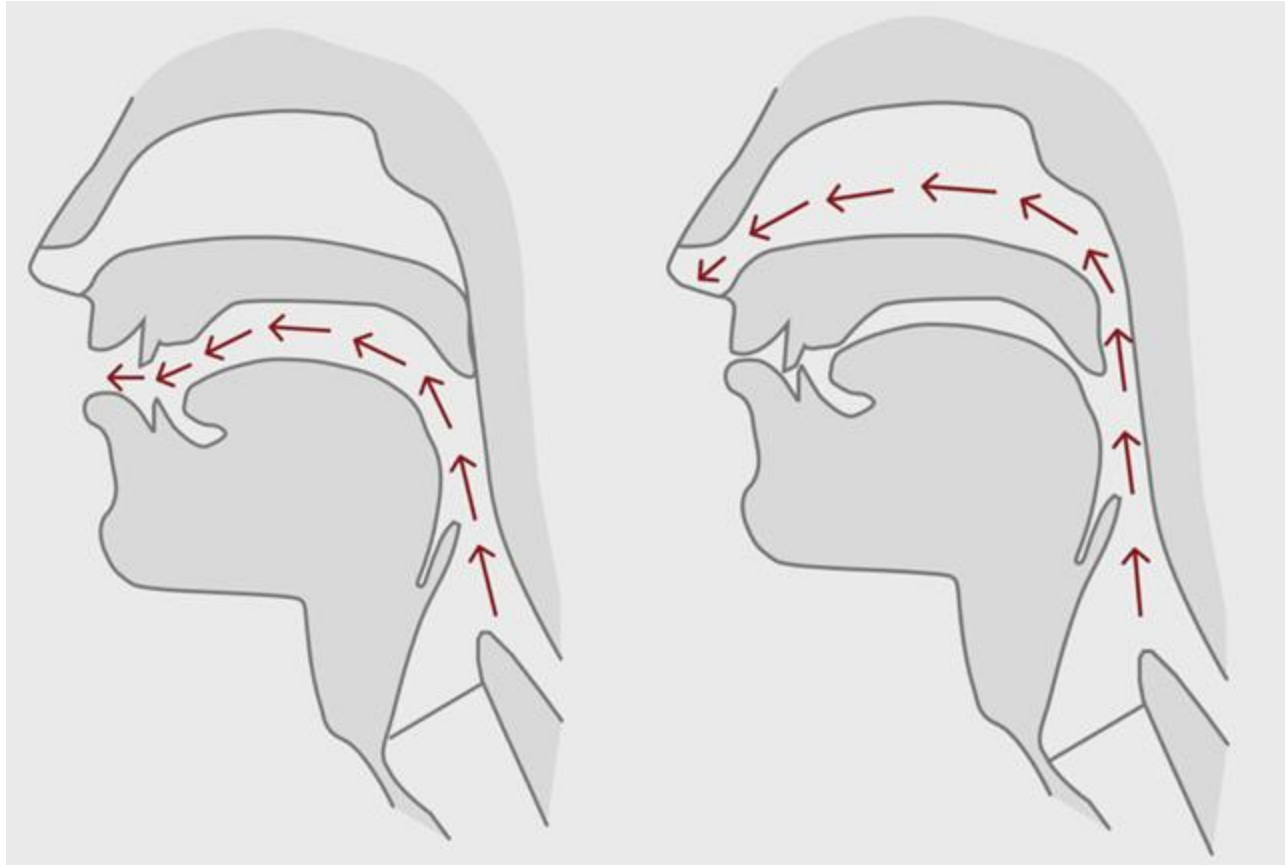


Manner of Articulation

- **Stop (Âm tắc):** complete closure of articulators, so **no air** escapes **through mouth**
- **Oral stop (Âm tắc miệng):** palate is raised, **no air** escapes **through nose**. Air pressure builds up behind closure, explodes when released
 - p, t, k, b, d, g
- **Nasal stop (Âm tắc mũi):** oral closure, but palate is lowered, **air escapes through nose**
 - m, n, ng

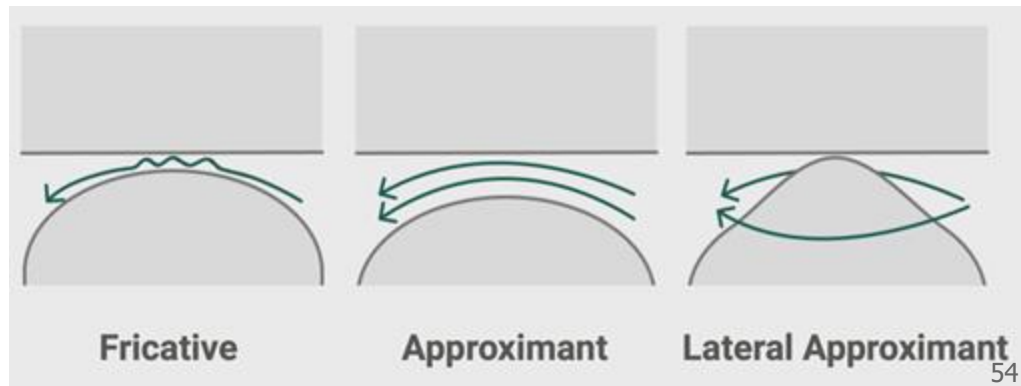


Oral vs. Nasal Sounds



Articulation Manner of Consonants

- **Fricatives (Âm xát):** close approximation of two articulators, resulting in turbulent airflow between them
 - f, v, s, z, th, dh
- **Approximant (Âm xấp):** not quite-so-close approximation of two articulators, so no turbulence
 - y, r
- **Lateral approximant (Âm xấp bên):** obstruction of airstream along center of oral tract, with opening around sides of tongue
 - l



Manner of Articulation



THE INTERNATIONAL PHONETIC ALPHABET (revised to 2020)

CONSONANTS (PULMONIC)

© 2020 IPA

	Bilabial	Labiodental	Dental	Alveolar	Postalveolar	Retroflex	Palatal	Velar	Uvular	Pharyngeal	Glottal
Plosive	p b		t d			ʈ ɖ	c ɟ	k ɡ	q ɢ		ʔ
Nasal	m	ɱ	n			ɳ	ɲ	ŋ	ɴ		
Trill	ʙ		r						ʀ		
Tap or Flap		ⱱ	ɾ			ɽ					
Fricative	ɸ β	f v	θ ð	s z	ʃ ʒ	ʂ ʐ	ç ʝ	x ɣ	χ ʁ	ħ ʕ	h ɦ
Lateral fricative			ɬ ɮ								
Approximant		ʋ	ɹ			ɻ	j	ɰ			
Lateral approximant			l			ɭ	ʎ	ʟ			

Symbols to the right in a cell are voiced, to the left are voiceless. Shaded areas denote articulations judged impossible.

**Manner of
Articulation**

Voiced

vs.

Unvoiced Consonants

Presence or absence of vocal cord vibration during the articulation of consonants.

- Vocal cords **vibrate** during pronunciation.
- Air from the lungs causes the vocal cords to rapidly open and close, producing sound waves.
- **Examples:**
 - **/b/** as in *bat*: vocal cords vibrate.
 - **/d/** as in *dog*.
 - **/z/** as in *zoo*.
 - **/v/** as in *van*.

- Vocal cords **do not vibrate** during pronunciation.
- Sound is produced solely by the airflow and contact between the articulators without vocal cord vibration.
- **Examples:**
 - **/p/** as in *pen*: vocal cords do not vibrate.
 - **/t/** as in *top*.
 - **/s/** as in *snake*.
 - **/f/** as in *fish*.

Voiced and Unvoiced Consonants



THE INTERNATIONAL PHONETIC ALPHABET (revised to 2020)

CONSONANTS (PULMONIC)

© 2020 IPA

	Bilabial		Labiodental		Dental		Alveolar		Postalveolar		Retroflex		Palatal		Velar		Uvular		Pharyngeal		Glottal	
Plosive	p	b					t	d			ʈ	ɖ	c	ɟ	k	g	q	ɢ			ʔ	
Nasal		m		ɱ				n				ɳ		ɲ		ŋ		ɴ				
Trill		ʙ						r										ʀ				
Tap or Flap				ⱱ				ɾ				ɽ										
Fricative	ɸ	β	f	v	θ	ð	s	z	ʃ	ʒ	ʂ	ʐ	ç	ʝ	x	ɣ	χ	ʁ	ħ	ʕ	h	ɦ
Lateral fricative								ɬ	ɮ													
Approximant				ʋ				ɹ				ɻ		j		ɰ						
Lateral approximant								l				ɭ		ʎ		ʟ						

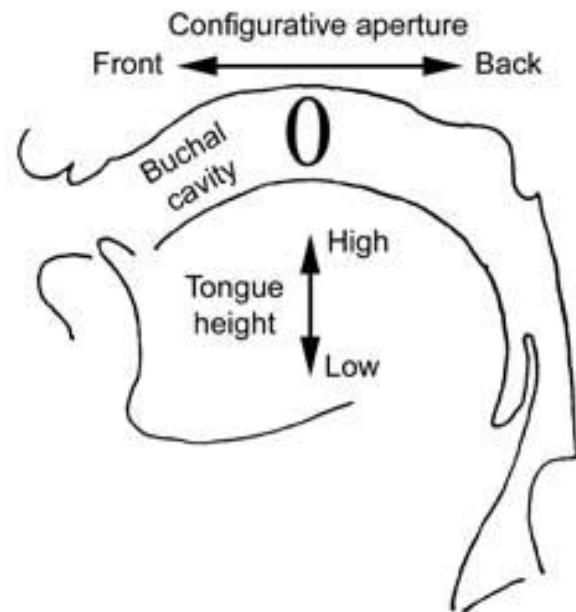
Symbols to the right in a cell are voiced, to the left are voiceless. Shaded areas denote articulations judged impossible.

Unvoiced: Left, Voiced: Right

Tongue Position of Vowels: Tongue Height

How high or low the tongue is in the mouth when producing a vowel:

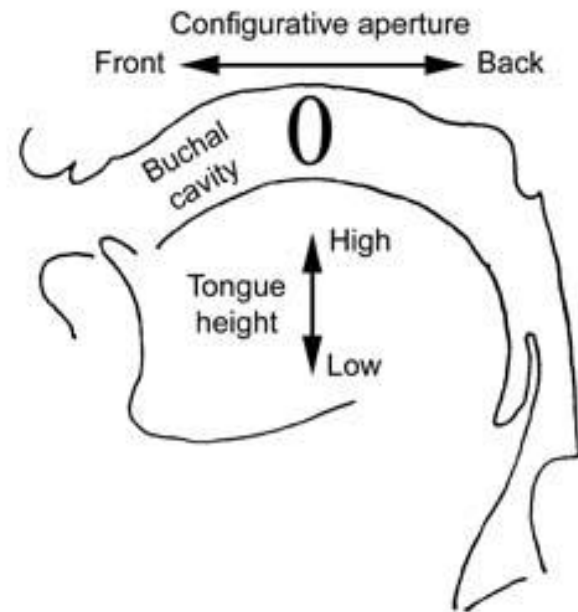
- **High vowels:** The tongue is raised close to the roof of the mouth.
 - Examples: /i/ (*see*), /u/ (*blue*).
- **Mid vowels:** The tongue is positioned midway between high and low.
 - Examples: /e/ (*bed*), /o/ (*go*).
- **Low vowels:** The tongue is lowered towards the bottom of the mouth.
 - Examples: /æ/ (*cat*), /a/ (*father*).



Tongue Position of Vowels: Tongue Backness

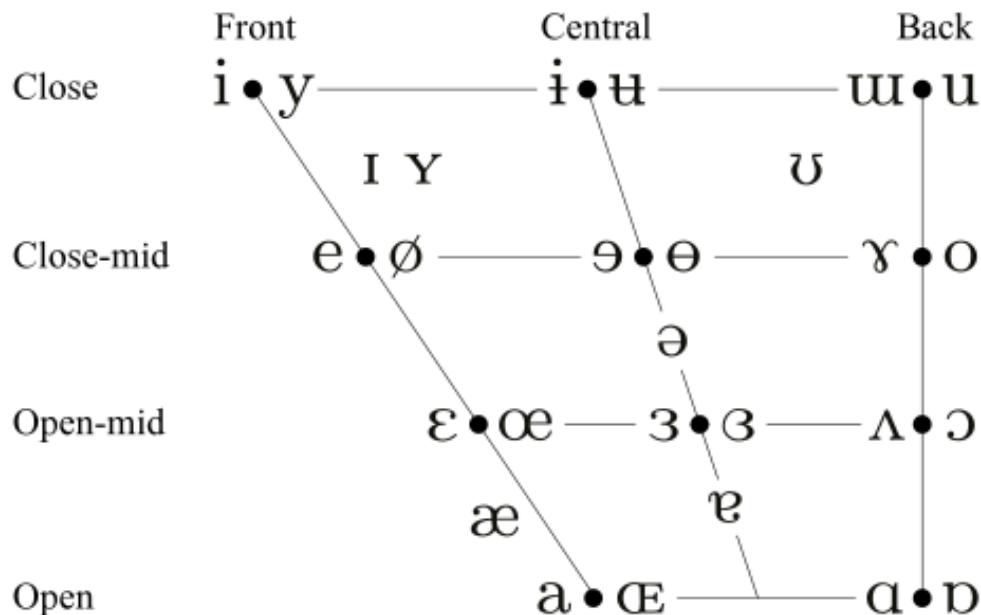
How far forward or backward the tongue is in the mouth:

- **Front vowels:** The tongue is positioned towards the front of the mouth.
 - Examples: /i/ (*see*), /e/ (*bed*).
- **Central vowels:** The tongue is centrally positioned.
 - Examples: /ə/ (*about*), /ʌ/ (*cup*).
- **Back vowels:** The tongue is positioned towards the back of the mouth.
 - Examples: /u/ (*blue*), /o/ (*go*).



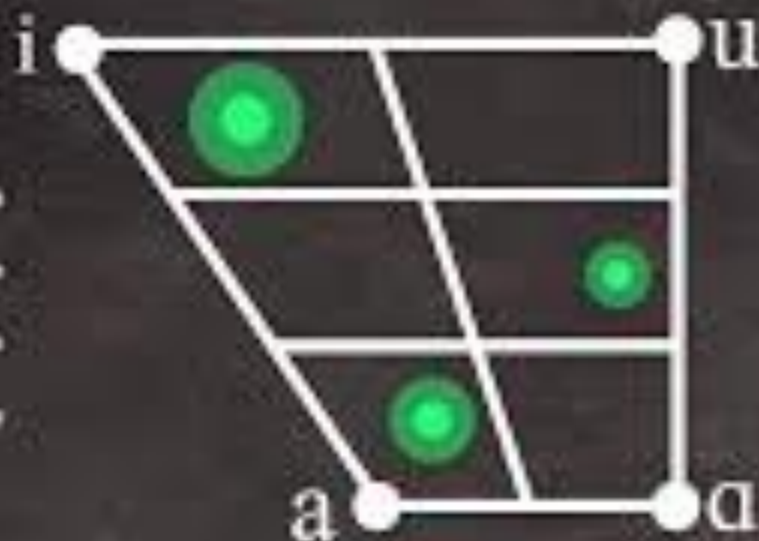
THE INTERNATIONAL PHONETIC ALPHABET (revised to 2020)

VOWELS



A E I O U

The VOWEL GRID explained!



What is Phonetics?

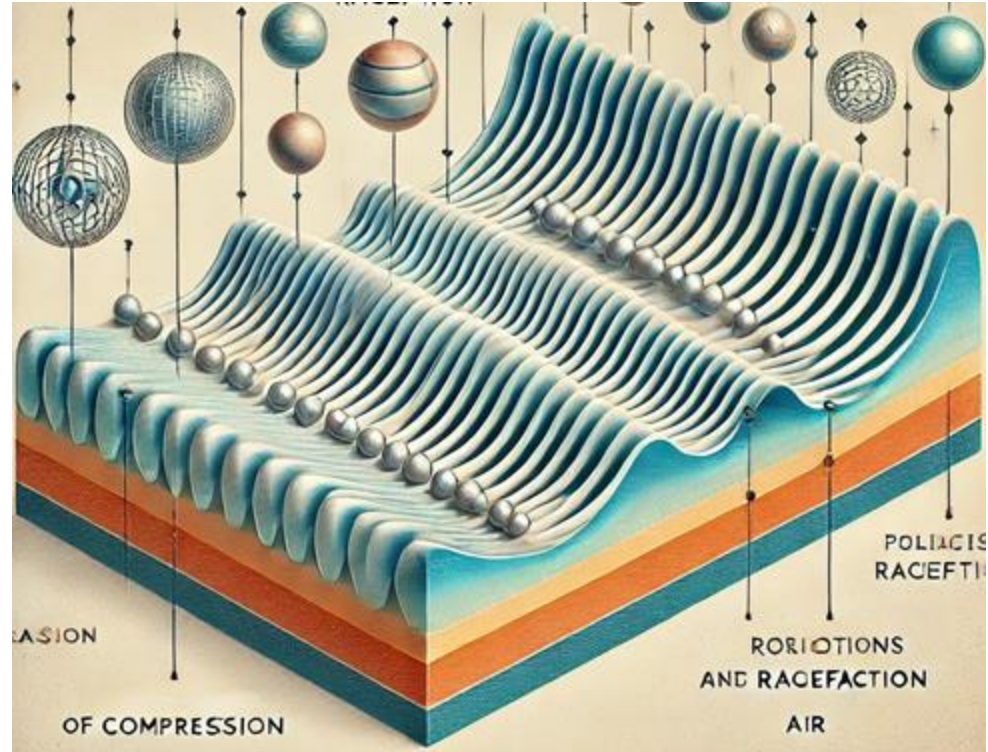
- **Definition:**
 - The study of speech sounds, their production, transmission, and perception.
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 - **Auditory Phonetics:** How speech sounds are perceived by the listener

Example: Different between /b/ and /p/ sounds



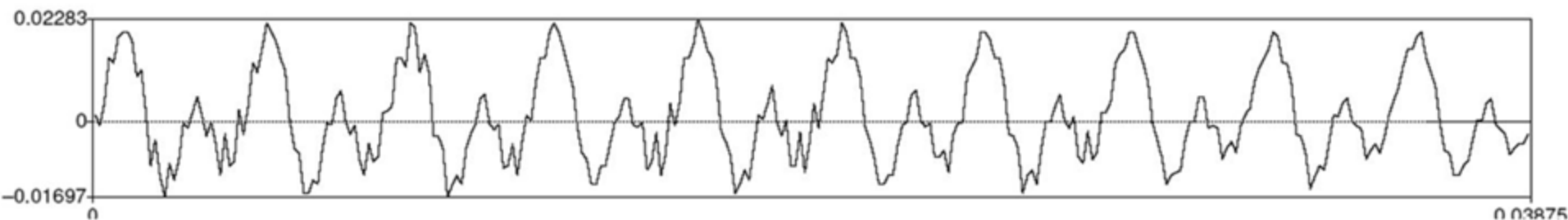
Sound Waves are Longitudinal Waves

- Particles of the medium vibrate parallel to the direction of the wave's propagation
 - e.g. Sound waves, P-waves, pressure waves....
- Repeating patterns described by **wavelength (bước sóng)**, **frequency**, **amplitude**, and **speed**



Back to Waves: Fundamental Frequency

- Waveform of the vowel /iy/



- Frequency: $10 \text{ repetitions} / 0.03875 \text{ seconds} = 258 \text{ Hz}$
- This is speed that vocal folds move, hence voicing
- Each peak corresponds to an opening of the vocal folds
- The low frequency of the complex wave is called the fundamental frequency of the wave or F0

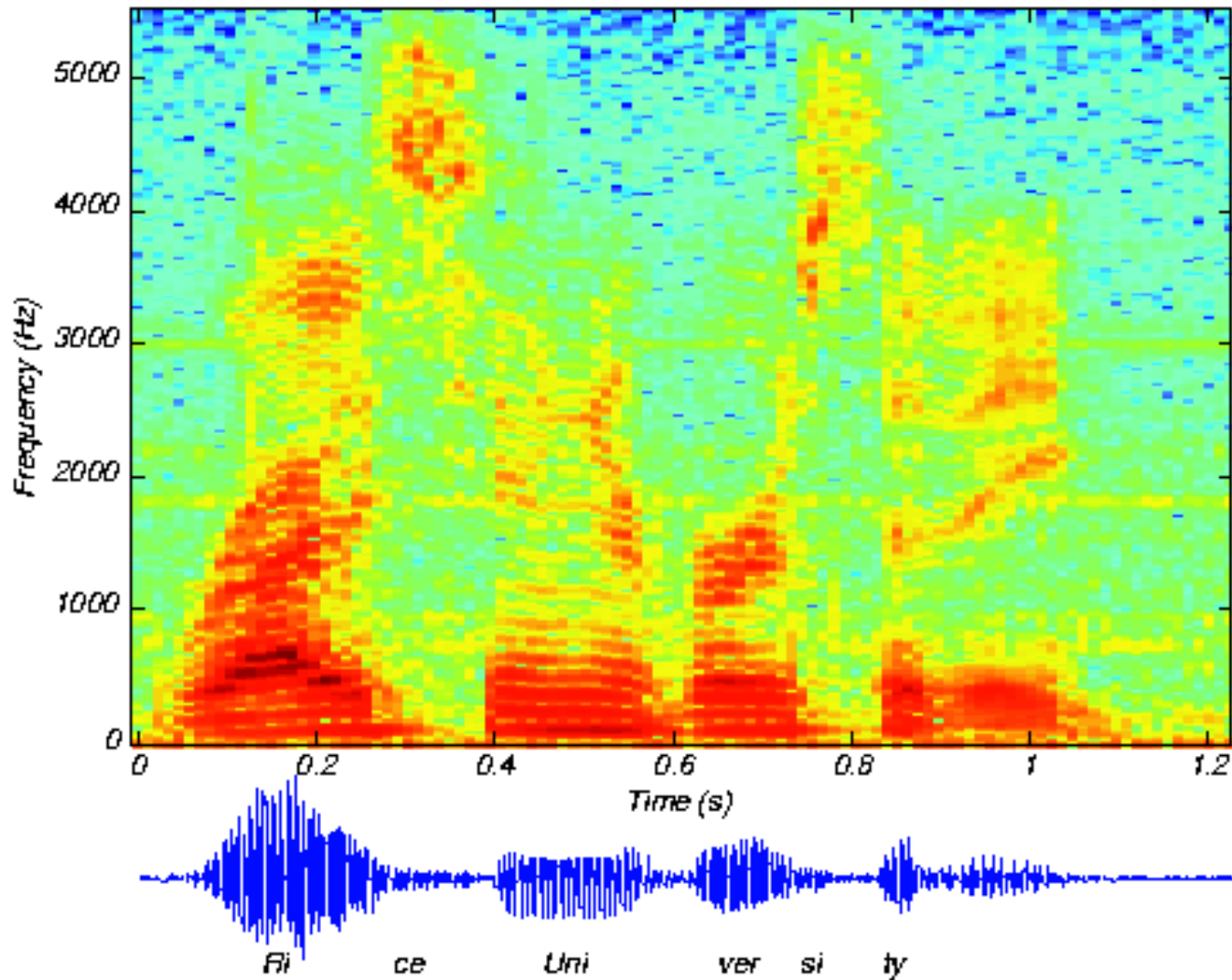
Fundamental Frequency

Fundamental Frequency



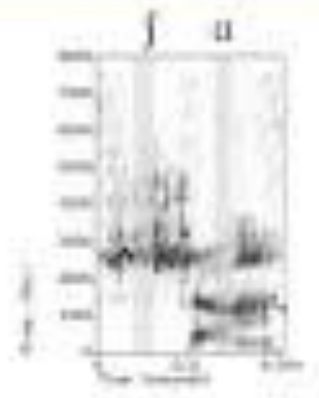
Spectrogram (Ảnh phổ)

A graph that displays the **strength** of a **signal** over **time** for a given **frequency** range



Waveforms and Spectrograms

Waveforms and Spectrograms



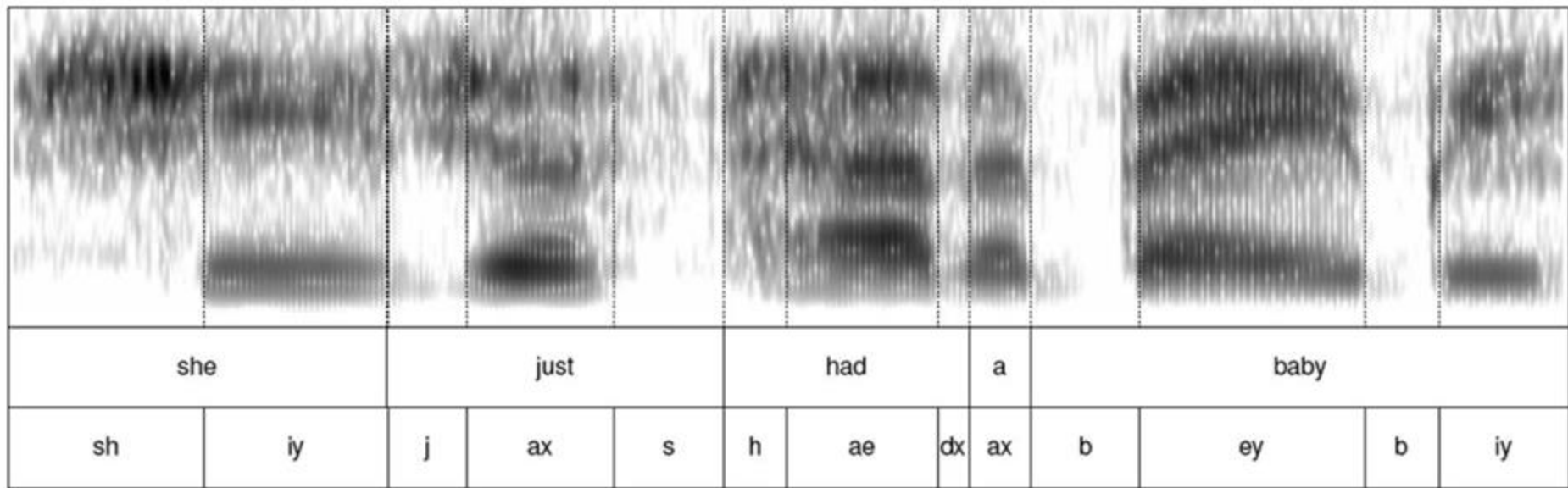
Spectrogram: Spectrum + Time Dimension

SHE

JUST

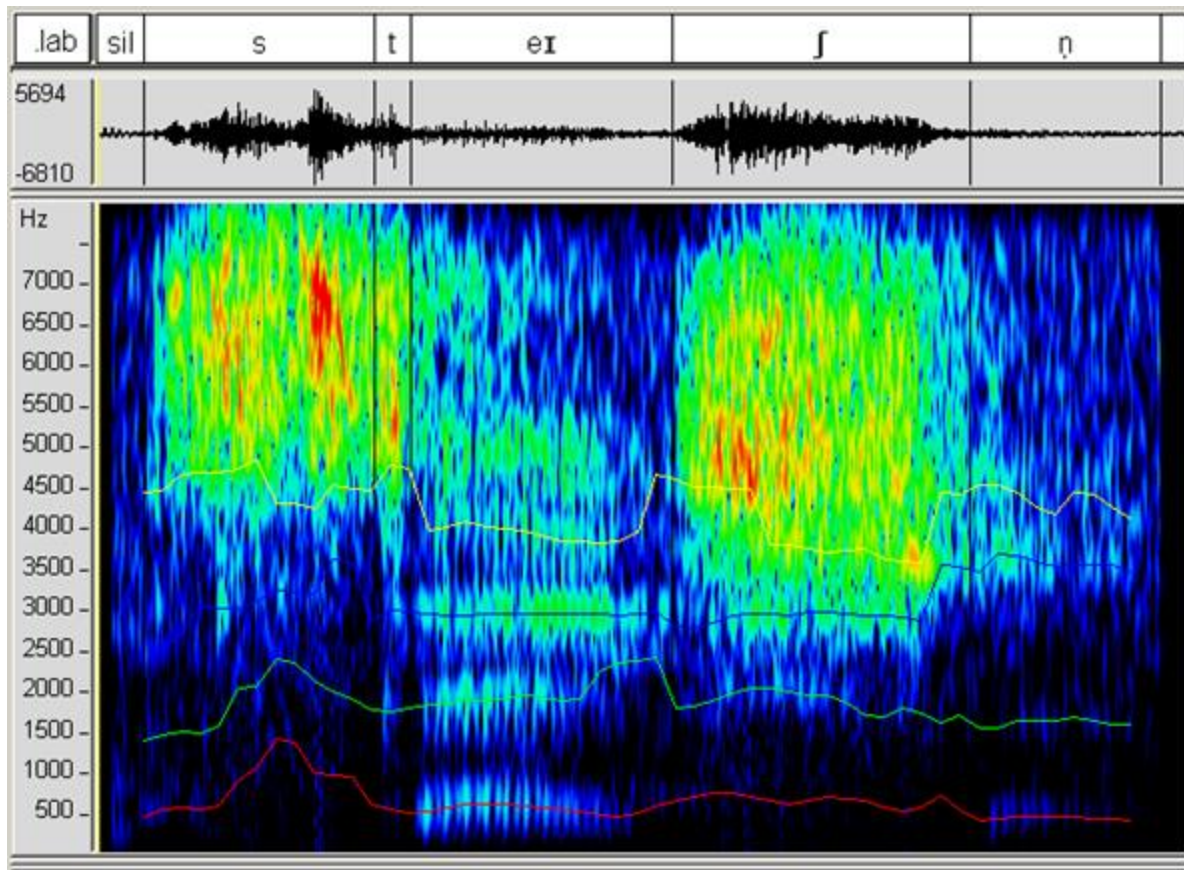
HAD

A



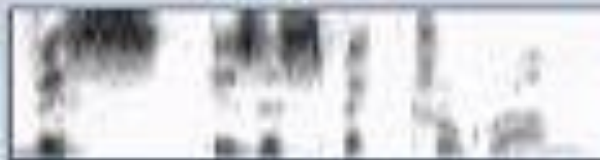
Consonant and Vowel Spectrograms

- **Vowels** have **full formant patterns**, based mostly on F1 and F2 and F3
- **Consonants**: No **clear formant patterns** themselves, but rather affect the patterns of the vowels surrounding them



How to read vowel spectrograms?

Reading Spectrograms a VLC Series by Prof. Dr. Jürgen Handke



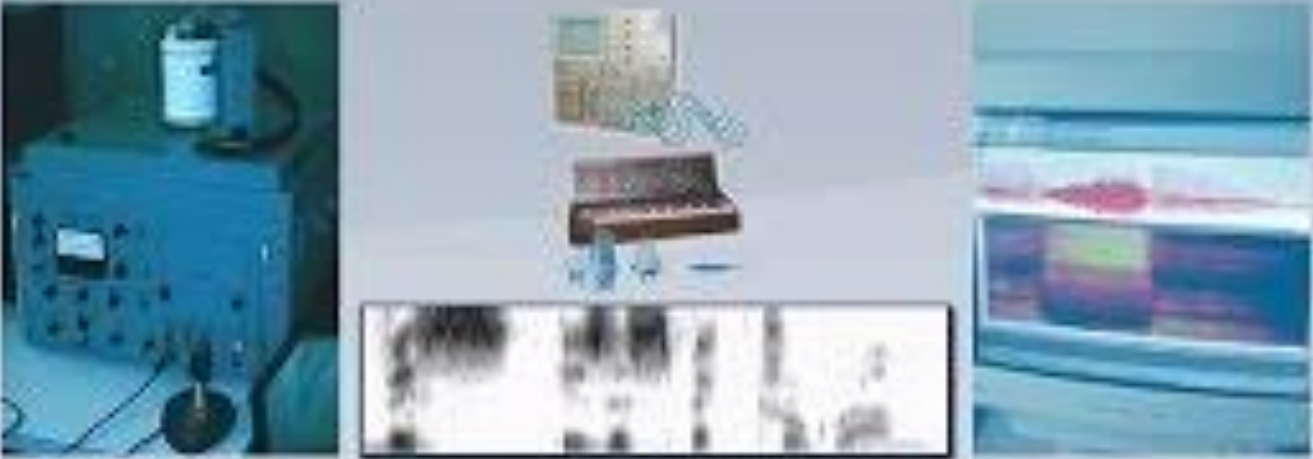
Vowels

How to read consonant spectrograms?

The Virtual Linguistics Campus

Reading Spectrograms
a VLC Series
by Prof. Dr. Jürgen Handke

TechSmith®



Consonants

Reading Spectrograms: Consonants © The Virtual Linguistics Campus, Marburg University, Germany

Review: Grapheme vs. Phoneme

- **Grapheme**

The smallest written unit representing a sound in speech.

Examples: "d", "th", "ngh", "a", "au".

- **Phoneme**

The smallest unit of sound in speech.

Examples: /d/, /th/, /ŋ/, /a/, /ăw/.

Review: IPA (International Phonetic Alphabet)

BẢNG PHIÊN ÂM TIẾNG ANH IPA

VOWELS	monophthongs				diphthongs		Phonemic Chart	
	i:	ɪ	ʊ	u:	ɪə	eɪ		
	leave	live	put	pool	year	paper		
	e	ə	ɜ:	ɔ:	ʊə	ɔɪ		
pen	a	girl	door	tourist	boy	phone	voiced unvoiced	
æ	ʌ	a:	ɒ	eə	aɪ	aʊ		
man	cup	car	hot	chair	fine	house	Designed by HapTOEC	
CONSONANTS	p	b	t	d	tʃ	dʒ	k	g
	pen	baby	table	door	chip	January	key	girl
	f	v	θ	ð	s	z	ʃ	ʒ
	fan	van	think	this	sun	zoo	shoes	vision
m	n	ŋ	h	l	r	w	j	
mouth	nose	ring	hear	letter	rain	window	yellow	

leave /li:v/
manly /'mænli/

Exercise

- Download Praat: www.praat.org
- Record the sentence: "She just had a baby"
- Create 2 tiers with TextGrid for annotating for the word (grapheme) and phoneme (using IPA) levels

Popular packages for signal analysis

- Praat (www.praat.org)
 - by Paul Boersma & David Weenink, Phonetics Amsterdam
- Wavesurfer (www.speech.kth.se/wavesurfer/)
 - by scientists at Stokholm Technical University - KTH
- Goldwave (www.goldwave.de)
 - commercial program, but available free for trial use

2.2 Vietnamese Phonology and Phonetics



Nguyen Thi Thu Trang

A traditional Vietnamese painting depicting a serene landscape. In the center, a multi-tiered pagoda with a brown roof stands on a small, grassy island in a calm body of water. The pagoda is surrounded by dense green trees and foliage. In the foreground, the water is filled with large, green lotus leaves and several pink lotus flowers in various stages of bloom. The scene is framed by dark, gnarled tree branches with small green leaves hanging down from the top. The overall style is characteristic of traditional Vietnamese art, with a focus on natural beauty and architectural elements.

**Why is Vietnamese
interesting?**

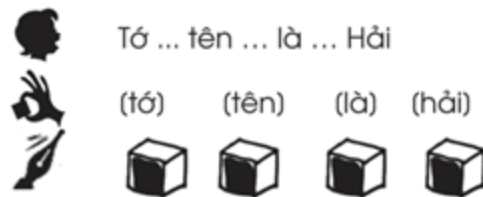
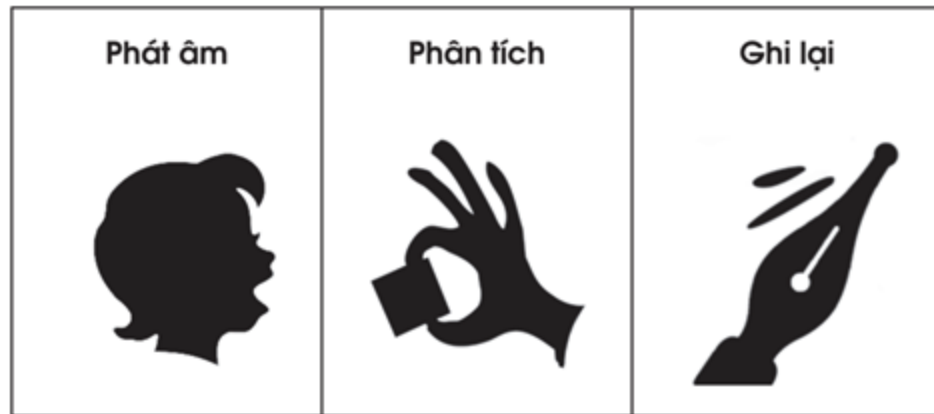
What is Vietnamese?

- Mon-Khmer sub-branch of the Austroasiatic (Nam Á) language family with mixture of Tai-Kadai (Tày-Thái)
- Official language of Vietnam
- ~100 millions speakers in the world



Tiếng Việt là ngôn ngữ đơn lập (monosyllabic)

- Tiếng/âm tiết (syllable) được phát âm và viết tách rời
 - Công ty vs. Company
- Từ (word) của tiếng Việt không biến đổi hình thái
 - Tôi **đã** xem phim này rồi.
- Trật tự từ quan trọng
 - Con gà vs. Gà con.



Tiếng Việt là ngôn ngữ có thanh điệu (tonal)

1 - Ngang (A1)

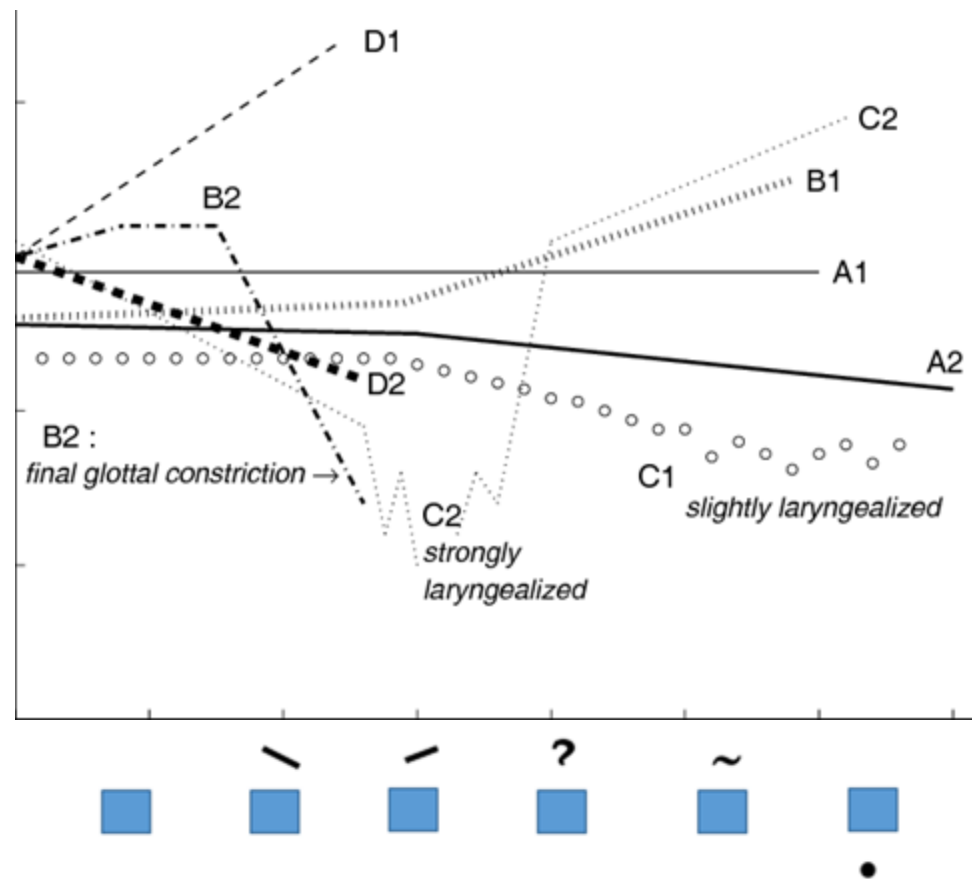
2 - Huyền (A2)

3 - Hỏi (C1)

4 - Ngã (C2)

5a, 5b - Sắc (B1, D1)

6a, 6b - Nặng (B2, D2)



Hanoi Vietnamese¹ tone system *(Michaud 2004)*

- 6 tones in sonorant-final syllables
- 2 tones in obstruent-final syllables (ending with /p t k/)

#	Name	Duration	Example
1	Level	Long	ba [ʃa-1] (three)
2	Falling	Long	bà [ʃa-2] (grand-mother)
3	Curve	Long	bả [ʃa-3] (poison)
4	Broken	Long	bã [ʃa-4] (residue)
5a	Rising	Long	bá [ʃa-5a] (to embrace)
5b	Rising*	Short	bác [ʃa-5b] (uncle)
6a	Drop	Short	bạ [ʃa-6a] (to strengthen)
6b	Drop*	Short	bạc [ʃa-6b] (silver)

¹*Standard Vietnamese (Thompson 1987)*

Michaud (2004). Michaud Alexis. *Final consonants and glottalization: new perspectives from Hanoi Vietnamese*. *Phonetica*.

Thompson (1987). Thompson Laurence C. *A Vietnamese Reference Grammar*. University of Hawaii Press.

Hanoi Vietnamese - G2P Rules for consonants

No.	Graph-eme	Position	Pho
			IPA
1	b	initial only	ɓ
2	đ	initial only	ɗ
3	x, s	initial only	s
4	g, gh	initial only	ɣ
5	kh	initial only	x
6	l	initial only	l
7	v	initial only	v
8	th	initial only	t ^h
9	d, gi, r	initial only	z
10	ph	initial only	f
11	tr, ch	initial only	tʃ
12	h	initial only	h
13	c, k, q	initial only	k

No.	Graph-eme	Position	Pho
			IPA
14	t	initial, final	t
15	p	initial, final	p
16	n	initial, final	n
17	ch	final after i, ê, a	k
18	c	final after u, o, ô	k ⁺ p
19	ch, c	final except 17, 18	k
20	m	initial, final	m
21	nh	final after i, ê, a	ŋ
22	nh	initial only	ɲ
23	ng, ngh	initial	ŋ
24	ng	final after u, o, ô	ŋ ⁺ m
25	ng	final except 24	ŋ

Hanoi Vietnamese - G2P Rules for vowels

Vowel type	Grapheme (Orthography)	Phoneme
		IPA
Long vowel	a	a
	e	ɛ
	ê	e
	i, y	i
	o, oo	ɔ
	ô, ôô	o
	ơ	ɤ
	u	u
	ư	ʊ

Vowel type	Grapheme (Orthography)	Phoneme
		IPA
Short vowel	ă, a (au, ay)	ă
	â	ɤ
	a (anh, ach)	ɛ
	o (ong, oc)	ɔ
Diphthong	ia, iê, yê, ya	iə
	ua, uô	uə
	ưa, ươ	ʊə

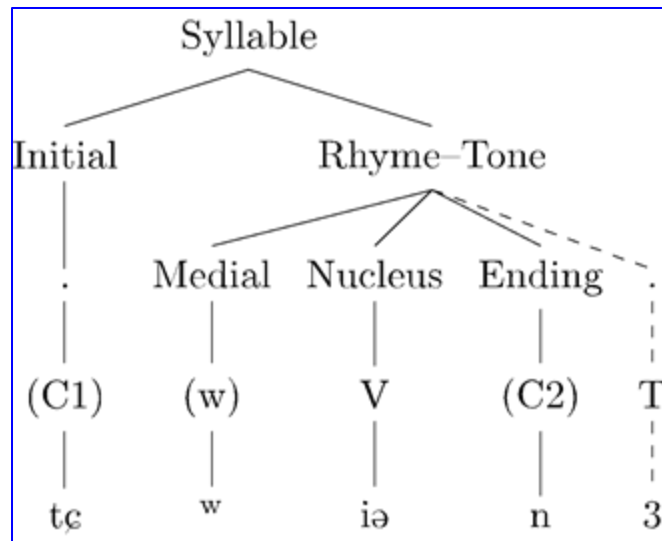
Vietnamese syllable structure

- Hierarchical structure

(Doan 1977) (Vogel 2004)

- **Segmental** elements
 - Initial consonant
 - Rhyme (medial, nucleus, ending)
- **Supra-segmental** element:
Lexical Tone (T)
 - mostly covers the rhyme of syllables (Tran 2005)

- Syllabic structure: (C1)(w)V(C2)-T



E.g. chuyên [tɕ²wiən-3] (*to move*)

Doan (1977). Doan Thien Thuat. *Vietnamese phonetics (in Vietnamese)*. Vacation School and University Publisher.

Doan (1999). Doan Xuan Kien. *Re-consider a problem of Vietnamese phonetics: Syllable structure (in Vietnamese)*. Hop Luu

Tran (2005). Tran Do Dat et al. *Influence of F0 on Vietnamese Syllable Perception*. Interspeech 2005. Portugal.

Vogel (2004). Vogel Irene et al. *Syllable structure in Vietnamese*. TEAL Workshop. Taiwan.

Structure-based syllable types

INITIAL	RHYME			Examples
	Medial	Nucleus	Ending	
		v		“a” /a-1/ (<i>ah</i>), “ủ” /ɤ-3/ (<i>keep wa</i>
v		v		“lá” /la-5a/ (<i>leaf</i>), “chờ” /tɕɤ-2/ (<i>wa</i>
	v	v		“oẹ” /we-6a/ (<i>retch</i>), “uy” /wi-1/ (<i>wa</i>
		v	v	“ích” /ik-5b/ (<i>helpful</i>), “ây” /ɤ̃-5a/ (<i>wa</i>
v	v	v		“loé” /lwe-5a/ (<i>flash</i>), “quá” /kwa-6a/ (<i>wa</i>
v		v	v	“trẹo” /tɕew-6a/ (<i>sprain</i>), “nhãn” /ɲwien-6a/ (<i>wa</i>
v	v	v	v	“nhuyễn” /ɲwien-4/ (<i>fine</i>), “soát” /sɔt-6a/ (<i>wa</i>

Allophone – []

- A variant of a phoneme
- The allophones of a phoneme form a set of sounds that:
 - Do not change the meaning of a word
 - Are all very similar to one another, and
 - Occur in phonetic contexts different from one another (for example, syllable-initial as opposed to syllable-final).
- The differences among allophones can be stated in terms of phonological rules.

Luật phiên âm tiếng Việt - Âm đệm

- Duy nhất 01 âm đệm **/w/** được biết bằng **“o”** hoặc **“u”**
- **“o”** đứng trước các nguyên âm rộng: a, ă, e
 - VD: toát, hoạ hoản, lòe xoè
- **“u”** đứng trước các nguyên âm: y, ê, ơ, â
 - VD: huy, thuở, huân, Huế

KHUYÊN /xwiən-1/			
kh /x/	u /w/	yê /iə/	n /n/
	1		

Exercise

- Phonology differences of Northern, Central and Southern Vietnamese
- Reference:
https://www.researchgate.net/profile/Deborah-Hwa-Froelich/publication/278524880_Characteristics_of_Vietnamese_Phonology/links/02e7e530f680f7361f000000/Characteristics-of-Vietnamese-Phonology.pdf

VD: Luật chính tả tiếng Việt cơ bản

Không tính các từ vay mượn

- Âm đầu **g/gh, ng/ngh, c/k**: Trước âm chính:
 - **Hẹp/hơi hẹp**: **i, ê, e** → **gh, ngh, k**
 - VD: kê, kia, nghe, ghe, nghênh, ké, ki bo, nghề nghiệp
 - **Rộng/hơi rộng**: **a, ă, â, o, ô, ơ, u, ư** → **g, ng, c**
 - VD: cá, ngốc, gặt gù, ngừ, cố, găm, ngủ, gặt, ngọt ngào
- Âm đầu **q/c/k** (đều có âm là /k/)
 - **q** luôn bao giờ cũng đi với âm đệm **u** để thành qu
 - VD: qua, quanh quẩn (*làm tròn môi âm chính*)
 - **c/k** như trên
- Âm cuối **c/ch** (đều có âm là /k/): Sau âm chính:
 - **Hẹp/rộng**: **i, ê, a** → **ch, còn lại là "c"**
 - VD: bích, ốc, ếch, khách, ích, lúc, bác, bức xúc

Exercise - Luật chính tả / phiên âm

- “gia” và “ghia”; “củ”, “cả”, “cua” và “qua”
- Quốc gia Qata, tỉnh Đắk Lắk



Discussion - Luật chính tả / phiên âm

- “gia” và “ghia”, “cua” và “qua”

gi		a	
----	--	---	--

gh		ia	
----	--	----	--

c		ua	
---	--	----	--

q	w	a	
---	---	---	--

- Quốc gia Qata, tỉnh Đắk Lắk

→ Từ vay mượn: Qatar và Dak Lak (Darlac)

Luật chính tả tiếng Việt cơ bản

Không áp dụng cho các từ vay mượn

- Âm chính: Nguyên âm y/i
 - Viết là “y”
 - Sau âm đệm “u”: huy, tuy, thúy, quy...
 - Sau nguyên âm ngắn a như: ây, ay
 - Đứng trước ê khi chữ đó không có âm đầu như: yêu, yết, yếm
 - Đứng một mình: y tế, ý nghĩ. Ngoại lệ: í ới
 - Viết “i/y” đều đúng:
 - Không có âm cuối, khuyến khích viết i: Châu Mỹ/Châu Mỹ, Địa lí/Địa lý, Bác sĩ/Bác sỹ
 - Còn lại là “i”
- Exercise
 - “bay” và “bai”, “tuy” và “tui”

Discussion - Luật chính tả / phiên âm

- “bay” và “bai”, “tuy” và “tui”

b		a	y
/b		ă	j/

t	u	y	
/t	w	i	/

b		a	i
/b		a	j/

t		u	i
/t		u	j/

Exercise

- Record in Praat for the following sentences:
“Dùng tay vuốt mái tóc bạch kim của em qua tai”
- Analyze and label at allophone level for those sentences

3. Digital Signal Processing

NGUYEN Thi Thu Trang (trangntt@soict.hust.edu.vn)

3.1. Sound representation



Analogue
Bit depth
Bit rate
Sample rate
Compression

Discussion

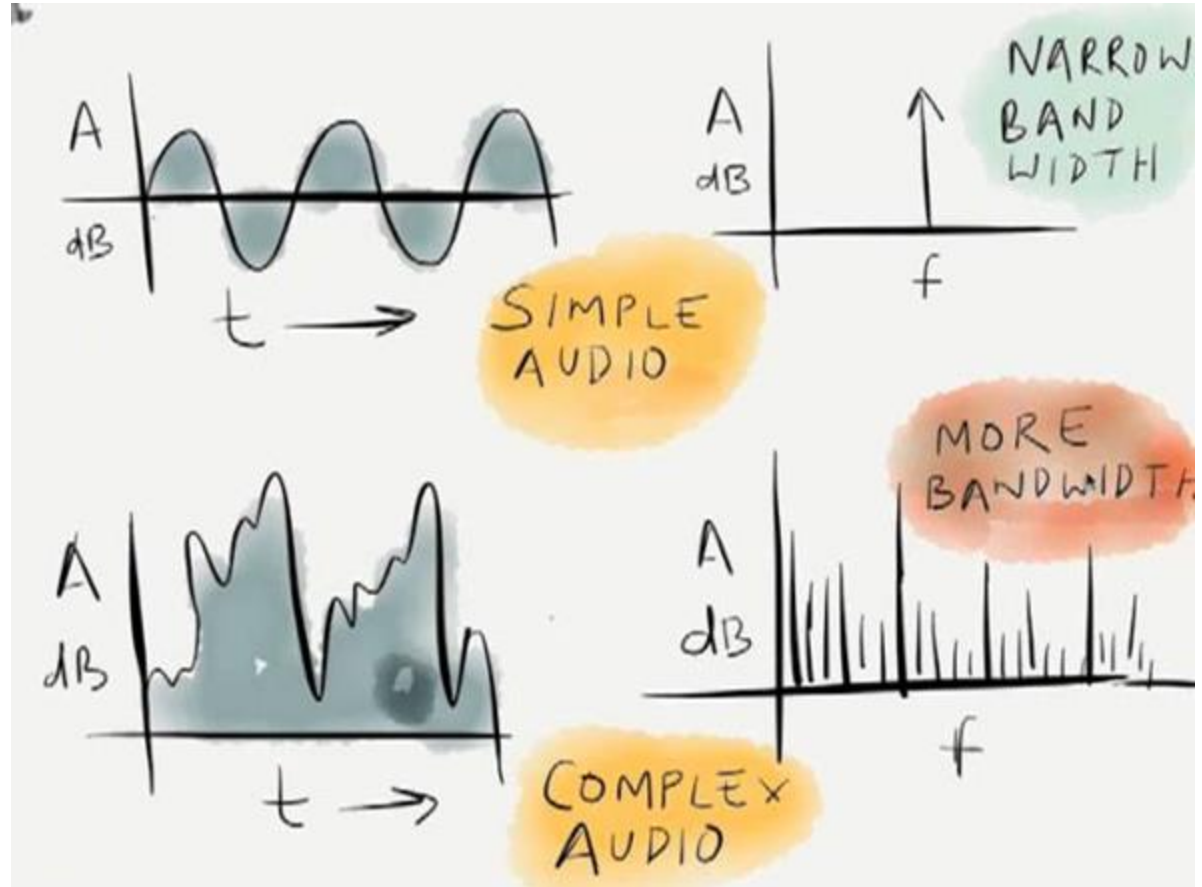
Which of the following has the best quality?

Record, CD or MP3?

Analogue Digital Converter (ADC)

- Sampling
- Quantization

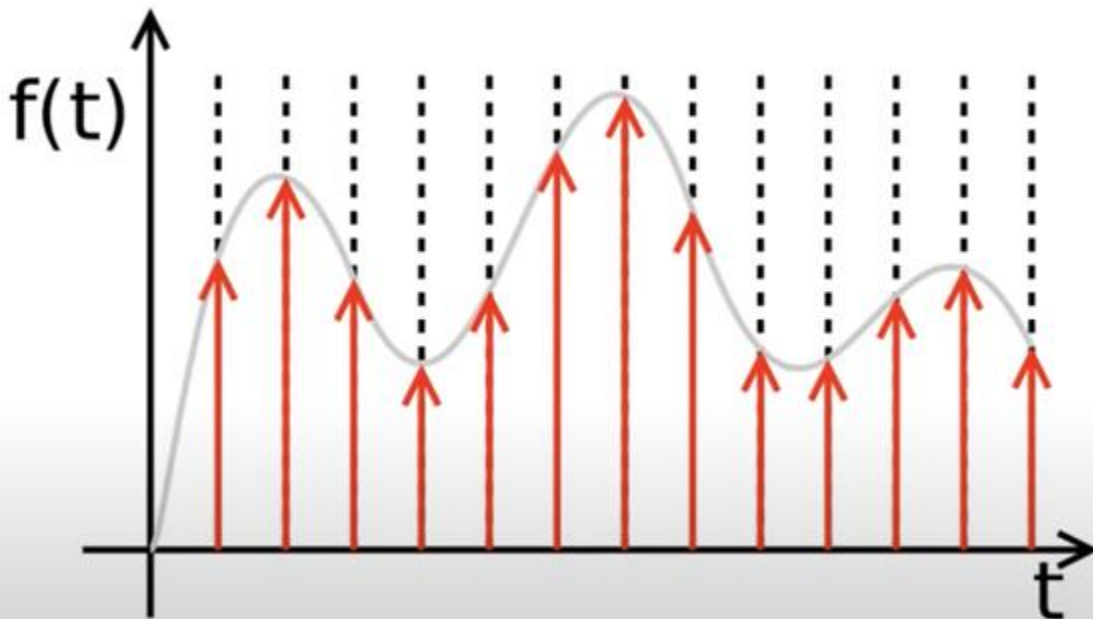
Analogue: Continuous changing values



3.1.1. Sampling

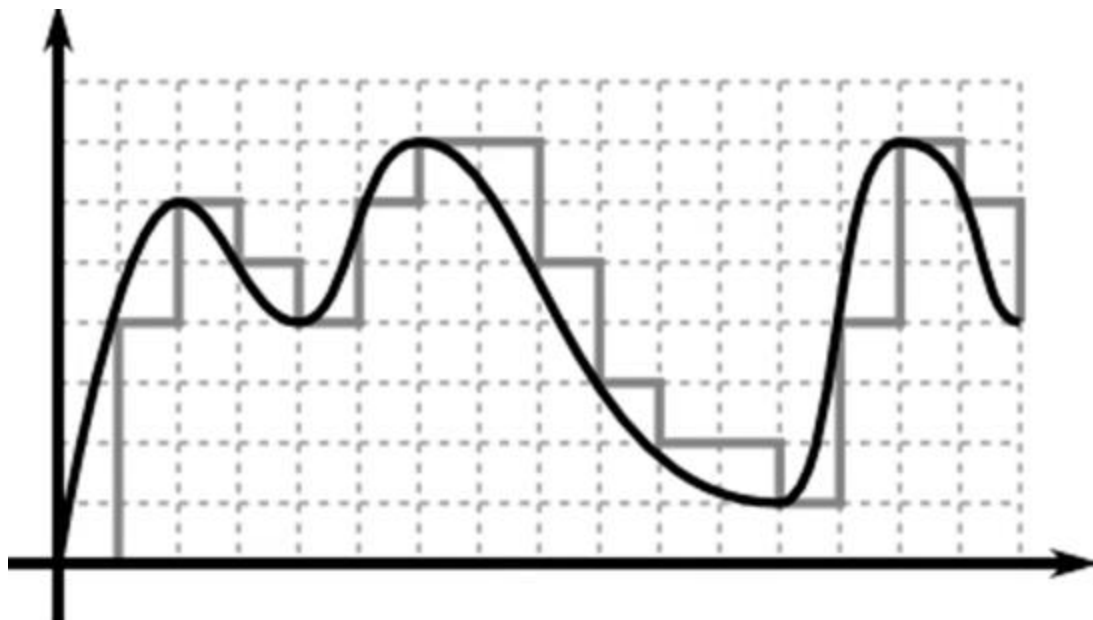
- **Sampling:** Converting a continuous analogue time signal into a discrete time representation
 - Carried out at key points → sampling intervals

A small **sampling interval** results which results in a better **quality sound** file and increases the **size** of the **audio** file



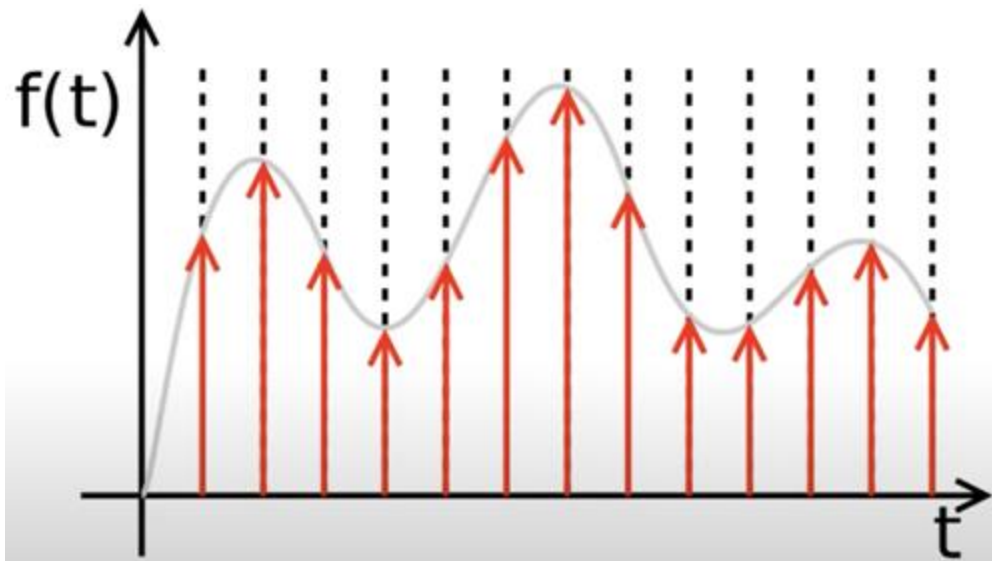
Approximate in Sampling

The sound quality will be poorer and less smooth because some bits are missing.

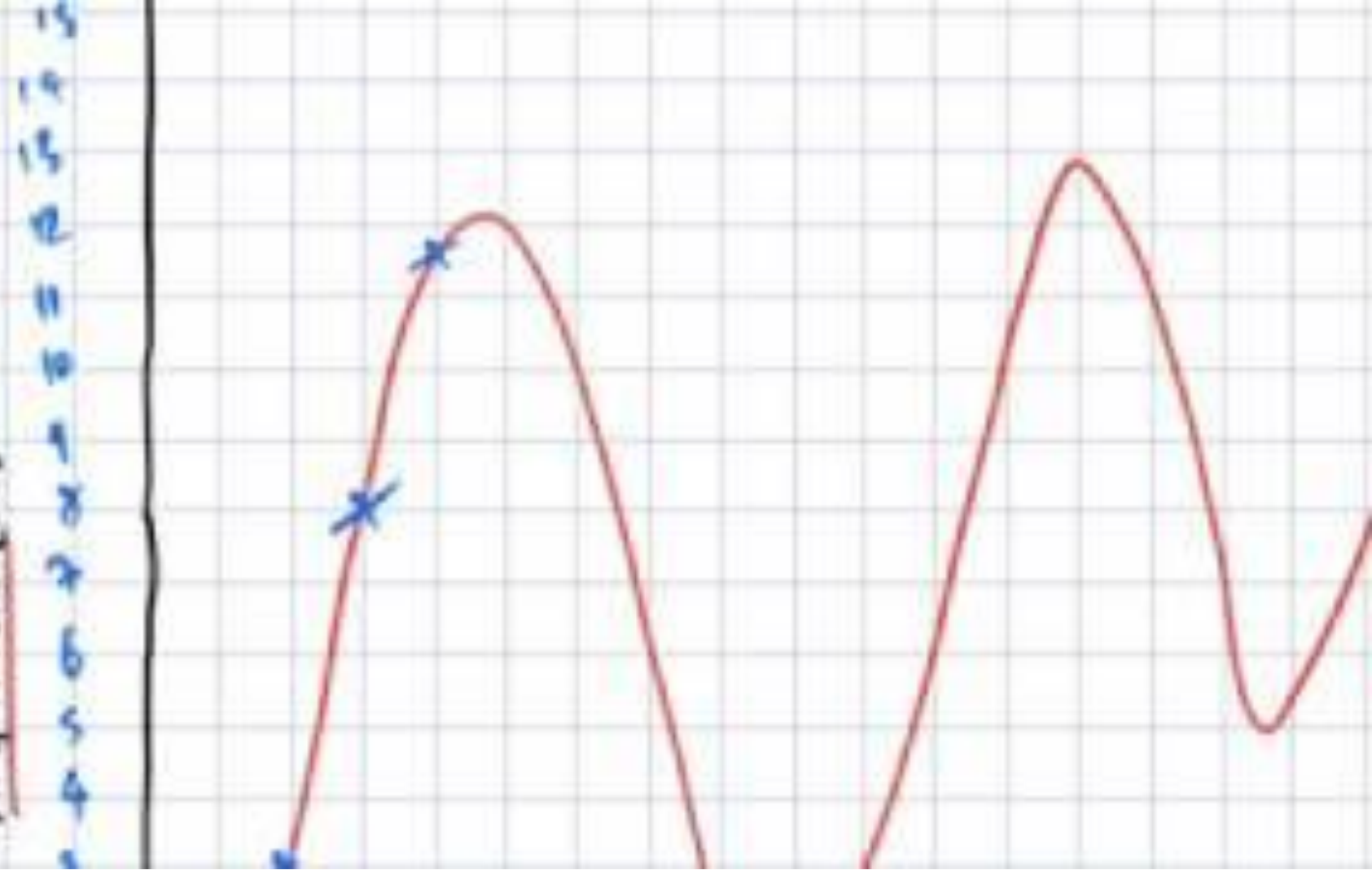


Sample rate

- Number of times analogue sound is sampled per second to transform to discrete digital signal
- Higher ~ more samples ~ Higher quality ~ bigger file size
 - Small sampling interval
- Common unit: **kHz**
(thousands of samples per second)
- CD: 44.1 kHz

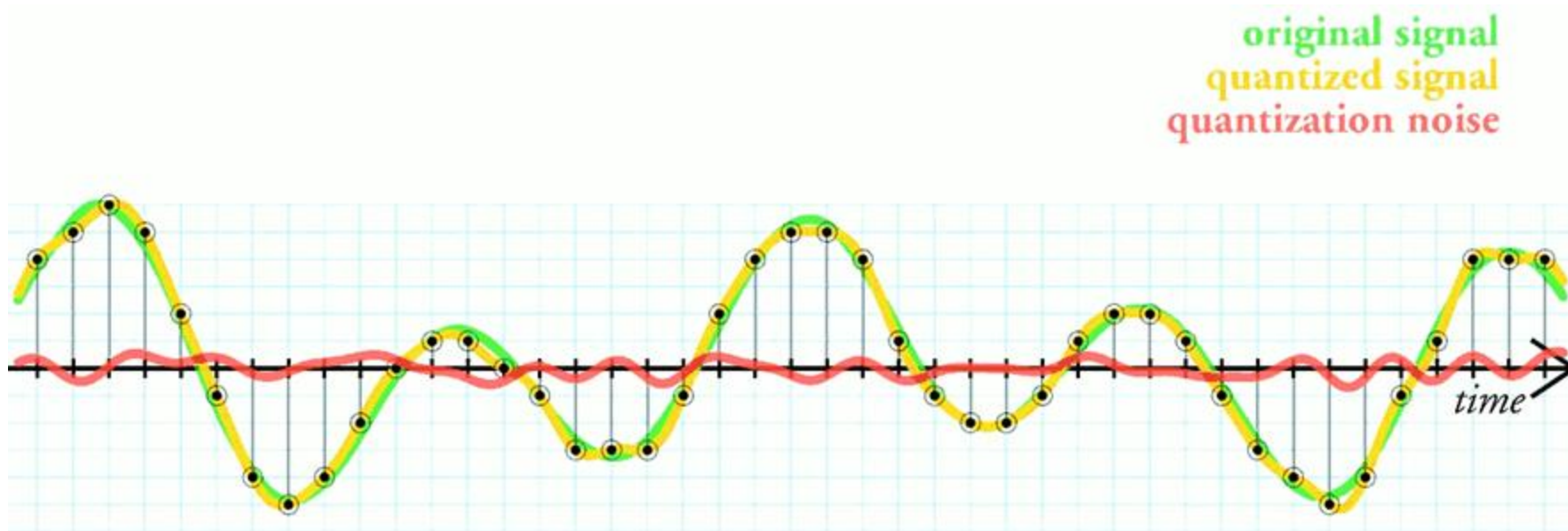


Amplitude (dB)



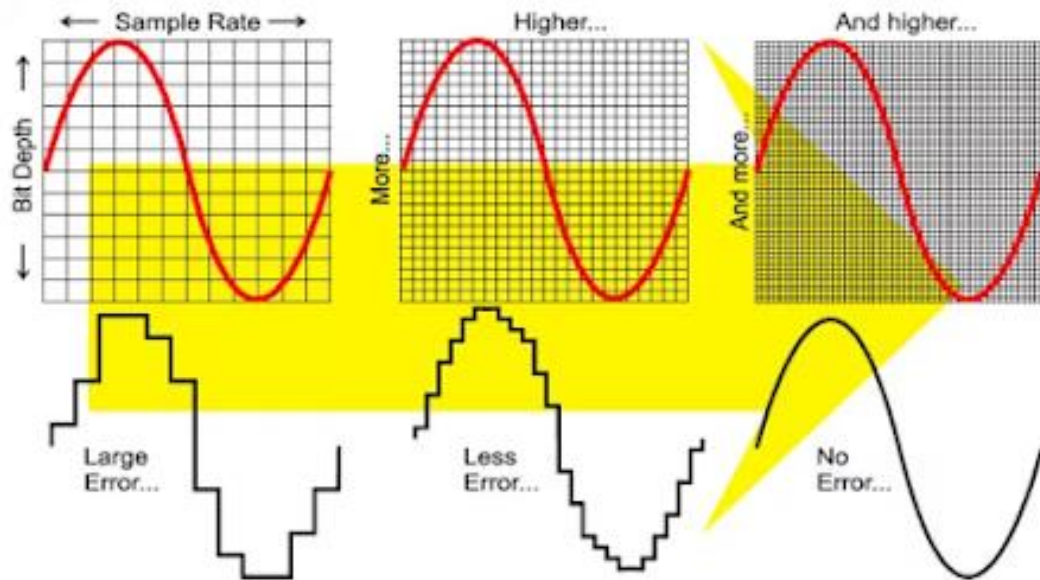
3.1.2. Quantization

- Mapping the **continuous amplitude values** of an **analog** audio signal into a **discrete** set of values in digital format, after sampling



Bit Depth vs Bit Rate

- **Bit depth** is the **number** of **bits** that are used to **represent each sample** of audio
- **Bit rate** is the **number** of **bits** used to **represent one second** of audio (bits per second)
 - **Bit rate = Bit depth**
x Sample rate
 - A higher bit rate means better audio quality and, thus, a larger file size.



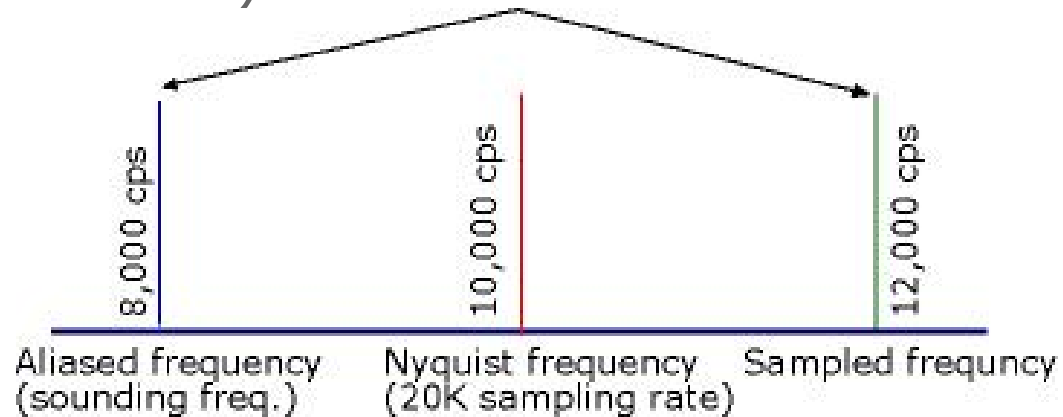
Nyquist Theorem

or **Nyquist-Shannon Sampling Theorem**

- fundamental principle in signal processing that explains how to **accurately** convert a continuous analog signal into a digital signal
- “to faithfully reconstruct a signal from its samples, the sampling rate must be at least twice the highest frequency present in the signal”.
 - E.g. Frequencies up to 20 kHz (highest), the sampling rate must be at least 40 kHz.

Nyquist Rate

- Maximum frequency that can be accurately represented, which is half the sampling rate
 - If not → Aliasing → Distortion
 - E.g. Audio CDs use a sampling rate of 44.1 kHz to capture frequencies up to 22.05 kHz, covering the full range of human hearing (20 Hz–20 kHz).



Hertz

1Hz = 1 Sample Per Second

1000Hz = 1000 Samples Per Second

44100Hz = 44,100 Samples Per Second



File Size and Sample Rate, Bit Depth

Calculating Sound File Size

Bits Per Second = Sample Rate x Bit Depth

File Size in Bits = Bits Per Second x Duration

File Size in Bytes = Number of Bits ÷ 8

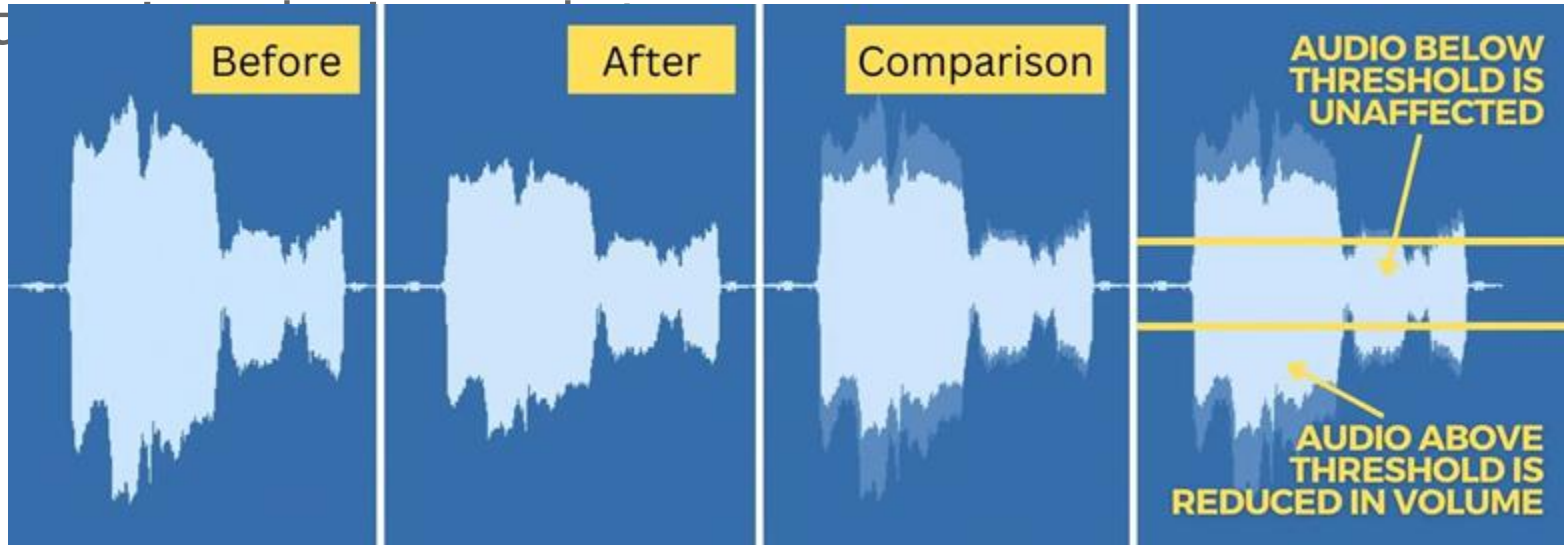
File Size in KB = Number of Bytes ÷ 1024

3.1.3. Compression

Feature	Lossy Compression	Lossless Compression	Uncompressed
File Size	70-90% small	30-70% smaller	Very large
Audio Quality	Reduced (depends on bitrate)	Original Quality	Original Quality
Use Case	Streaming, mobile storage	Archiving, professional use	Audiophile listening, professional use
E.g.	MP3, AAC, OGG, Opus	FLAC, ALAC	WAV, AIFF

Lossy Compression of Audio

- Permanently **removing** certain audio **information** that is deemed **less important** or **less noticeable** to the human ear.
- Balances file size and sound quality, making it ideal for streaming



Interest Facts on Bit Rate

1. Music Streaming

- 96 kbps: Low quality, often used for streaming on mobile devices with limited data
- 128 kbps: Standard quality, commonly used in MP3 files for general music playback
- 256 kbps: High quality, used in services like Apple Music or high-quality MP3 files
- 320 kbps: Very high quality, near-CD quality, preferred by audiophiles for MP3 playback

2. Lossless Audio (Uncompressed Formats)

- 1,411 kbps (CD quality): Audio from CDs uses a bitrate of 1,411 kbps (44.1 kHz sample rate, 16-bit depth, stereo)
- 4,608 kbps: High-resolution lossless audio, typically 96 kHz sample rate, 24-bit

Interest Facts on Bit Rate

3. Video Streaming Audio

- 96-160 kbps: Used for audio tracks in standard-definition (SD) video streaming
 - e.g. 128kbps for YouTube SD, 96 kbps for basic stereo audio track on Netflix
- 192-256 kbps: Common for high-definition (HD) streaming audio tracks
 - e.g. 192kbps for YouTube HD
- 384 kbps or higher: Used in ultra-high-definition (4K) video streaming for enhanced audio quality
 - e.g. 768 kbps or higher for Dolby Atmos surround sound.

Quiz

What is the **sample rate** of an audio file that captures 96,000 samples per second?

- A. 44.1 kHz
- B. 48 kHz
- C. 96 kHz
- D. 192 kHz

Quiz

Which of the following correctly describes **bit depth** in digital audio?

- A. The number of samples captured per second
- B. The number of amplitude levels captured per sample
- C. The number of bits used for file compression
- D. The file size in bits

Quiz

If an audio file has a **bit depth** of **16** bits, how many possible **amplitude levels** can it represent?

- A. 16 levels
- B. 256 levels
- C. 65,536 levels
- D. 1,048,576 levels

Quiz

An **audio file** has a sample rate of 44.1 kHz, a bit depth of 16 bits, and 2 channels (stereo). What is the **file size** for 1 minute of uncompressed audio?

- A. 176.4 KB
- B. 10.584 MB
- C. 5.292 MB
- D. 1.41 MB

Quiz

An audio file with a **bit rate** of 128 kbps has a **duration** of 3 minutes. What is the **file size**?

- A. 2.88 MB
- B. 3.84 MB
- C. 4.32 MB
- D. 5.76 MB

Quiz

An audio file has a **sample rate** of 96 kHz, a **bit depth** of 24 bits, **stereo** channels, and a **file size** of 50 MB. What is the **duration** of the audio?

- A. 3 minutes 5 seconds
- B. 4 minutes 10 seconds
- C. 5 minutes 30 seconds
- D. 6 minutes

Quiz

A 5-minute audio file sampled at 44.100 samples per second. The amplitudes of the samples are quantized to 1.024 levels. Determine the size of the audio file?

- A. 132.3 Mb
- B. 13.23 Mb
- C. 2.205 Mb
- D. 220.500 bits

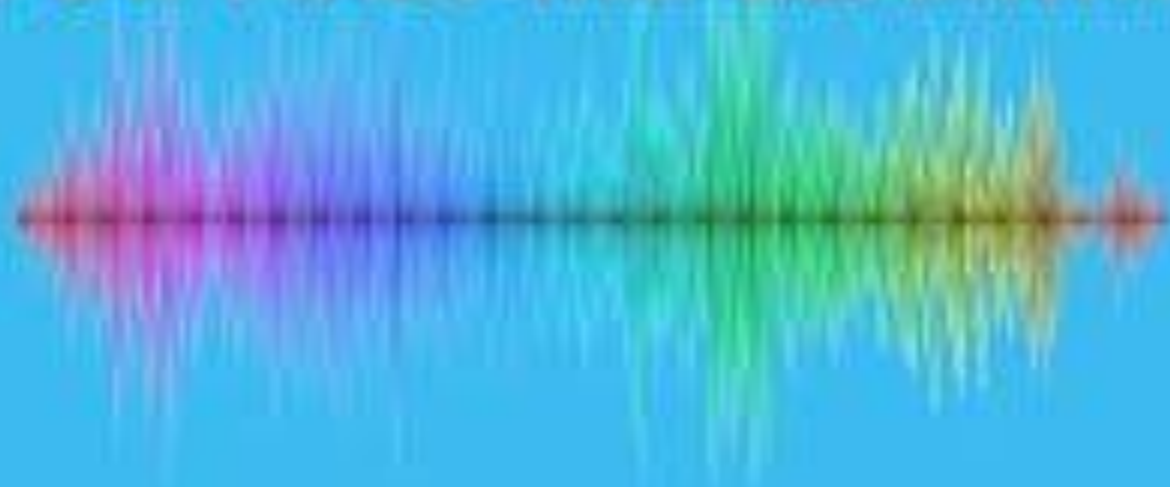
3.2. Signal Representation



Waveform Basic Information

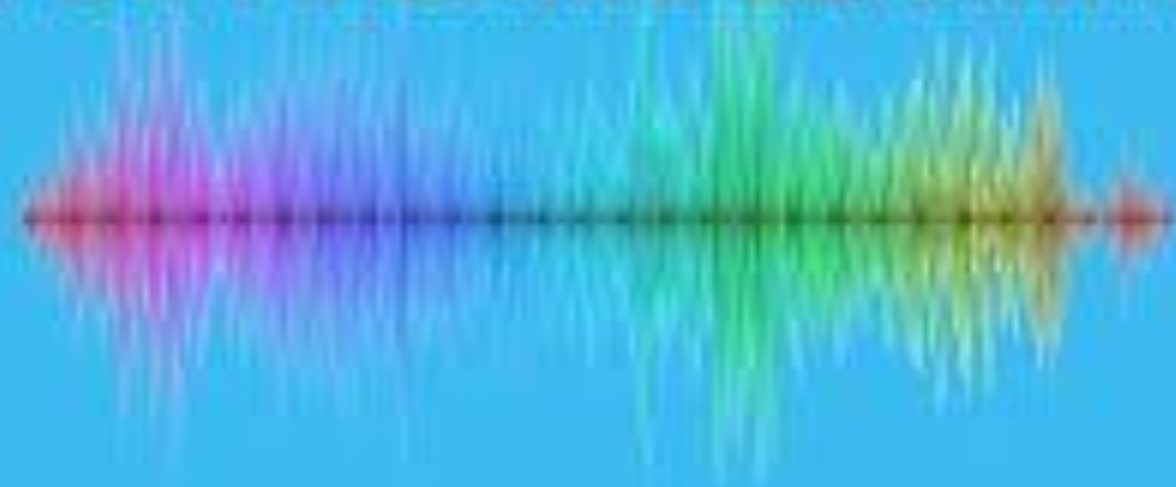
- Frequency
 - Refers to how "high" or "low" a sound is.
- Intensity
 - Refers to how "loud" or "soft" a sound is.
- Timbre
 - Refers to the unique "color" or "quality" of a sound that makes it distinguishable, even if two sounds have the same frequency and intensity.

AUDIO SIGNAL PROCESSING FOR MACHINE LEARNING



2- SOUND AND WAVEFORMS

AUDIO SIGNAL PROCESSING FOR MACHINE LEARNING

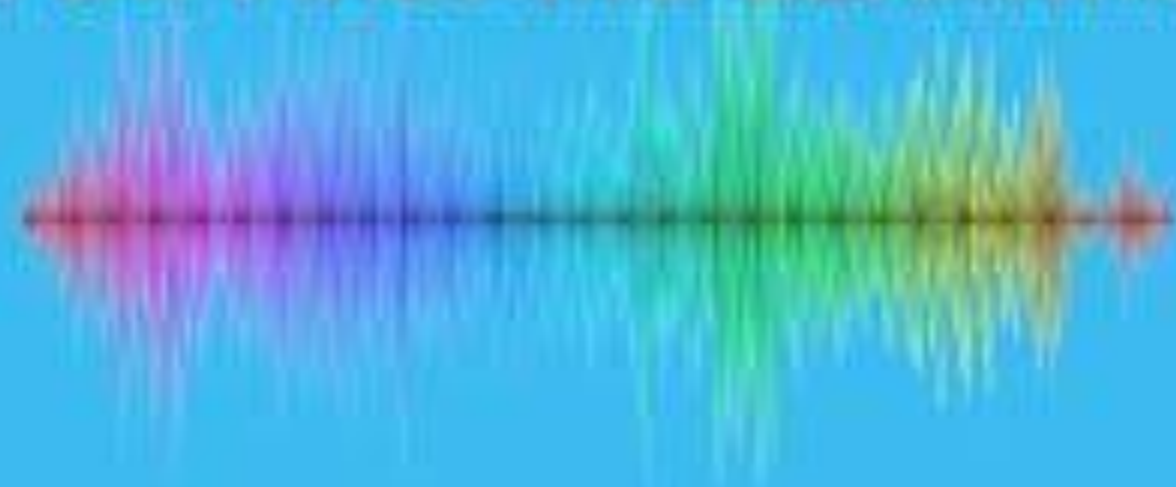


3- LOUDNESS AND TIMBRE

Audio Signal Representation

- **Time** Domain Representation:
 - Represents a signal as it varies over time (e.g., an audio waveform)
- **Frequency** Domain Representation:
 - Describes a signal in terms of its frequency components using methods like the Fourier Transform.
- **Time-Frequency** Domain Representation:
 - Determines the signal's strength and timing characteristics

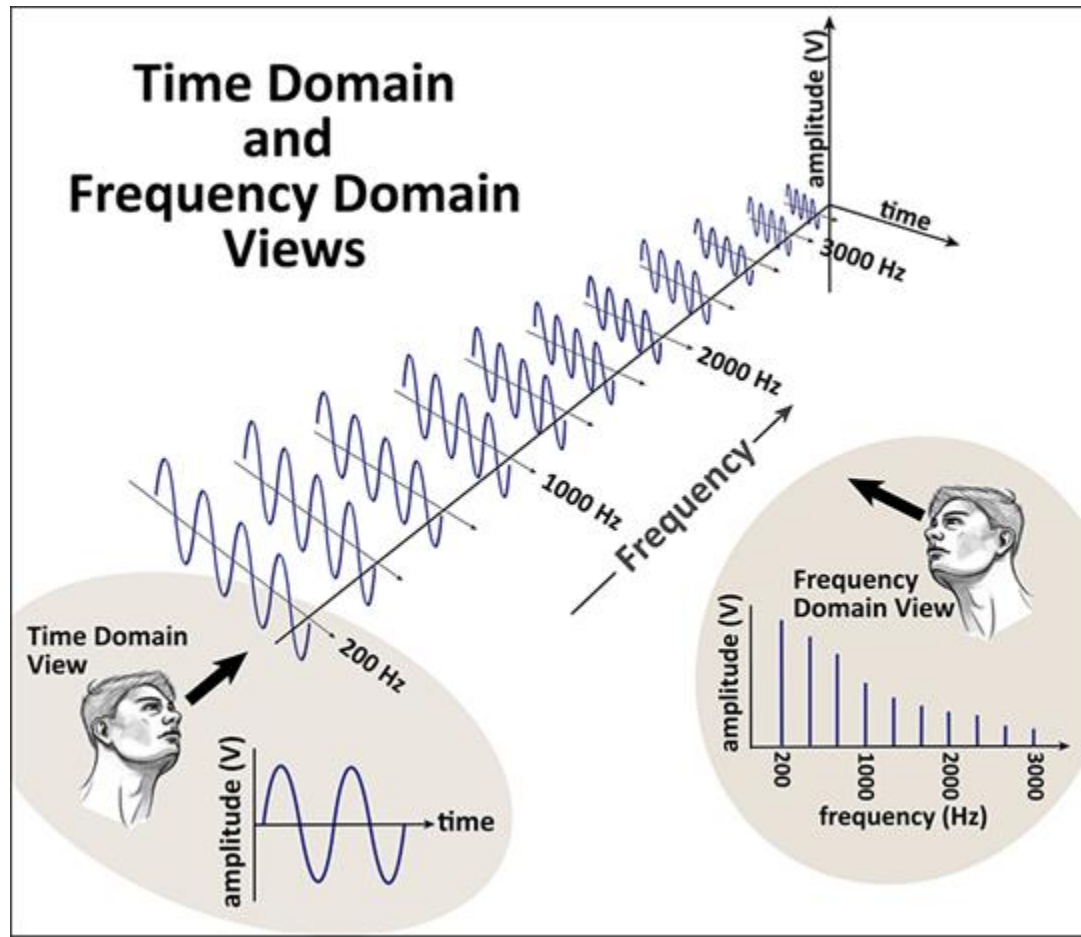
AUDIO SIGNAL PROCESSING FOR MACHINE LEARNING



5- AUDIO FEATURES

Time-Frequency Domain Representation

- Analyze and represent **signals** (e.g., audio) in both the **time** domain and the **frequency** domain **simultaneously**
 - How the **frequency content** of a **signal evolves**



Time-Frequency Domain Methods

Method	Resolution	Applications	Strengths	Weaknesses
STFT	Fixed time-frequency	Speech, music analysis	Simple, widely used	Limited resolution trade-off
Wavelet Transform	Adaptive resolution	Biomedical, transient signals	Good for transient signals	Computationally complex
Wigner-Ville Dist.	High resolution	Radar, quantum mechanics	High time-freq resolution	Cross-term interference
Mel Spectrogram	Mel scale resolution	Speech recognition, AI	Matches human perception	Limited raw frequency info
Constant-Q Transform	Logarithmic resolution	Music transcription	Suited for pitch analysis	Computationally intensive

Short-Time Fourier Transform (STFT)

- Divides the signal into **small, overlapping time windows (segments)** and applies the **Fourier Transform** to each window
 - This creates a spectrogram
 - Provides a time-varying frequency spectrum
- **Pros:**
 - Easy to implement
 - Widely used and understood.
- **Cons:**
 - Fixed resolution: It uses the same window size for all frequencies, which limits the trade-off between time and frequency resolution.

Signal



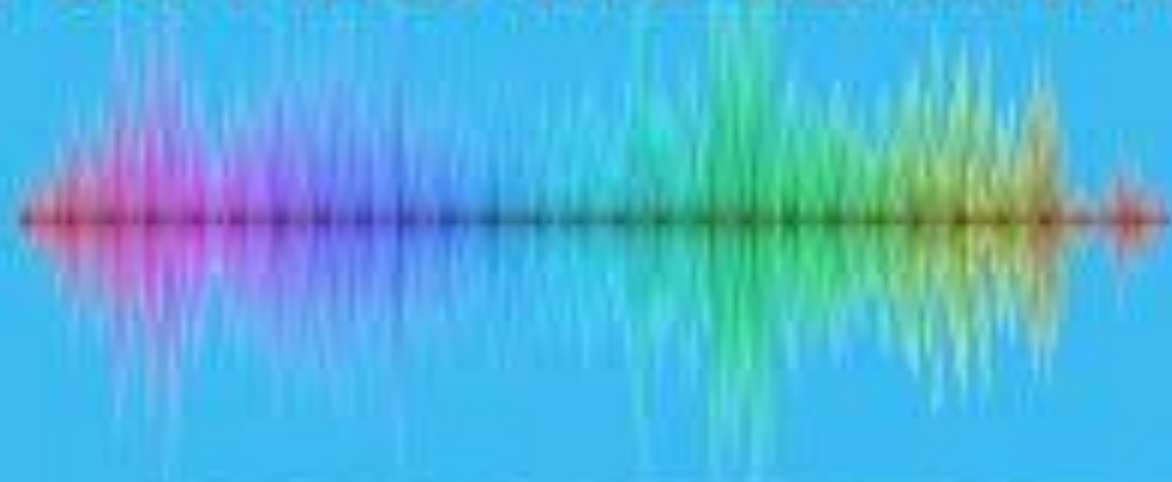
Winding



Transform



AUDIO SIGNAL PROCESSING FOR MACHINE LEARNING

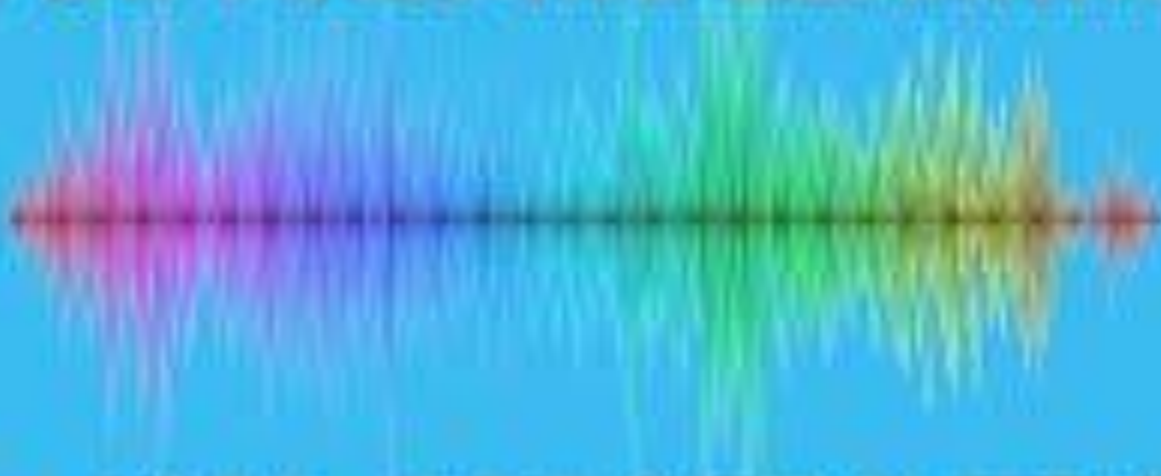


6- AUDIO FEATURES EXTRACTION PIPELINE

Mel Spectrogram

- A **variant** of the **spectrogram** that maps frequencies onto the **Mel scale**, which **approximates** how **humans perceive pitch**.
 - A time-frequency representation where the y-axis is on the Mel scale
- Pros:
 - Matches human auditory perception.
 - Compact and efficient for machine learning tasks
- Cons:
 - Does not preserve raw frequency information.

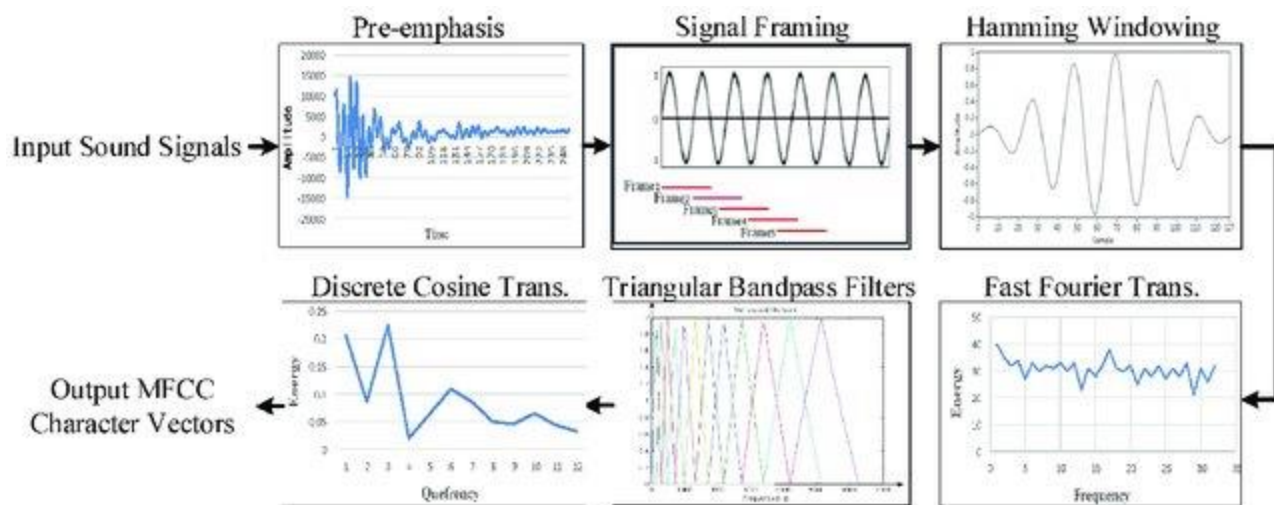
AUDIO SIGNAL PROCESSING FOR MACHINE LEARNING



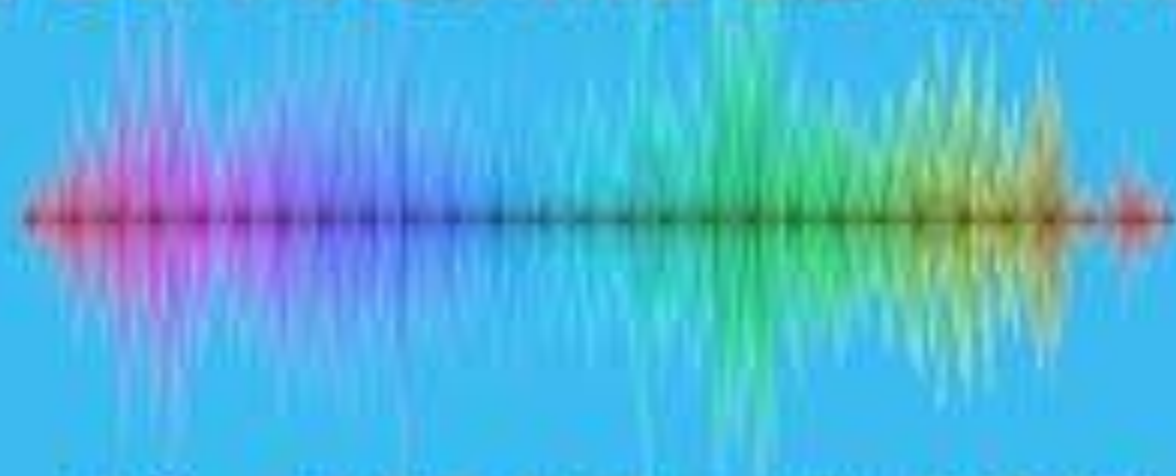
17 - MEL SPECTROGRAMS

Mel-Frequency Cepstral Coefficients (MFCC)

- **Compact** representation of the **power spectrum** of an audio signal and are specifically designed to approximate how humans perceive sound (Mel scale)
 - Reduce high-dimensional audio data into **meaningful** and compact features aligned with **human perception**



AUDIO SIGNAL PROCESSING FOR MACHINE LEARNING



19 - MFCCs: A KEY AUDIO FEATURE

Homework

- Extract Mel Spectrogram using Python: https://youtu.be/TdnVE5m3o_0?si=dio_XClWn-trMjWX
- Extract MFCCs using Python: <https://youtu.be/WJI-17MNpdE?si=ptIJBZG6qPzldLWu>

4. Speech Dataset



Nguyen Thi Thu Trang

Speech Dataset

1. Introduction
2. Dataset Building
3. Dataset Annotation
4. Data Processing
5. Benchmarking & Evaluation

1. Introduction

- Why we need Speech Datasets?
 - Foundation for training and evaluating speech technology models
 - Main tasks: ASR, TTS, Speaker Recognition, Speaker Diarization, Speech Emotion Recognition, Multimodal Generative AI, Mispronunciation Detection and Diagnosis...

Type of Speech Datasets

- Controlled vs. Spontaneous speech
 - Controlled: in a reading style, clear pronunciation (audio books, reading datasets, etc.)
- Monologue vs. Dialogue vs. Conversational
 - Single, two-, multi-speaker

Type of Speech Datasets

- Scripted vs. improvised
 - For emotion detections
- Clean vs. noisy
 - Some fields cannot be recorded in noise-free environment (air traffic control)

Example: VOiCES Dataset



Popular Speech Datasets

1. ASR

- LibriSpeech, CommonVoice, TEDLIUM, VoxPuli

1. Speaker Recognition & Diarization

- VoxCeleb, DIHARD, AMI Meeting, CNCeleb

1. TTS & Speech Synthesis

- VCTK, LibriTTS

1. Speech Emotion Recognition

- IEMOCAP, MSP-Podcast, MELD

2. Speech Dataset Building

Collection methodologies:

- Recording (expensive, not diverse)
- Crawling (unlabeled/weakly labeled)

Speech Data Collection

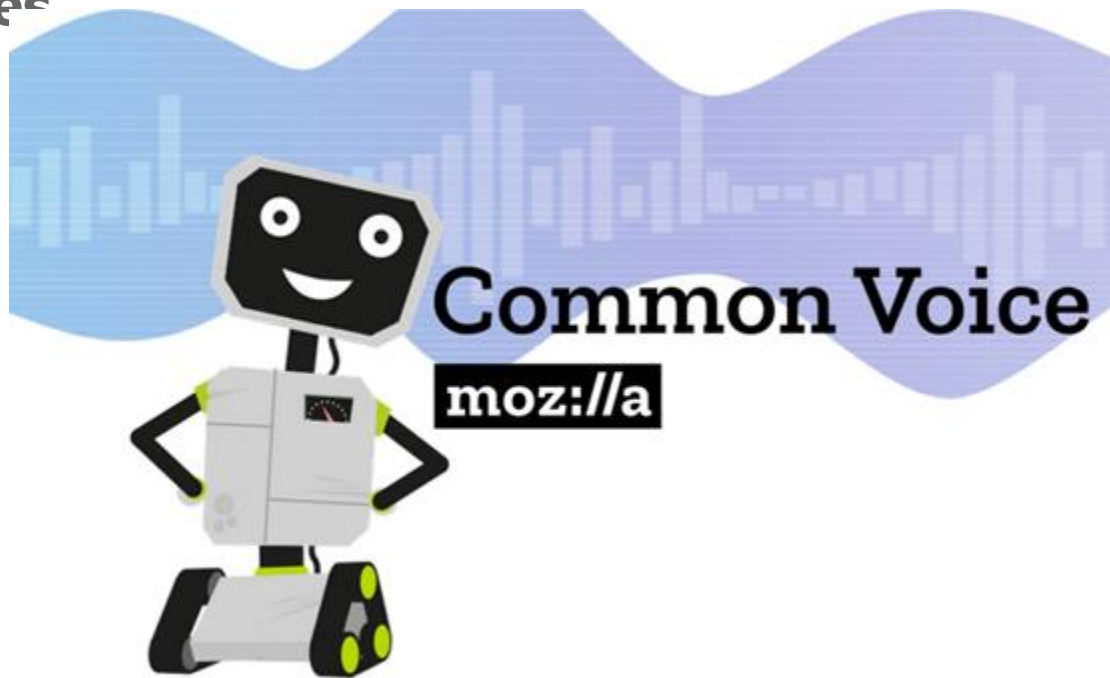
Method	Pros	Cons
Studio recording	- High-quality, clean speech - Controlled conditions (consistent microphone & environment) - Easier transcription & labeling	- Expensive & time-consuming - May not represent real-world conditions - Limited speaker diversity
Crowdsourced recording	- Large-scale collection possible - Speaker diversity - Cheaper than studio recording	- Variable recording quality - Risk of noisy or low-quality samples - Requires extensive quality control

Noise-free Recording in Studio



CommonVoice: Crowd-sourced Recording

Visit <https://commonvoice.mozilla.org/> and **donate** some utterances



CommonVoice: Crowd-sourced Recording



CommonVoice - more detail



Speech Data Collection

Method	Pros	Cons
Crawling	- Huge amount of data available - Natural, real-world speech - Useful for domain-specific datasets	- Licensing & legal issues - Poor transcription quality (requires re-annotation) - Inconsistent audio quality
Synthesis	- Unlimited data generation - Can simulate rare accents or low-resource languages	- Lack of natural prosody - Can introduce biases - May not generalize well to real-world scenarios

Synthetic Speech Data Generation

- **Step 1:** Define Data Requirements
- **Step 2:** Select a TTS Model for Speech Synthesis
- **Step 3:** Prepare Text Data for Speech Synthesis
- **Step 4:** Generate Synthetic Speech Data
- **Step 5:** Post-Processing & Data Augmentation

Step 1: Define Data Requirements

- Which task?
- Diverse in terms of speaker, speed, and intonation?
- How many hours of data are required?

Task	Minimum Hours	Optimal Hours	Diversity Requirements
ASR (Speech-to-Text)	100-500h	5,000-10,000h	Multiple speakers, accents, background noise, different speaking speeds
Voice Cloning (1-shot/few-shot)	1-5h per speaker	10-20h per speaker	High-quality recordings, same microphone conditions, same speaking style
Text-to-Speech (TTS) Training	10-50h	500-1000h	Multiple voices, different emotions, balanced phoneme distribution
Speaker Recognition	1-5h per speaker	500-1000h	Various microphones, environments, speaking styles
Emotion Recognition	50-100h	500-1000h	Different emotional tones, gender balance, age diversity

Step 2: TTS Models for Speech Synthesis

- **FastSpeech 2, Tacotron 2 + WaveGlow/WaveRNN:** Natural-sounding speech but requires fine-tuning.
- **VITS (End-to-End Vocoder + TTS):** High-quality speech with fast synthesis speed.
- **F5-TTS:** multi-speaker & multilingual, high-quality, expressive , but requires fine-tuning for best quality.
- **Neural TTS API (Microsoft Azure, Google Cloud, Amazon Polly):** Fast synthesis, high-quality speech but limited customization.

TTS Models for Speech Synthesis

Model	Key Features	Advantages	Disadvantages
FastSpeech 2, Tacotron 2 + WaveGlow/WaveRNN	Natural-sounding speech, prosody modeling	High-quality, expressive speech	Requires fine-tuning for optimal performance
VITS (End-to-End Vocoder + TTS)	GAN-based architecture, end-to-end synthesis	High-quality speech with fast synthesis speed	May require high computational power
F5-TTS	Multi-speaker & multilingual, expressive speech	High-quality, expressive speech, adaptable for multiple languages	Requires fine-tuning for best results
Neural TTS API (Microsoft Azure, Google Cloud, Amazon Polly)	Pre-trained commercial API, optimized synthesis	Fast synthesis, high-quality speech, low latency	Limited customization, predefined voices

Step 3: Prepare Text Data for Speech Synthesis

- **Ensure a diverse text corpus** (covering phonemes, vocabulary, and sentence structures)
- **Use proper punctuation and spacing** to improve synthesis quality
- **Balance content** to avoid dataset bias
- **Apply text augmentation** techniques to increase diversity

Best practices for text corpus selection:

- Sentences covering different tonal variations (for tonal languages like Vietnamese)
- A mix of short, long, declarative, and interrogative sentences
- Dialogues, news texts, audiobook transcripts, or movie scripts

Step 4: Generate Synthetic Speech Data

- Run **TTS synthesis** to generate **WAV (or FLAC) files**.
- **Combine multiple voices** to create a more balanced dataset.

Example for SER task:

- Use an expressive TTS model (e.g., F5-TTS, VITS, or StyleTTS2) that supports emotion synthesis.
- Balance the dataset so that each emotion is equally represented.

Step 5: Post-Processing & Data Augmentation

- **Add background noise:** Helps ASR models handle real-world conditions
- **Modify speech rate, pitch, and emotion:** Improves robustness to different speaking styles
- **Simulate different microphone effects:** Improves model generalization

Useful tools for augmentation:

- **SoX, ffmpeg** for modifying speech characteristics
- **Room Impulse Response (RIR)** for simulating recording environments
- **WaveAugment, SpecAugment** for perturbing audio signals.

Step 5: Example

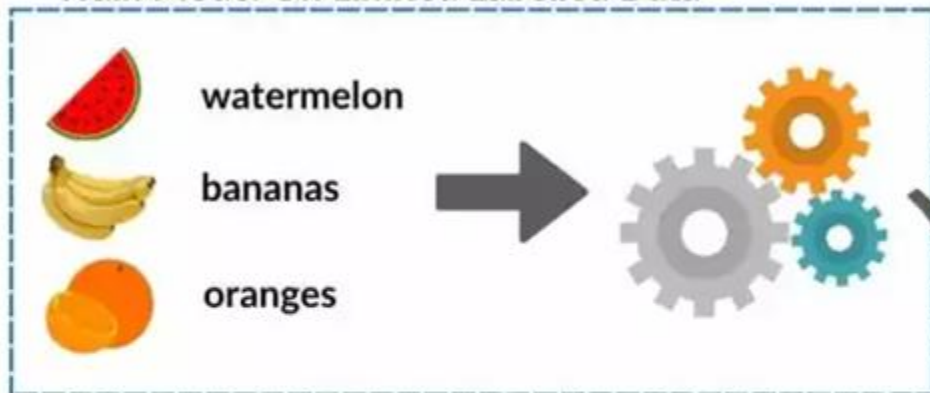
- **ASR:**
 - Adding **background noise** (e.g., cafe noise, street noise, reverberation).
 - Applying **speed perturbation** (slightly speeding up or slowing down speech).
 - Mixing **low-quality and high-quality samples** (e.g., phone call audio vs. studio recordings).
- **Voice Cloning**
 - **Speaking speed** (to simulate different conversational styles).
 - **Pitch & intensity** (to add natural variation).
- **SER:** Generate speech samples with different emotions, ensuring:
 - Variations in pitch, intensity, and speaking rate.
 - Different speakers expressing the same emotion (to prevent speaker bias).

3. Speech Data Annotation

- Human annotation: more correct, but requires large-scale quality control, and expensive
- Weak labels: mostly generated by larger models of the same task
 - Auto-generated transcriptions as labels for ASR models (Whisper)
 - Clustering for speaker labels
- Semi-supervised annotations

How To Expand A Partially-Labelled Dataset With Semi-Supervised Learning

Train Model On Limited Labelled Data

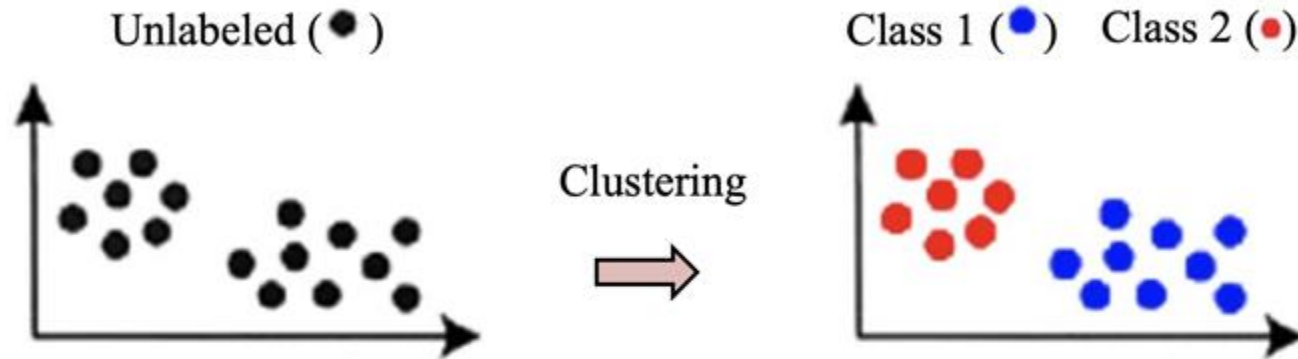


Use The Trained
Model To Make
Predictions

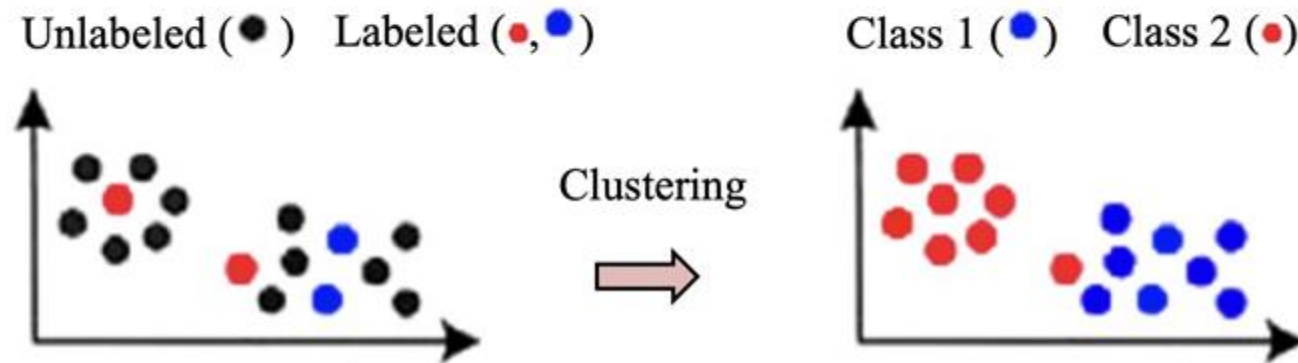
Predict Labels For Unlabelled Data



Unsupervised clustering



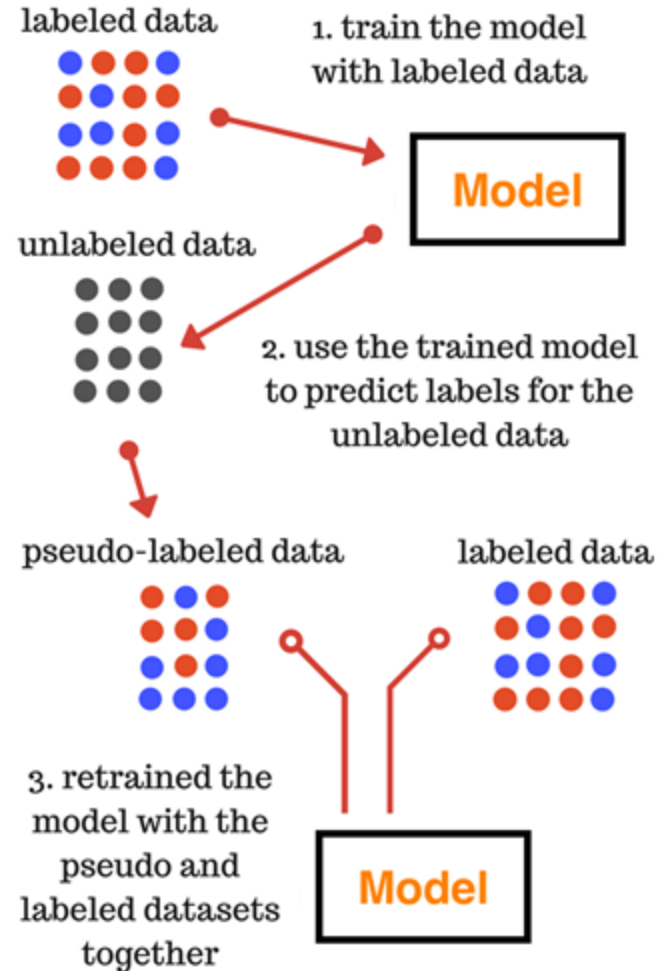
Semi-supervised clustering



Shi, X., Yue, C.,
Quan, M. *et al.* A
semi-supervised
ensemble
clustering algorithm
for discovering
relationships
between different
diseases by
extracting cell-to-
cell biological
communications. *J
Cancer Res Clin
Oncol* 150, 3 (2024)

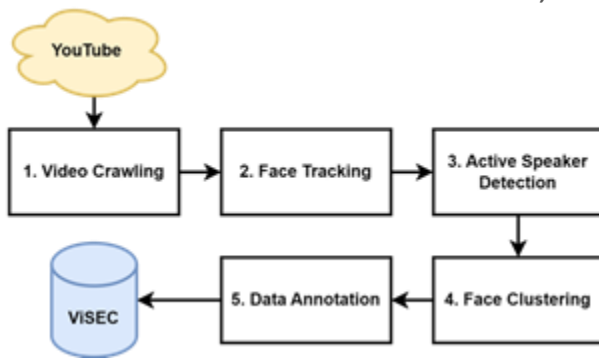
Pseudo Labeling

1. Train a model with labelled data
2. Use the trained model to predict labels for the unlabeled data, which creates pseudo-labeled data.
3. Retrain the model with the pseudo-labeled and labeled data together.



Example: Vietnam-Celeb PIPELINE

- Crawling from public media
 - Visual-audio construction pipeline
 - 100% automatic or semi-auto labeling
 - Tasks: Speaker Recognition, SER
- Recording in spontaneous environments
 - Crowded sourcing with validation process or expert annotation
 - Tasks: MD&D, SLU



(Pham et al.) Pham, V.T., Nguyen, X.T.H., Hoang, V., Nguyen, T.T.T. (2023) Vietnam-Celeb: a large-scale dataset for Vietnamese speaker recognition. Proc. INTERSPEECH 2023, 1918-1922, doi: 10.21437/Interspeech.2023-1989

Visual-aid for Speaker Identification

Face Tracking



- With the crawled videos, face tracking is performed to extract videos of only facial areas.
- Use the S3FD (Zhang et al. 2017) model for detecting faces in a video. This is a model that can run in real-time and can detect faces of various sizes and scales.
- **Face stream videos are produced by aggregating consecutive facial bounding boxes where the Intersection over Union (IOU) score is higher than 0.5.**

Visual-aid for Speaker Identification

Face Detection

- Performing face detection using RetinaFace (Deng et al. 2020)
- Extracting face embeddings / representations using ArcFace (Deng et al. 2019)
- Removing invalid images with K-means clustering



Xuân Bắc
(actor/comedian)

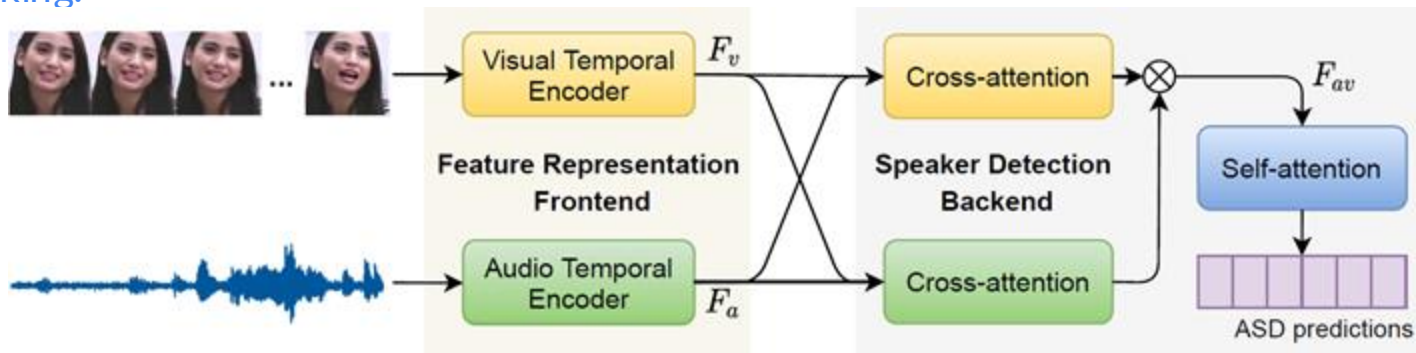
(Deng et al. 2020) Deng, J., Guo, J., Ververas, E., Kotsia, I., & Zafeiriou, S. (2020). Retinaface: Single-shot multi-level face localisation in the wild. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 5203-5212).

(Deng et al. 2019) Deng, J., Guo, J., Xue, N., & Zafeiriou, S. (2019). Arcface: Additive angular margin loss for deep face recognition. In Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (pp. 4690-4699).

Visual-aid for Speaker Identification

Active Speaker Verification

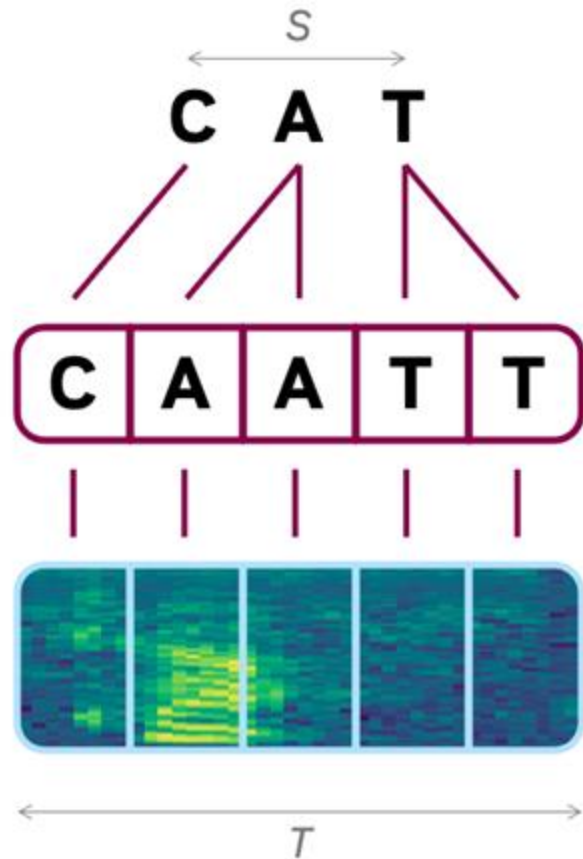
- Use the TalkNet (Tao et al. 2021) model to verify if the person in a video is really speaking
- For each video input, TalkNet outputs a **sequence of confidence scores** corresponding to the frames of the video.
- This sequence of scores is then used to mark out the videos where the person is actually speaking.



(Tao et al. 2021) Tao, R., Pan, Z., Das, R. K., Qian, X., Shou, M. Z., & Li, H. (2021, October). Is someone speaking? exploring long-term temporal features for audio-visual active speaker detection. In Proceedings of the 29th ACM International Conference on Multimedia (pp. 3927-3935).

Introduction

- Process of **automatically aligning speech audio with its corresponding phoneme, word, or sentence-level transcription**
 - Determines **the precise timestamps** (start and end times) of words or phonemes in a given speech recording.



Why we need forced alignment?

- **Speech Dataset Annotation:** Helps align transcriptions with speech data, enabling better supervised learning for ASR, TTS... models
- **Phonetic and Prosody Analysis** – Used in linguistic research to analyze pronunciation, stress, and intonation.

If I put it in my batter

Common Alignment methods

Method	Example Tools	Strengths	Weaknesses
(A) HMM-GMM Based Forced Alignment	HTK, Kaldi, MFA (GMM mode)	Precise phoneme alignment , time-proven	Requires lexicons, struggles with noisy speech
(B) Deep Learning-Based Forced Alignment	MFA (DNN mode), Gentle, Deep Phonetics Aligners	More robust to speaker variations and accents	Requires large datasets for training
(C) CTC-Based Forced Alignment (Hybrid)	Kaldi (Hybrid HMM-DNN), DeepSpeech, Wav2Vec 2.0	Faster than HMM, does not require frame alignment during training	Less precise than HMM for phoneme-level segmentation
(D) End-to-End ASR-Based Forced Alignment	Whisper, Aeneas, DeepSpeech	Handles real-world speech well , good for word alignment	Lacks precise phoneme-level segmentation

HMM-GMM-based Forced Alignment

- Traditional method based on **acoustic models trained with HMM-GMM (Gaussian Mixture Models)**.
- Works by **aligning phonemes to the most probable time frames** based on the model's acoustic scores.
- **Used in:** CMU Sphinx, Kaldi, HTK.
- **Limitation:** Less accurate for spontaneous speech due to **rigid phoneme modeling**.

Deep Learning-Based Forced Alignment

- Uses **Deep Neural Networks (DNNs)**, **Convolutional Neural Networks (CNNs)**, or **Transformers** for better acoustic modeling.
- More robust to **speaker variations, accents, and background noise**.
- **Used in:** MFA (Montreal Forced Aligner), Gentle (Kaldi-based).
- **Limitation:** Requires **more computational resources** compared to HMM-based methods.

CTC-Based Forced Alignment (Hybrid)

- Uses **Connectionist Temporal Classification (CTC)** to align speech with text **by optimizing over possible paths, allowing for flexible time alignment at the word or phoneme level rather than requiring strict frame-wise labeling.**
- Can handle **variable-length speech**, making it **faster than HMM-based alignment** while maintaining reasonable accuracy.
- Works well for **phoneme, word, or character-level alignment** depending on the task.
- Used in: **Kaldi (Hybrid HMM-DNN mode), DeepSpeech, Wav2Vec 2.0.**
- **Limitation:** Less precise than **HMM-based forced alignment**, especially for fine-grained phoneme boundaries.

End-to-End ASR-based Forced Alignment

- Uses **state-of-the-art ASR models (e.g., Whisper)** for alignment.
- Can handle **spontaneous speech, conversational speech, and noisy environments**.
- **Used in:** Aeneas, Whisper-based Forced Alignment.
- **Limitation:** Can be **less precise at phoneme-level alignment** compared to dedicated phonetic models.

Common Tools for Forced Alignment

Tool	Methodology	Features	Use Cases
Montreal Forced Aligner (MFA)	HMM + Deep Learning	Supports multiple languages, high accuracy, word/phoneme-level alignment	ASR dataset preparation, phonetic analysis
Gentle Forced Aligner	Kaldi-based	Robust for English, easy to use, handles noisy data	ASR pre-processing, word alignment
Aeneas	ASR-based (DeepSpeech, Kaldi)	Aligns text with speech at word/sentence level	Audiobooks, TTS alignment
Prosodylab-Aligner	HMM-GMM	Lightweight, works well with clean speech	Linguistic research, phoneme alignment
HTK (Hidden Markov Toolkit)	HMM-based	Older but widely used in speech research	ASR training, phonetic segmentation
Whisper Forced Aligner	Transformer-based ASR	High accuracy, zero-shot alignment	Automatic transcription, ASR tuning

4. Speech Data Processing

- Clean and well-segmented datasets improve the accuracy of speech recognition, speaker identification, and other AI-driven speech applications.
- Speech augmentation for more variances

Speech Processing

- **Speaker Diarization:** Identify and label different speakers in conversations.
- **Speech Segmentation:** Split long recordings into manageable utterances (using VAD)
- **Noise Reduction:** Remove background noise and reverberation if needed.
- **Volume Normalization:** Standardize audio levels for consistency.
- **Resampling:** Convert all recordings to a common sample rate (e.g., 16kHz for ASR).
- **Silence Trimming:** Remove excessive silence at the beginning and end.

Speech Data Augmentation

- **Noise augmentation, room simulation**
 - Background noise addition
 - Simulating reverberation effects with room impulse responses
- **Speed perturbation:** changing playback speed while preserving pitch
- **SpecAugment:** masking parts of the spectrograms

Noise Reduction in Datasets Preprocessing

- Unwanted sounds that interfere with the clarity and intelligibility of speech
- Noise reduces the accuracy of speech recognition and speaker verification models.
- Helps in creating a standardized, high-quality dataset for training AI models.

Types of Noises in Speech Datasets

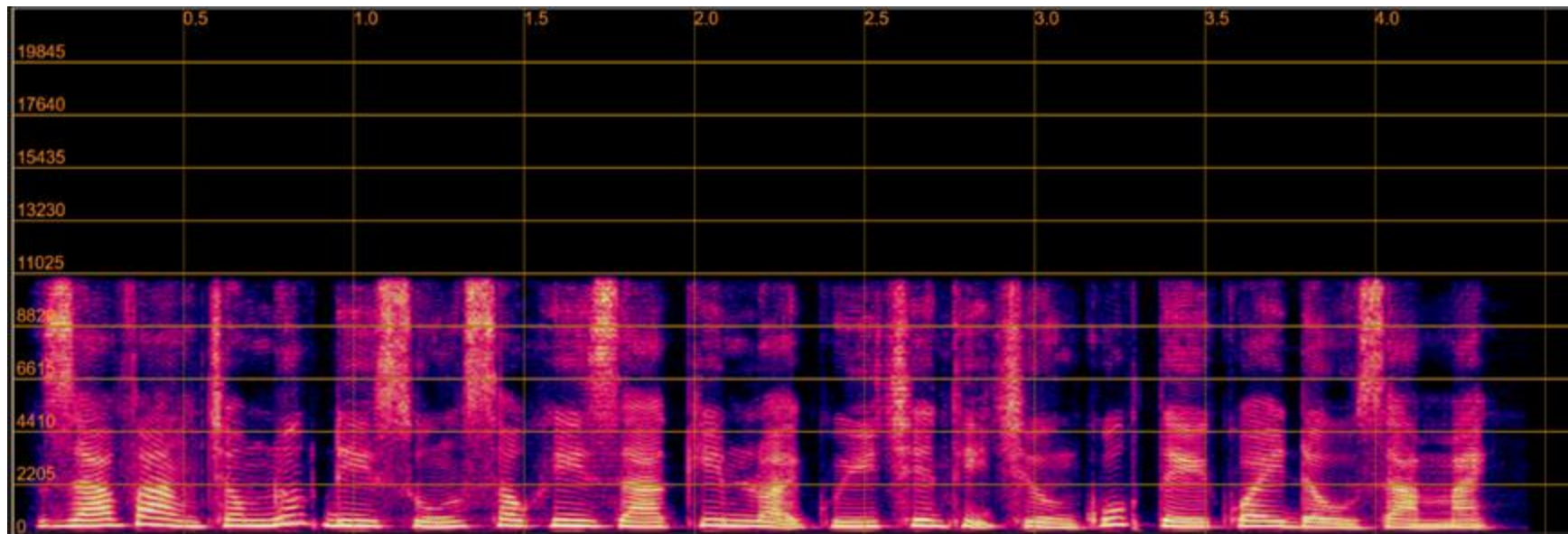
- **Background disturbances in recordings:**
 - Environmental sounds (traffic, fan noise, crowd chatter).
 - Microphone artifacts (hissing, electrical hum).
- **Transient Noise:** Short-duration sounds, including clicks, pops, or microphone handling noises.
- **Reverberation:** Reflections of the speech signal off surfaces, causing a prolonged and muffled sound.

Types of Noises in Speech Datasets

Clean audio examples:



Melspectrogram:

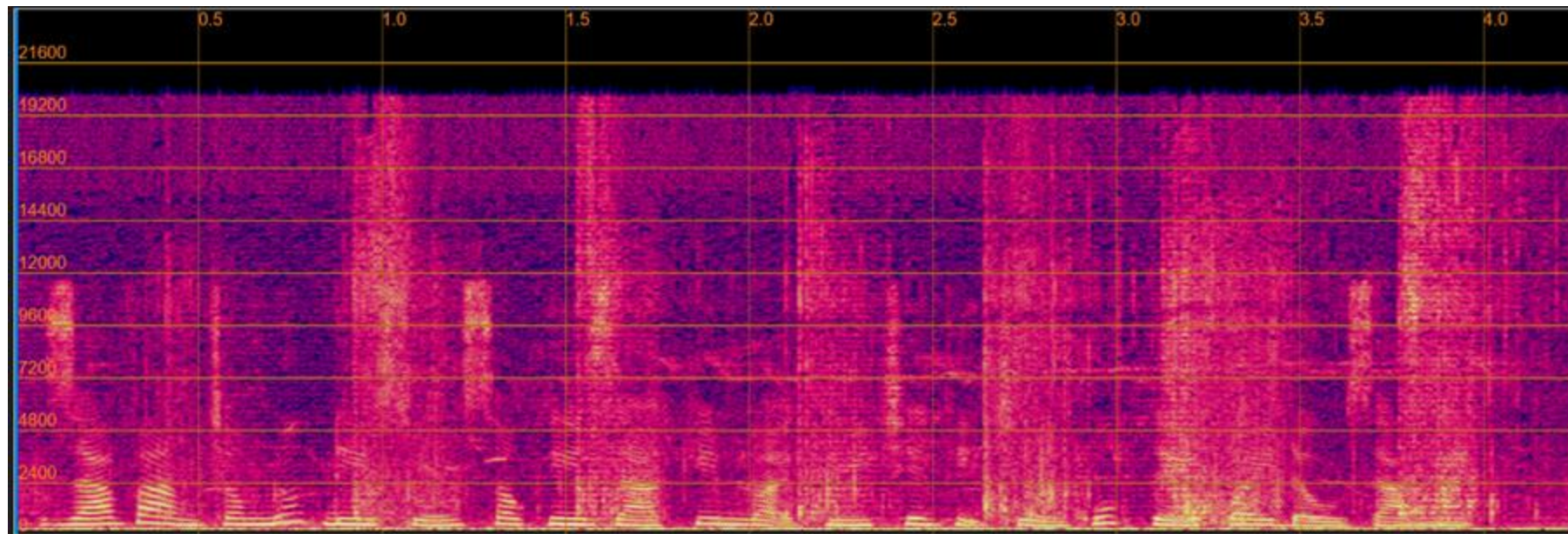


Types of Noises in Speech Datasets

Audio with background noise examples:



Melspectrogram:

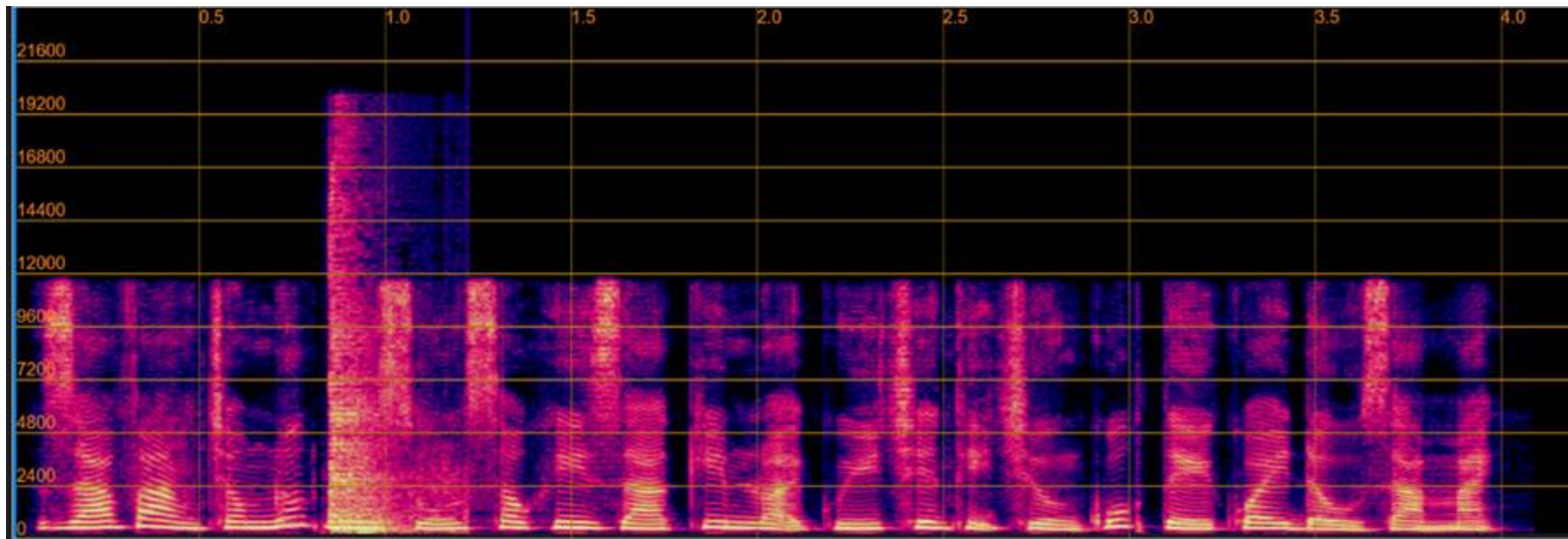


Types of Noises in Speech Datasets

Audio with transient noise examples:



Melspectrogram:

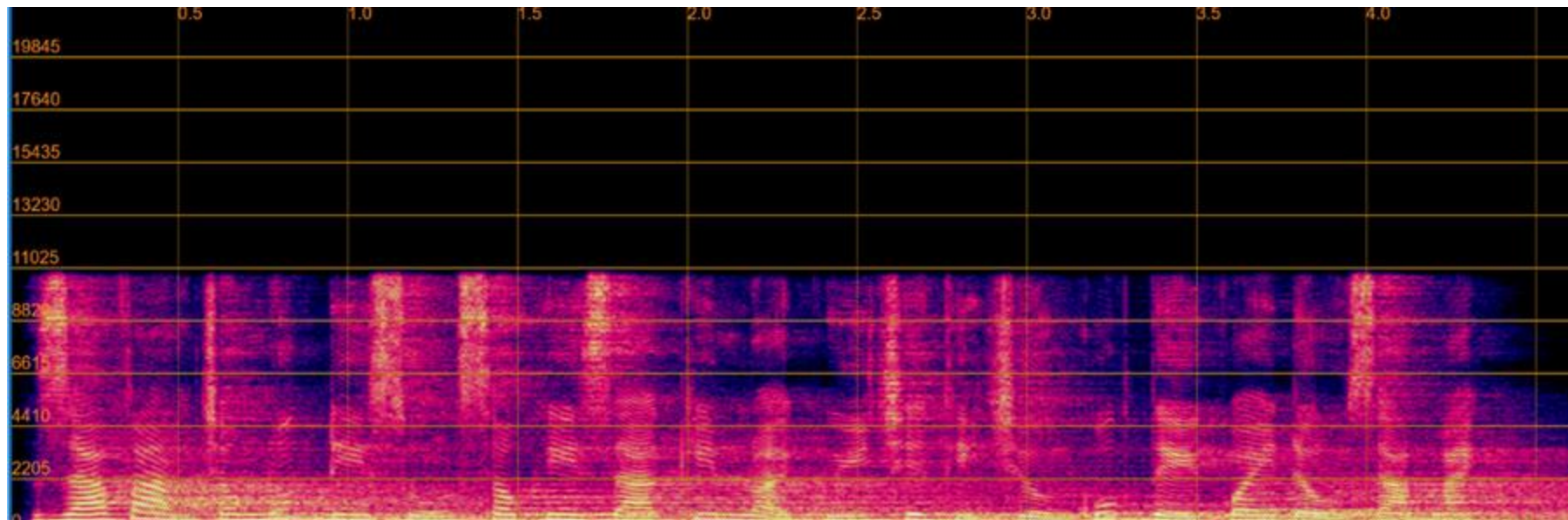


Types of Noises in Speech Datasets

Reverberation audio examples:



Melspectrogram:



Noise Removal Techniques

- **Spectral Subtraction** – Identifies and subtracts background noise from speech signals.
- **Adaptive Filtering** – Uses algorithms like Wiener filters to suppress unwanted noise.
- **Deep Learning-based Denoising** – Neural networks (e.g., U-Net, Denoising Autoencoders) trained to filter out noise.

Silence Trimming in Datasets Preprocessing

What is Silence in Speech Data?

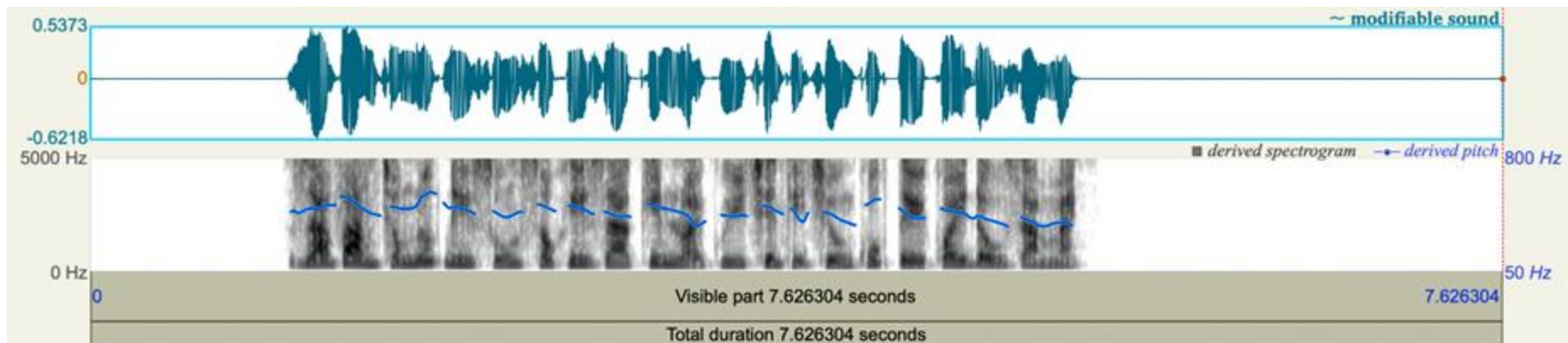
- **Intentional Silence:** Pause between words/sentences
- **Unintentional Silence:** Long gaps due to recording errors, microphone delay, or background interference.

Why Trim Silence?

- Reduces dataset size and training time.
- Prevents silent frames from biasing the model.
- Makes speech recognition models more responsive.

Silence Trimming Examples

	Before trimming	After trimming
Duration	7s	4s
Size	660Kb	400Kb



Silence Trimming Techniques

- **Energy-Based Detection:** Removes frames with very low amplitude.
- **Thresholding Methods:** Sets a dB threshold to define silence.
- **Voice Activity Detection (VAD):** Machine learning-based detection of speech vs. silence.

Practical Tools for Silence Trimming

Pydub



Speech Segmentation in Datasets Preprocessing

- Speech segmentation is the process of dividing continuous speech into distinct, meaningful units such as sentences, words, or phonemes.
- This process is essential in various speech-related applications, including automatic speech recognition (ASR), speaker diarization, and language modeling.
- Effective segmentation enhances the performance of these applications by providing structured and analyzable speech data.

Applications of Speech Segmentation

- **Automatic Speech Recognition (ASR):** Accurate segmentation aids in identifying word boundaries, improving transcription quality.
- **Speaker Diarization:** Determines "who spoke when," facilitating the analysis of multi-speaker recordings.
- **Language Modeling:** Segmentation into smaller units allows for the study of linguistic patterns and the development of more robust language models.

Speech Segmentation Techniques

- **Voice Activity Detection (VAD):** Distinguishes between speech and non-speech segments, focusing processing on relevant data.
- **Speaker Diarization:** Partitions audio based on speaker identity, essential for multi-speaker environments.
- **Phoneme Segmentation:** Breaks down speech into individual phonetic units, crucial for phonetic analysis and speech synthesis.

Voice Activity Detection (VAD)

Purpose: Identifies segments containing speech, filtering out silence and background noise.

Methods:

- **Energy-Based Approaches:** Analyze signal energy to detect speech presence.
- **Machine Learning Approaches:** Utilize trained models to differentiate speech from non-speech segments.

Speaker Diarization

Definition: The process of partitioning an audio stream into homogeneous segments according to the identity of each speaker.

Techniques:

- **Clustering Algorithms:** Group similar voice characteristics to identify distinct speakers.
- **Neural Network-Based Methods:** Employ deep learning models for more accurate speaker identification.

Applications: Useful in transcribing meetings, interviews, and analyzing conversational dynamics.

5. Benchmarking & Evaluation

- Large-scale speech datasets are crucial for ASR, TTS, Speaker Profiling, and Voice Cloning.
- Poor data quality can lead to biased or inaccurate models.
- Key evaluation criteria: Diversity, Quality, Balance, Labeling Accuracy

Data Diversity & Representativeness

- Ensuring that the dataset covers a **wide range of speakers, accents, and speaking conditions** is crucial.
- **Metrics:**
 - **Speaker Count & Distribution:** Ensure a balanced dataset across different genders, age groups, and accents.
 - **Phonetic Coverage:** Measure how well phonemes in the dataset match real-world speech patterns.
 - **Linguistic Coverage:** Compute word frequency distributions to detect bias toward certain vocabulary or sentence structures.
 - **Environment Variety:** Analyze different acoustic conditions (e.g., background noise levels, room reverberation).

Data Quality & Cleanliness

- Low-quality or noisy speech data can negatively impact model performance, making quality assessment crucial.
- **Metrics:**
 - **Signal-to-Noise Ratio (SNR):** Measures how much speech is present relative to background noise. (Higher SNR = better quality).
 - **Perceptual Evaluation of Speech Quality (PESQ):** Quantifies speech quality as perceived by humans.
 - **Short-Time Objective Intelligibility (STOI):** Measures how understandable speech is, especially in noisy environments.
 - **Word-Level and Phoneme-Level Annotation Accuracy:** Checks alignment errors in labeled datasets.

Dataset Balance & Bias Analysis

- An imbalanced dataset can cause bias in models, leading to poor generalization.
- Metrics:
 - **Gender & Age Distribution:** Ensure dataset has fair representation across different demographics.
 - **Accent & Language Distribution:** Check if the dataset is biased toward a specific accent or dialect.
 - **Outlier Analysis:** Identify if certain speaker groups are underrepresented.

Labeling Accuracy & Annotation Quality

- For supervised learning, high-quality labels are essential. Misaligned or noisy labels can degrade performance.
- **Metrics:**
 - **Labeling Consistency Score:** Measures variation in transcription across annotators.
 - **Alignment Accuracy (ASR datasets):** Checks if word/phoneme timestamps match actual speech.
 - **Inter-Annotator Agreement (IAA):** (Cohen's Kappa Score, Fleiss' Kappa) – Measures how much agreement exists between human annotators.
- A recent trend: LLM as a judge

Best practices for Dataset Evaluation

- **Use Automated Tools for Quality Checks**

- Noisiness & Speech Quality → SoX, librosa, PESQ, SNR estimators
- Diversity Analysis → t-SNE visualization of speaker embeddings
- Bias Detection → Gender, Age, Accent Distribution Analysis

- **Run Benchmark Models on Subsets**

- Evaluate Word Error Rate (WER), Character Error Rate (CER), and Speaker Verification EER before using the full dataset.

Best practices for Dataset Evaluation

- **Human-in-the-Loop Validation**

- Use a small subset for manual annotation verification.

- **Monitor Dataset Drift**

- If adding new data, check for shifts in distribution (e.g., changes in speaker demographics, recording quality).

5. Model Evaluation



Nguyen Thi Thu Trang

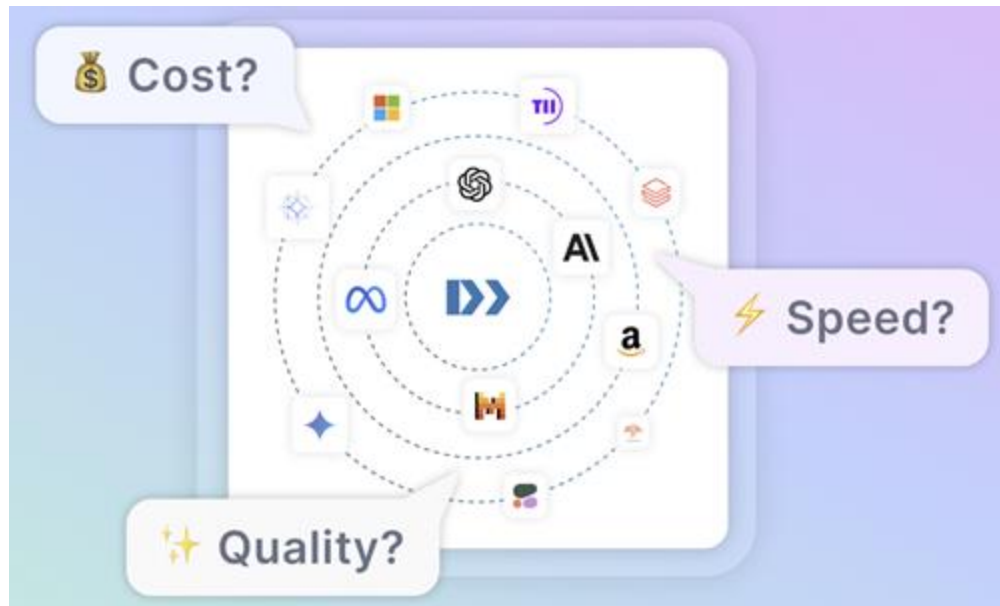
5. Model Evaluation

1. Model Evaluation
2. Evaluation for Automatic Speech Recognition (ASR)
3. Evaluation for Text-to-Speech (TTS) & Voice Cloning
4. Challenges and Future Directions

5.1. Model Evaluation

- Assess how well a model performs on **unseen data**
- Measure progress and compare different models or algorithms

=> A poorly evaluated model can lead to usability issues, mistrust in AI systems, or even harm in critical



Dataset

- Train: Seen
- Dev / Validation: Seen
 - Parameter Optimization
 - Error Analysis
- Test: Unseen

Importance of Model Evaluation

- Direct impact on user experience (e.g., intelligibility of TTS or accuracy of ASR)
- Application domains are sensitive to errors, such as healthcare (speech profiling for diseases) or customer service (ASR for call centers).
- Real-time applications demand not just accuracy but also efficiency and robustness.

Discussion

- Why might traditional metrics like accuracy or F1-score be insufficient for evaluating speech models?
- Can you think of any real-world applications where evaluating speech models poorly could have significant consequences?

Discussion

- Loss of Temporal Information
 - Speech is inherently sequential and time-dependent.
 - Accuracy and F1-score do not capture alignment errors, which are crucial in Automatic Speech Recognition (ASR) and Text-to-Speech (TTS).

Common metrics for different tasks

- **STT (ASR):** Word Error Rate (WER), Syllable Error Rate (SylER), Character Error Rate (CER), Phoneme Error Rate (PER)
- **TTS:** Mean Opinion Score (MOS), Intelligibility (WER), Mel Cepstral Distortion (MCD)
- **Voice Cloning:** Similarity Score, WER, Mean Opinion Score (MOS), Signal-to-Noise Ratio (SNR)
- **Voice Conversion:** Similarity Score, WER
- **Speaker Verification:** EER (Equal Error Rate)
- **Speaker Identification:** Accuracy, Top-k accuracy, Confusion Matrix

5.2. Evaluation for ASR Task

- Automatic Speech Recognition (ASR) refers to converting spoken language into text.
- Accurate ASR is critical for applications like virtual assistants, transcription services, and voice-controlled systems.
- Challenges in ASR Evaluation:
 - Variability in accents, dialects, and background noise.
 - Vocabulary size and domain specificity.
 - Balancing accuracy and real-time performance.

Word Error Rate (WER)

Definition and Significance

- Word Error Rate (WER) is the most commonly used metric to evaluate ASR systems.
- It measures the percentage of words incorrectly predicted by the ASR system compared to a ground truth reference.

Calculation Process: $WER = (S + D + I) / N$

where:

- S = Substitutions (words replaced incorrectly)
- D = Deletions (words omitted)
- I = Insertions (extra words added)
- N = Total number of words in the reference

Word Error Rate (WER)

Reference Sentence:



Tôi thích ăn phở gà.

ASR Output:



Tôi yêu ăn gà

bò.

Reference	ASR Output	Error Type
Tôi /toj/	Tôi	 Correct
Thích /tʰik/	yêu	 Substitutions
ăn	ăn	 Correct
phở	—	 Deletions
gà	gà	 Correct
—	bò	 Insertions

Word Error Rate (WER)

WER Calculation:

- Substitutions (S): 1 (“thích” → “yêu”)
- Deletions (D): 1 (“phở” is missing)
- Insertions (I): 1 (“bò” were added)
- Total words in reference (N): 5

$$\text{WER} = (S + D + I) / N = (1 + 1 + 1) / 5 = 0.6$$

Word Error Rate (WER)

Variants of WER

- Character Error Rate (CER): Measures errors at the character level, useful for languages with complex scripts.
- Syllable Error Rate (SylER): Measures errors at the syllable level, useful for languages with complex scripts.
- Sentence Error Rate (SER): Measures the percentage of sentences with at least one error.

Word Error Rate (WER)

Strengths and Limitations

- Strengths: Widely understood, easy to calculate, and directly related to ASR quality.
- Limitations:
 - Does not consider semantic correctness.
 - Penalizes minor errors equally to major ones.
 - Sensitive to out-of-vocabulary words and proper nouns.

Other ASR Metrics

Accuracy and Word Accuracy (WAcc)

- Accuracy = (Correctly recognized words / Total words) × 100%.
- Word Accuracy (WAcc) adjusts WER by accounting for correctly recognized words:
$$\text{WAcc} = 1 - \text{WER}$$

Word Recognition Rate (WRR)

- Measures the proportion of words recognized correctly out of the total spoken words.

Time to Transcription (Latency)

- Measures the time taken by the ASR system from receiving the audio input to generating the text output.
- Critical for real-time applications like live captioning or voice assistants.

Human vs. Machine Performance

- Human transcription is often used as a gold standard, but even human transcribers have error rates, typically around 5-6%.
- Recent ASR systems, especially those using deep learning, have approached and even matched human-level accuracy in controlled environments.
- Case Study: Google's ASR system achieved a WER of 5.1% on the Switchboard dataset, comparable to human transcription performance.

ASR Evaluation Best Practices

- **Dataset Considerations**

- Use large, diverse datasets covering multiple accents, speaking rates, and background conditions.
- Common benchmarks include LibriSpeech, Switchboard, and Common Voice.

- **Speaker Diversity and Acoustic Conditions**

- Include speakers from different age groups, genders, and regional accents.
- Evaluate performance in noisy environments, telephone channels, and varying microphone qualities.

- **Continuous Evaluation in Real-World Deployments**

- Regularly evaluate ASR systems with live data to adapt to evolving language use, slang, and domain-specific vocabulary.
- Employ user feedback and error analysis to improve system accuracy over time.

5.3. Evaluation for TTS/Voice Cloning

- **Objective** (automatic, computable)
 - Mel Cepstral Distortion (MCD)
 - Latency and real-time factor (RTF)
 - Speaker Similarity
- **Subjective** (based on human perception)
 - MOS, CMOS, SMOS
 - ABX or A/B Test
- **Intelligibility**: Word Error Rate (WER)

Perception Test

A perception test in Text-to-Speech (TTS) systems evaluates how well a synthetic voice performs in terms of its naturalness, intelligibility, and listener preferences. These tests aim to assess the quality of TTS output from the perspective of human listeners.

Common types of perception tests used in TTS evaluation:

- Mean Opinion Score (MOS)
- ABX Test
- AB Test
- Intelligibility Test
- DMOS
- CMOS / SMOS
- Degradation Category Rating (DCR)
- Preference Test

Goals of Perception Testing in TTS:

- Benchmarking: Compare the system against competitors or human recordings.
- Fine-tuning: Identify areas for improvement, such as pronunciation, intonation, or fluency.
- User acceptance: Ensure the TTS system meets the expectations of its target

Mean Opinion Score (MOS): Difference aspects

- Standard human evaluation method (subjective)
- Listeners rate audio on a **scale from 1 to 5** based on **naturalness, intelligibility, or any other aspects of quality.**
- Ground-truth should be included
- Common score scale:
 - 5 = Excellent
 - 4 = Good
 - 3 = Fair
 - 2 = Poor
 - 1 = Bad

Mean Opinion Score (MOS): Naturalness

Naturalness refers to how closely the synthesized speech resembles human speech in terms of sound quality, absence of synthetic artifacts, and overall pleasantness.

Xin chào, Lê Minh Nguyễn

Đổi mật khẩu

Đăng xuất

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Đánh giá chất lượng giọng nói

5 điểm - Rất tự nhiên, giống như giọng người thật.

4 điểm - Tương đối tự nhiên, khá giống giọng người thật

3 điểm - Hơi tự nhiên, khá nhiều yếu tố nhân tạo

2 điểm - Kém tự nhiên, rất nhiều yếu tố nhân tạo

1 điểm - Rất kém tự nhiên, hoàn toàn nhân tạo

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Kết thúc

Câu tiếp >

Mean Opinion Score (MOS): Naturalness

How to Perform a MOS Test?

Prepare the Test Materials

- Select a representative dataset of text prompts to synthesize. These should include diverse sentence structures, word complexities, and prosody variations.
- Generate audio samples using the TTS system being tested.

Design the Test Setup

- Choose a rating scale (typically 1-5). You can also use finer scales (e.g., 1-10) or anchor-based rating (e.g., comparing against a reference).
- Provide clear instructions to the participants on how to rate the samples.

Recruit Participants

- Use human evaluators who are fluent in the target language and familiar with the nuances of natural speech.
- Ensure a diverse pool of evaluators to reduce biases (e.g., native vs. non-native speakers, gender balance).

Conduct the Test

- Randomize the order of the audio samples to prevent bias.
- Use a blind test setup, where listeners are not aware of which system generated the audio.
- Provide headphones or a controlled environment to ensure audio quality consistency.
- Ask participants to rate each sample based on specific criteria (e.g., naturalness).

Collect Data

- Gather the ratings from all participants for all audio samples.
- If applicable, collect additional qualitative feedback or notes.

MOS: Perceptive Results

Calculate Results

Aggregate Scores: For each audio sample, compute the mean score by averaging the ratings across all participants.

- Example: If 10 participants rate a sample as [4, 4, 5, 3, 4, 5, 4, 5, 3, 4], the mean score is: $MOS = (4 + 4 + 5 + 3 + 4 + 5 + 4 + 5 + 3 + 4) / 10 = 4.1$

Calculate Overall MOS: Average the mean scores of all the audio samples to get the overall MOS for the TTS system.

Statistical Analysis:

- Compute the standard deviation (SD) to understand the variability in scores.
- Use confidence intervals (CI) to assess the reliability of the MOS. A common confidence level is 95%.

Different metrics for MOS

- **MOS**: "How good is this voice?"
- **DMOS**: "How degraded is this compared to the original?"
- **CMOS**: "Which one is better?"
- **SMOS**: "Does it sound like this speaker / emotion / style?"

Metric	Full Name	Purpose	Type
MOS	Mean Opinion Score	General naturalness/quality	Absolute
DMOS	Degradation MOS	Perceived degradation vs reference	Relative
CMOS	Comparison MOS	Preference between two samples	Relative
SMOS	Style/ Speaker MOS	Style similarity or speaker identity	Absolute

DMOS (Degradation MOS)

- DMOS evaluates how much worse a **degraded signal** is compared to a reference (clean) version.
- Listeners compare original vs. distorted speech, and rate the perceived degradation.
- **Scale:**
 - Usually also from 1 (highly degraded) to 5 (no degradation).
- **Use cases:**
 - Used in speech enhancement, denoising, codec evaluation, or low-bitrate synthesis.
 - Common in telecom, VoIP, or **noisy TTS** outputs.

CMOS (Comparative MOS)

- Listeners **compare** two TTS samples (A vs B) and judge which one sounds better.
- Instead of absolute scoring, CMOS captures **relative preference**.
- **Scale:** From -3 to +3:
 - +3: A is much better than B
 - 0: No preference
 - -3: B is much better than A
- **Use cases:**
 - Directly compare two TTS systems (e.g., your model vs baseline)
 - Useful in A/B testing for model iterations

SMOS – Style MOS (or Speaker MOS)

- **Style MOS**

- Evaluates how well the TTS conveys speaking style, such as emotion, expressiveness, or intonation.
- Listeners rate: “Does this voice sound expressive, angry, happy, neutral...?”

- **Speaker MOS**

- Evaluates speaker similarity in voice cloning.
- Question to listener: “Does this voice sound like the target speaker?”

Intelligibility: Word Error Rate (WER)

Definition and Relevance

- **Intelligibility** measures how easily the synthesized speech can be understood.
- **Word Error Rate (WER)** is commonly used to evaluate speech recognition accuracy. For TTS, you can measure WER by having listeners transcribe the synthesized speech and comparing their transcriptions to the original text.

Intelligibility: Word Error Rate (WER)

Full-Transcript Approach:

- Provide participants with synthesized speech samples.
- Ask them to write down what they heard.
- Compare their transcription to the ground truth.
- Calculate $WER = (\text{Substitutions} + \text{Deletions} + \text{Insertions}) / (\text{Total Words}) \times 100\%$.

Semantically Unpredictable Sentences (SUS):

- SUS are sentences constructed to minimize semantic context, forcing the listener to rely purely on acoustic-phonetic cues (e.g., "The green snake liked a singing pencil.").
- Useful for isolating articulation clarity from prior language knowledge.

Intelligibility: Word Error Rate (WER)

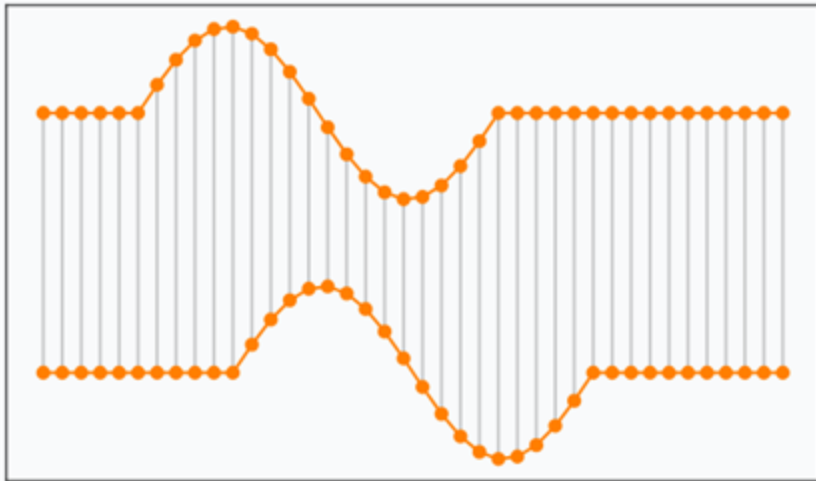
Considerations

- Listener Fatigue: Keep tests manageable in length.
- Vocabulary Familiarity: Use a representative vocabulary set relevant to the application domain if not using SUS.
- Reliability: Larger test sets and more raters improve reliability.

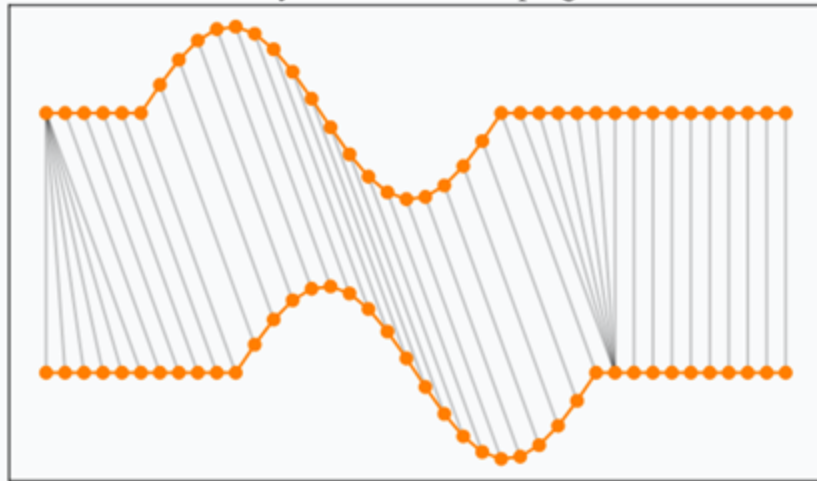
Mel Cepstral Distortion (MCD)

- Measures the distance between the mel-cepstral coefficients of real vs. synthetic audio.
 - Lower MCD = better quality
- Common in parallel TTS setups with ground truth (Euclidean distance or DTW)

Euclidean distance



Dynamic Time Warping



Latency and Real-Time Factor (RTF)

Importance in TTS Systems

- Beyond quality metrics, practical deployment requires evaluating speed and latency.
- Latency refers to the time it takes from receiving input text to generating the corresponding audio output.
- Real-Time Factor (RTF) is a common objective metric defined as:
 - **$\text{RTF} = (\text{time taken to synthesize the speech}) \div (\text{duration of the generated audio})$**
 - **If $\text{RTF} < 1$, the system is faster than real time (e.g., 0.5 RTF means 2x real-time speed).**

Latency and Real-Time Factor (RTF)

Benchmarks:

- Synthesize a representative set of text inputs of varying lengths (short sentences, paragraphs, etc.).
- Record total computation time and output speech duration.

Hardware Considerations:

- Report the hardware used (CPU, GPU, RAM) since performance can differ significantly.
- Compare different system configurations to see how they affect RTF.

Trade-offs

- Faster synthesis might reduce quality if model size or complexity is limited.
- Some methods (e.g., streaming or chunk-based synthesis) can reduce perceived latency but may affect audio continuity.

TTS vs. Voice Cloning

Aspect	TTS (Text-to-Speech) 🗣️	Voice Cloning 🗣️👤
Definition	Converts text into synthetic speech.	Reproduces a specific person's voice.
Purpose	Generate natural, intelligible speech for general use.	Generate speech that sounds like a specific speaker .
Training Data	Large datasets from multiple speakers .	Requires data from one or a few target speakers .
Customization	Uses pre-trained voices , limited speaker selection.	Can replicate any speaker's voice.
Data Required	100+ hours (traditional), 1–10 hours (modern).	Few-shot learning (minutes to hours) for cloning.
Output Quality	Highly natural , but generic voice .	Highly realistic , captures unique speaker traits .
Speaker Consistency	Voice remains generic across all outputs.	Mimics speaker's tone, style, and pitch .

Evaluation for Voice Cloning

- Ensuring high-quality, natural-sounding speech.
- Maintaining consistency and intelligibility.
- Avoiding ethical and security concerns related to synthetic voices.

Key Challenges in Voice Cloning Evaluation

- Balancing speaker similarity and intelligibility.
- Handling variability in emotions, speaking styles, and recording conditions.
- Measuring how well a cloned voice generalizes across different linguistic content.

Speaker Similarity Metrics

Subjective Evaluation: Mean Opinion Score (MOS) for Similarity

- Listeners rate how similar the cloned voice is to the target speaker on a 5-point or 7-point MOS scale.
- Example Scale:
 - 1 = Not Similar
 - 3 = Somewhat Similar
 - 5 = Very Similar
- Test setup: Randomized listening sessions where raters compare original vs. cloned speech.

Speaker Similarity Metrics

Objective Metrics: Speaker Embedding Distance

- Cosine Similarity: Measures how close the speaker embeddings of the cloned and original voices are.

$$\textit{Similarity Score} = \frac{E_{\textit{cloned}} \cdot E_{\textit{original}}}{||E_{\textit{cloned}}|| ||E_{\textit{original}}||}$$

- Higher values (closer to 1) indicate higher similarity.
- Euclidean Distance: Measures the absolute difference in feature vectors between cloned and original voices.
 - Lower values indicate greater similarity.

Speaker Similarity Test

VLSP Vietnamese Text To Speech

Xin chào, **Nguyễn Hoàng Kỳ** [Đổi mật khẩu](#) [Đăng xuất](#)

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☐ Hoàn toàn không giống (1 điểm)

☐ Hơi giống nhau (2 điểm)

☐ Khả giống nhau (3 điểm)

☐ Giống hoàn toàn (4 điểm)

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Speaker Verification

- False Acceptance Rate (FAR): Rate of accepting an impostor (false positive)
- False Rejection Rate (FRR): Rate of rejecting a genuine speaker (false negative)
- **Equal Error Rate (EER)**: The point where $FAR = FRR$
 - Lower EER = Better system
- **Example**
 - Suppose you have:
 - 1,000 test trials (speech, claimed ID pairs)
 - 500 genuine (true speaker)
 - 500 impostor (wrong speaker)
 - SV System returns similarity scores from 0 to 1

Speaker Verification

Threshold	FAR (Impostor accepted)	FRR (Genuine rejected)
0.2	35%	5%
0.4	20%	10%
0.5	10%	10%
0.6	6%	15%
0.8	3%	28%

Speaker Identification

- Accuracy = (Number of correct predictions / Total samples) × 100%
- Example
 - You have 100 test utterances from 10 known speakers
 - The system correctly identifies 85 of them → Accuracy = 85%
- Optionally: Top-k Accuracy
 - Measures if the correct speaker is in the top k predicted candidates

Naturalness and intelligibility metrics

- **MOS for Naturalness**

- Similar to TTS evaluation, naturalness is rated on a 1-5 or 1-7 scale.
- Factors influencing naturalness:
 - Smoothness of prosody
 - Presence of robotic artifacts
 - Speech discontinuities

- **Relationship Between Intelligibility and Cloning Quality**

- A cloned voice that is highly natural but low in intelligibility is problematic.
- Ideally, WER should be close to that of the original speaker.

Ethical and Security Considerations

Detectability of Cloned Voices

- Can human listeners or AI models detect synthetic speech?
- Develop metrics to measure the likelihood of cloned speech being misidentified as human.

Misuse Risks and Watermarking Strategies

- Risks:
 - Fraud, misinformation, impersonation attacks.
- Watermarking and Traceability:
 - Embedding inaudible signals to indicate synthesized content.
 - AI-based forensic models to detect cloned voices.

Ethical Guidelines in Voice Cloning

- Ensuring consent before cloning a person's voice.
- Clearly marking AI-generated speech to avoid deception.

Challenges and Future Directions

- Cross-metric trade-offs and balancing priorities
- Subjectivity in Human Judgments
- Ethics in evaluation: Bias and fairness in speech models.
- Emerging trends in speech model evaluation (e.g., explainable AI, multimodal evaluation).

06. Speech Synthesis



Nguyen Thi Thu Trang

Outline

6.1. Overview

6.2. Speech Synthesis Techniques

6.3. System Architecture

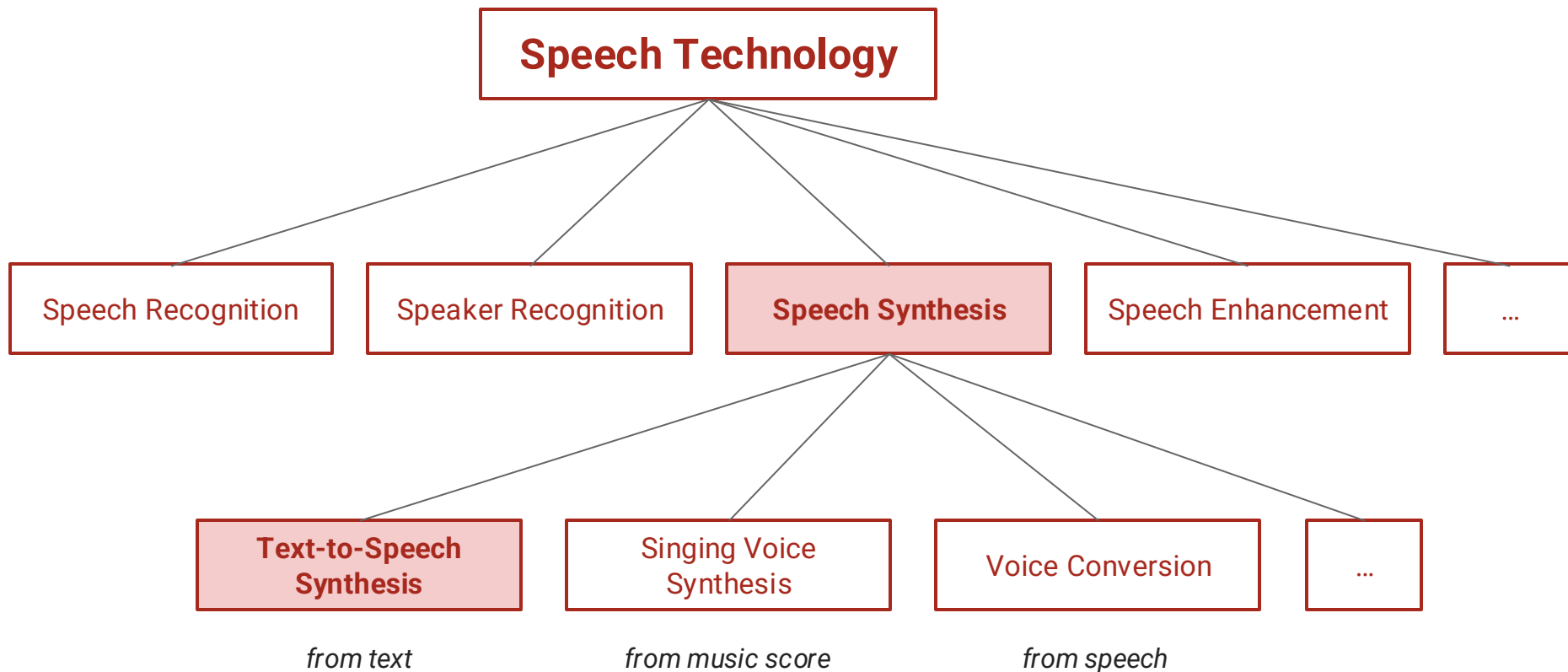
6.4. Challenges and Solutions

6.1 Overview

- ❖ Speech Synthesis - Tổng hợp tiếng nói
 - Quá trình nhân tạo sinh ra tiếng nói của con người
- ❖ Text-To-Speech (TTS) - Chuyển từ văn bản thành tiếng nói
 - Một loại tổng hợp tiếng nói
 - Quá trình chuyển đổi từ văn bản ngôn ngữ tự nhiên sang tiếng nói đầu ra

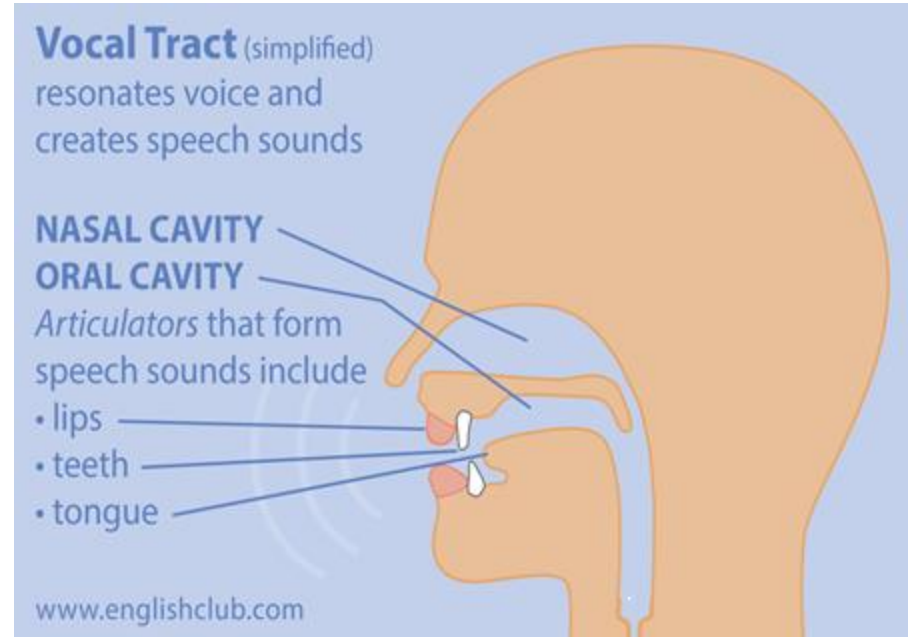
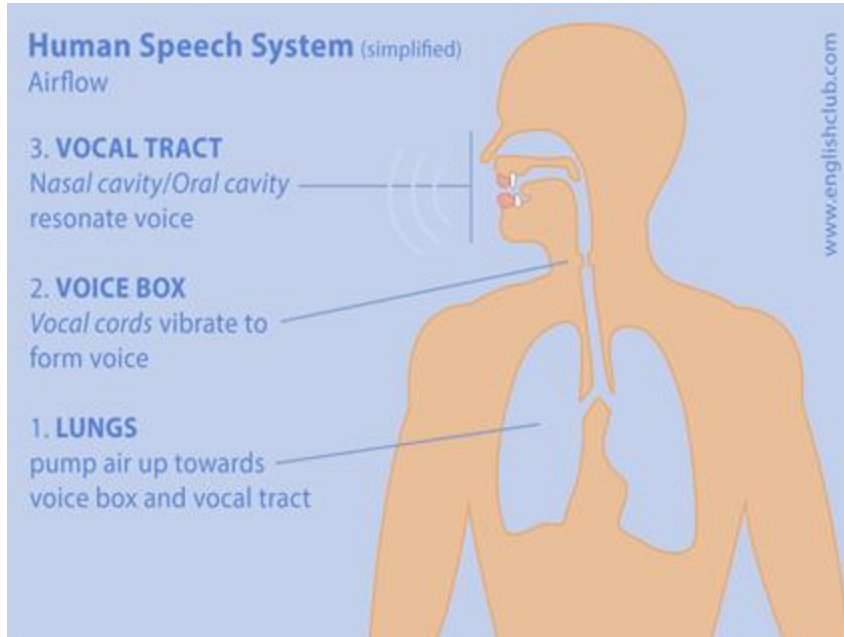


6.1. Overview



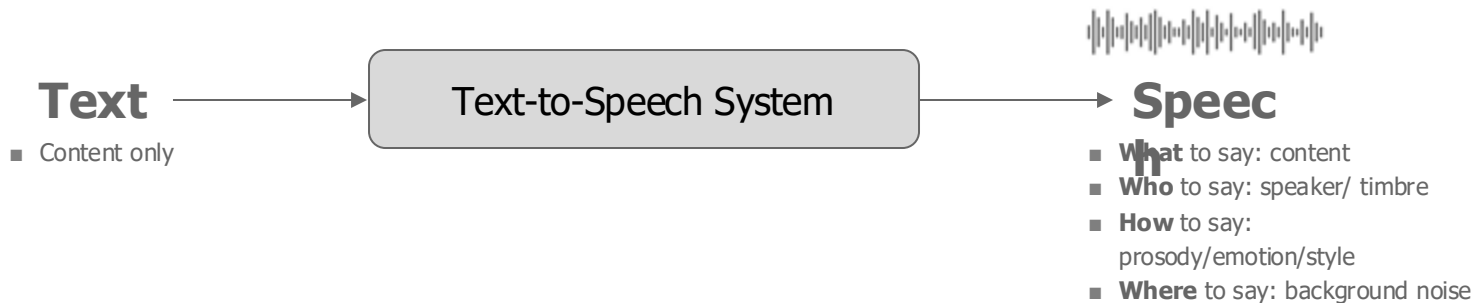
Text-to-Speech Synthesis

- The generation of human-like speech from text.
 - Human speech system

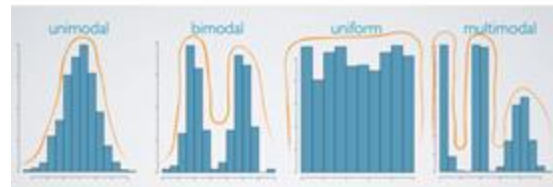


Text-to-Speech Synthesis

- The generation of **human-like speech** from text
 - Text-to-Speech (TTS) System



- Text-to-Speech Mapping is **One-to-Many**
 - Usually not **single-modal**, but **multi-modal**



History of Speech Synthesis

- 1003 - Máy móc bắt chước giọng nói đầu tiên ra đời (chế tạo bởi Gerbert ở Aurillac)
- 1779 - Mô hình bắt chước giọng nói người với năm nguyên âm ([a], [e], [I], [o] và [u])
- 1791 - Cải tiến tạo ra mô hình của lưỡi và môi, tạo ra phụ âm thêm vào nguyên âm
- 1857 - M.Faber chế tạo máy 'Euphonia'
- 1930 - Máy VOCODER, phân tích và tổng hợp giọng nói điều khiển bằng bàn phím, phát âm rõ ràng
- 1950 - Các hệ thống tổng hợp giọng nói bằng kỹ thuật điện tử ra đời
- 1961 - John Larry Kelly, Jr dùng máy tính IBM 704 để tổng hợp giọng nói
- Hiện nay - Hàng trăm hệ thống tổng hợp giọng nói, tiến hóa nhanh với chất lượng giọng nói đạt tới ngang người thật

Applications



1. Voice assistant



2. Content creation



3. Movie Dubbing



4. Text narration

Ứng dụng thực tế

- ❖ Hệ thống thông báo tự động
- ❖ Số hoá dữ liệu
 - Báo nói, sách nói
 - Số hoá bài giảng
 - Thuyết minh phim
 - Số hoá nội dung
- ❖ Giao tiếp bằng tiếng nói
 - Tổng đài tự động
 - Trợ lý ảo
 - Voicebot
 - Vạn vật kết nối
- ❖ Hỗ trợ người khuyết tật
 - Người khiếm thị
 - Người mất khả năng nói, diễn đạt...



6.2. Speech Synthesis Techniques

- **Articulatory Synthesis**

- Complicated, very low quality

- **Concatenative Synthesis**

- High intelligibility, but requires huge database, less natural and emotionless

- **Statistical Parametric Synthesis**

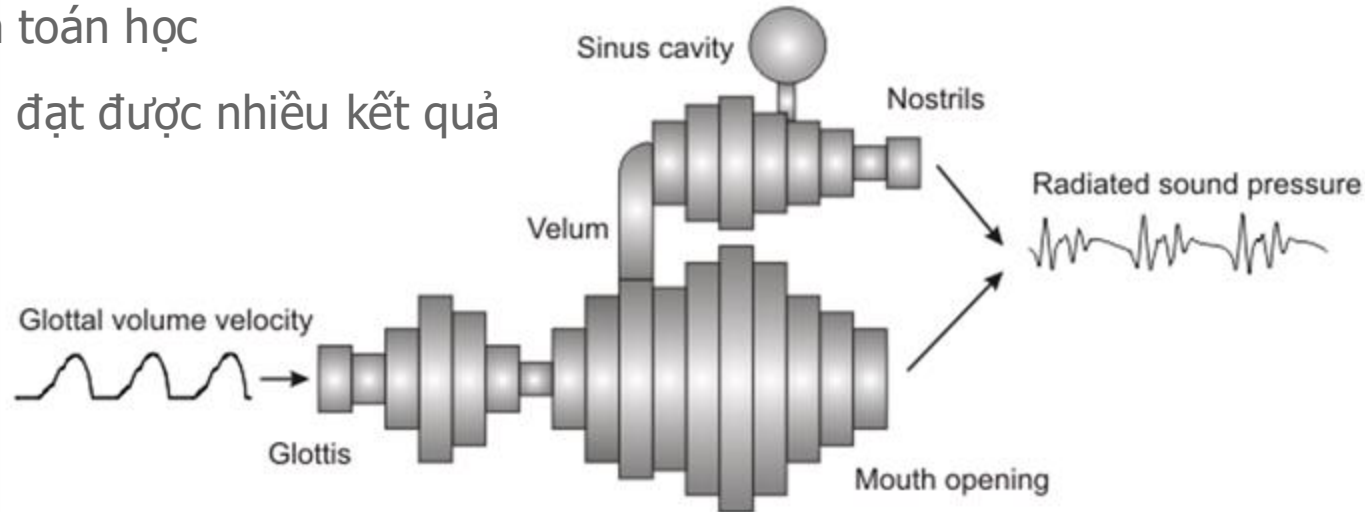
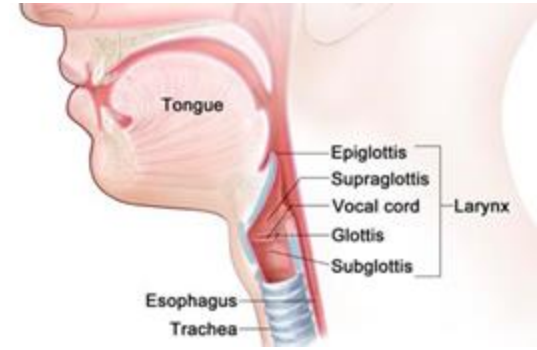
- Lower data cost and more flexible, but lower quality and robotic

- **Neural-based End-to-end Synthesis**

- Huge quality improvement, less human preprocessing and feature development.

Mô phỏng bộ phận cấu âm (1958)

- Replicate the movement of the human vocal tract to generate speech
- Mô phỏng các thành phần của bộ máy phát âm con người bằng các mô hình toán học
- Phức tạp → Chưa đạt được nhiều kết quả



Thảo luận

❖ Anh chị biết đơn vị âm (speech unit) nào?

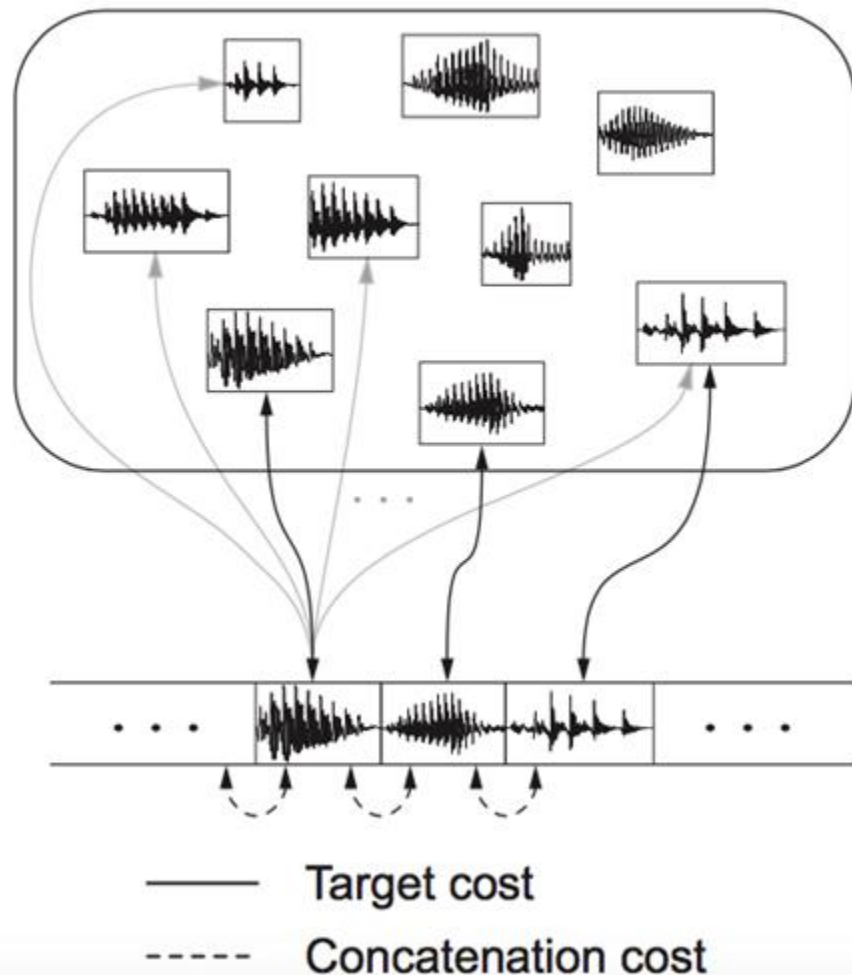
Thảo luận – Các đơn vị âm (speech unit)

- ❖ Câu nói – Utterance
- ❖ Đoạn tiếng nói – Prosodic Phrase
- ❖ Từ – Word
- ❖ Tiếng – Syllable
- ❖ Âm vị – Phoneme: Smallest perceptible segment

Concatenative Synthesis

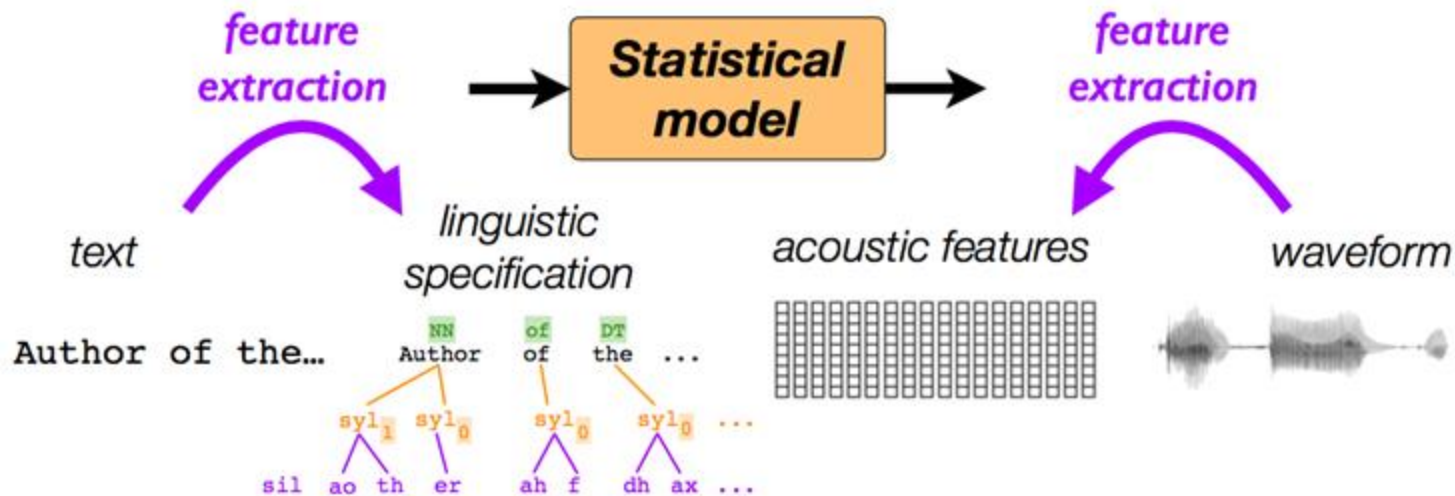
- Builds speech by concatenating segments of recorded speech (phones, syllables, or words).
- High naturalness, but limited flexibility
- Types:
 - Unit selection synthesis
 - Diphone synthesis

All segments

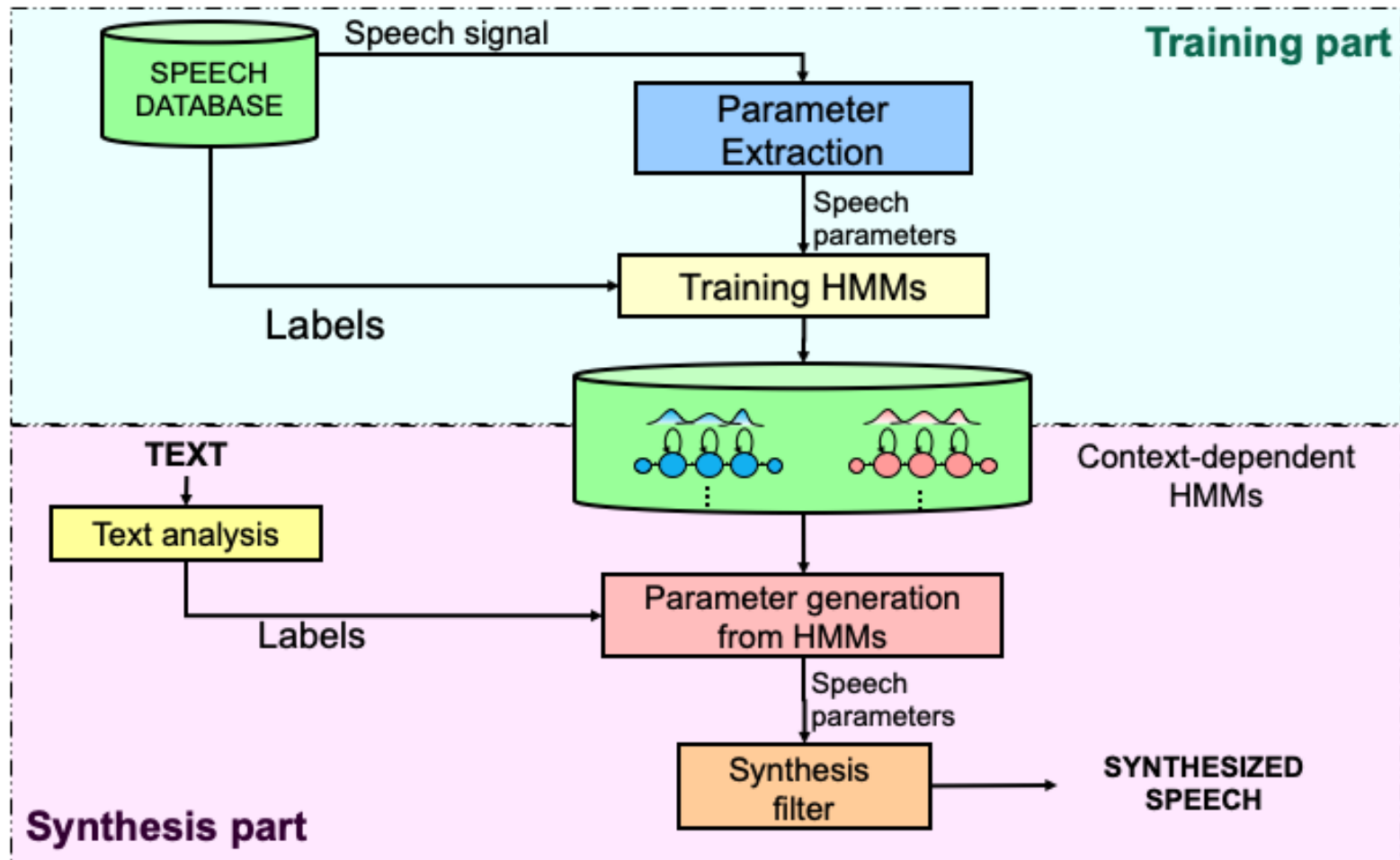


Statistical Parametric Synthesis

- Models speech acoustics and prosody statistically (e.g. HMMs)
 - Generating the average of some sets of similarly sounding speech segments (Zen 2009)
- Smaller footprint, more flexible, but less natural

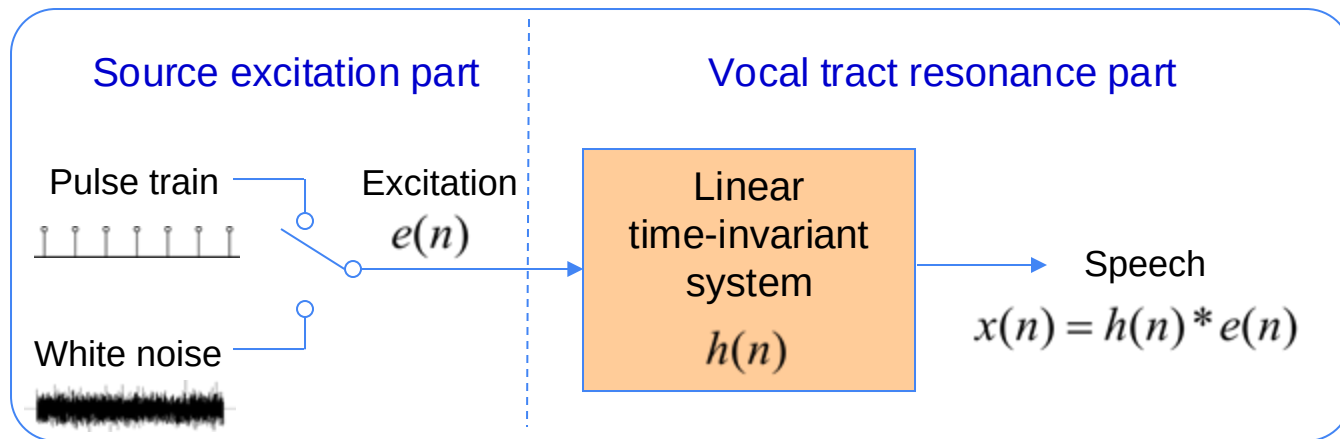


HMM
based
Speech
Syn-
thesis



Yoshimura (2002). Yoshimura Takayoshi. *Simultaneous Modeling of Phonetic and Prosodic Parameters, and Characteristic Conversion for HMM-Based Text-To-Speech Systems*. PhD thesis, Nagoya Institute of Technology, Japan, 2002.

Source-filter model (Vocoder)



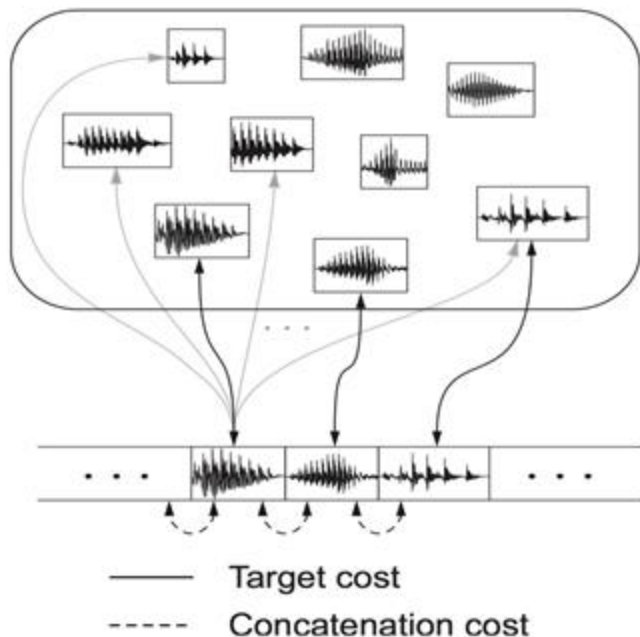
$$x(n) = h(n) * e(n)$$

↓ Fourier transform

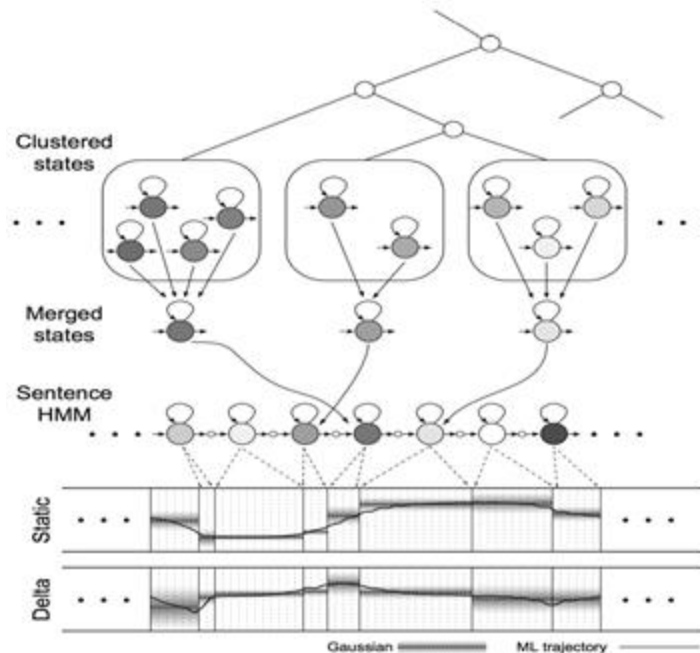
$$X(e^{j\omega}) = H(e^{j\omega})E(e^{j\omega})$$

$H(e^{j\omega})$ should be defined by the state-output vector of HMMs
(e.g., mel-cepstral coefficients, LSP coefficients)

Concatenation and Statistical Parametric Synthesis



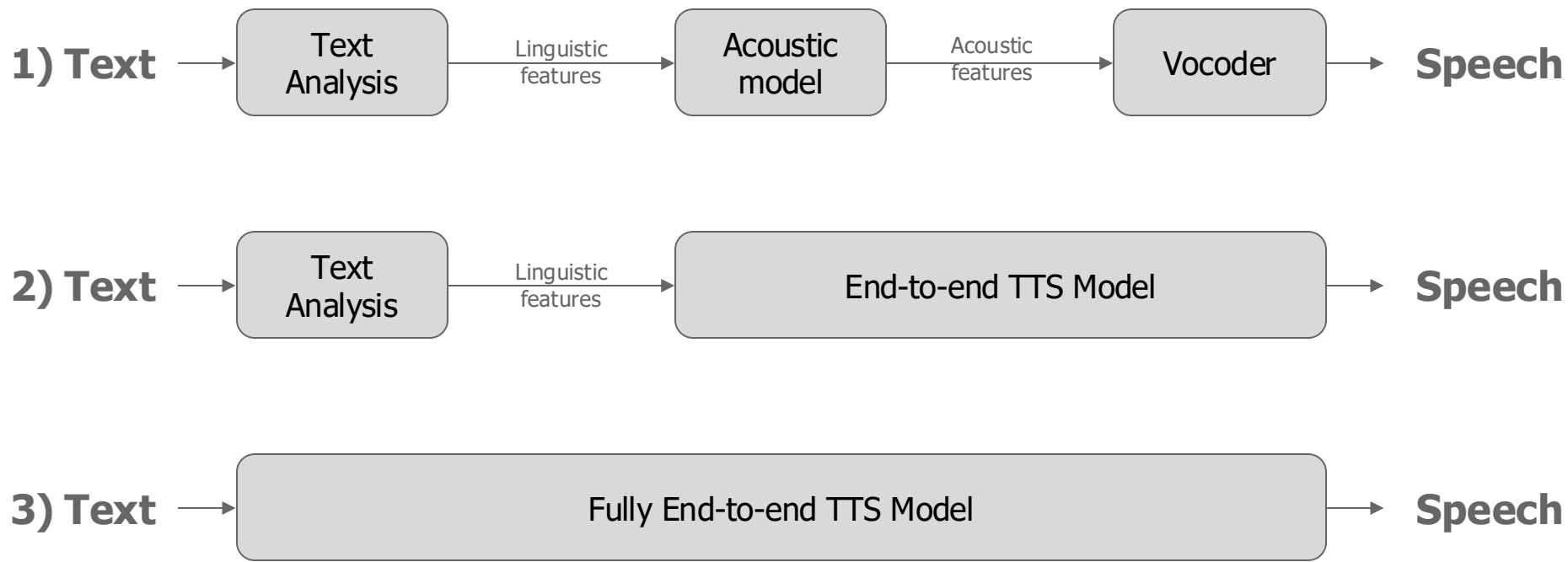
From **concatenation** to
unit selection synthesis



From **vocoding** to
statistical parametric synthesis

Zen (2009). Zen Heiga et al. Statistical parametric speech synthesis. Speech Communication.

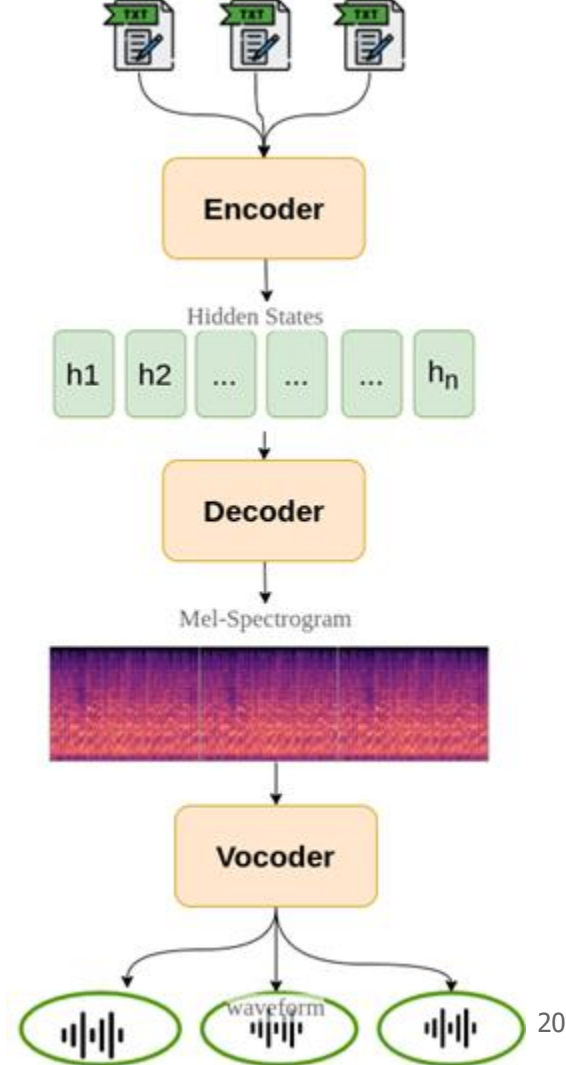
Neural-based Text-to-Speech



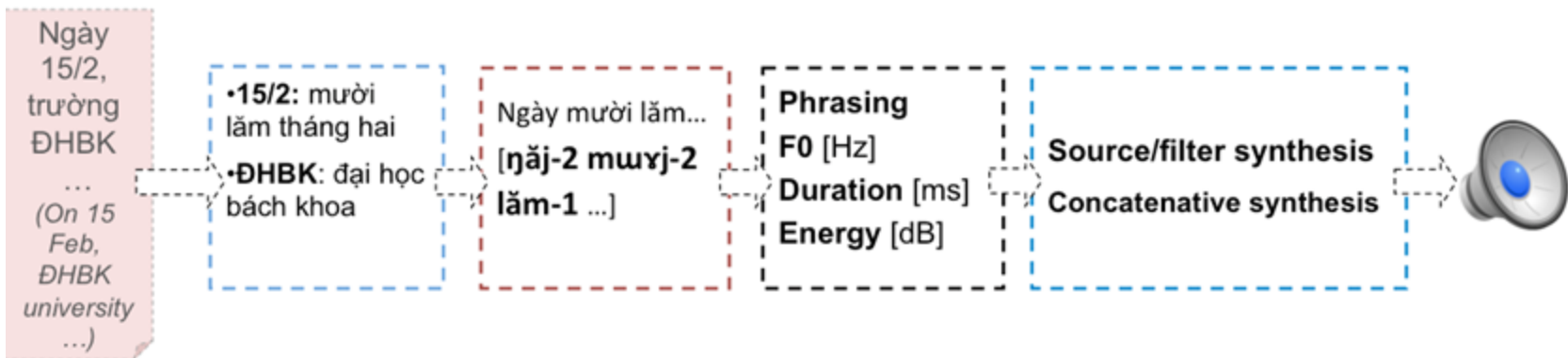
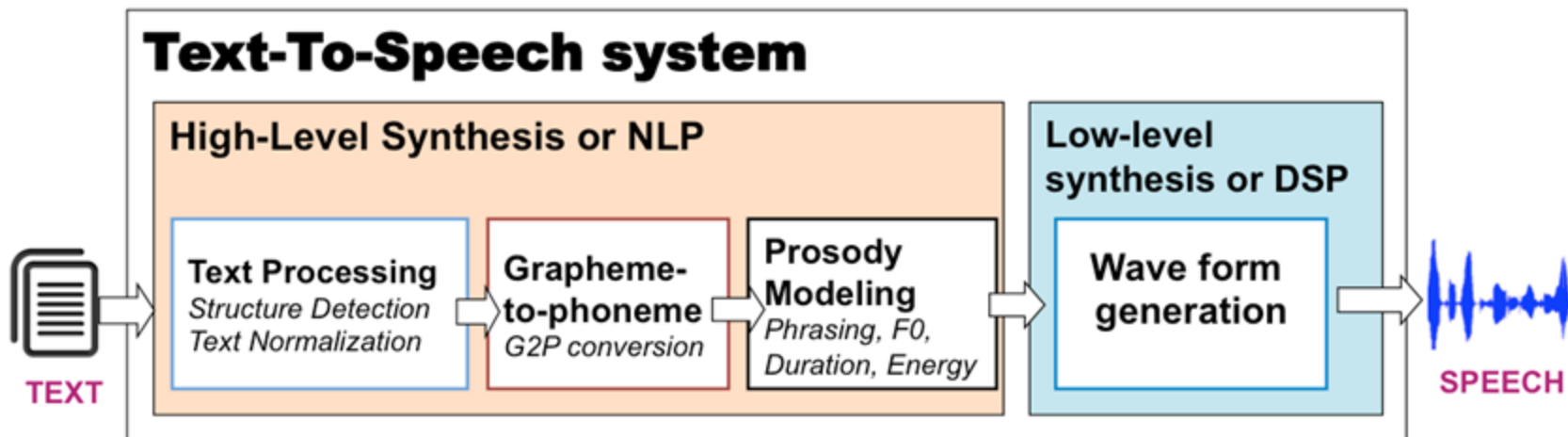
Towards a Fully end-to-end TTS model

Neural-based Speech Synthesis

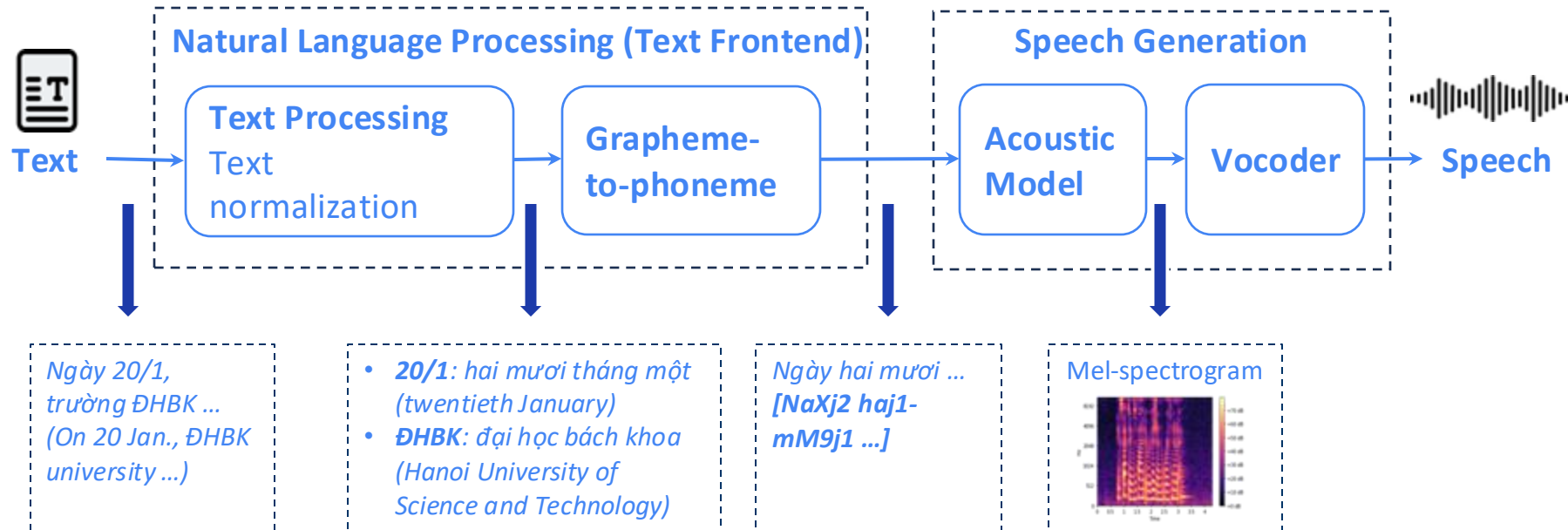
- **Encoder:**
 - Transforms the input text into a sequence of real-valued vectors and encodes them into hidden representations suitable for training deep neural networks.
- **Decoder:**
 - Decodes the hidden representations predicted by the neural network into acoustic features, typically in the form of mel-spectrograms.
- **Vocoder:**
 - Converts the mel-spectrogram into a time-domain audio waveform (speech signal).



6.3. TTS System Architecture - Traditional



6.3. TTS System Architecture - Modern

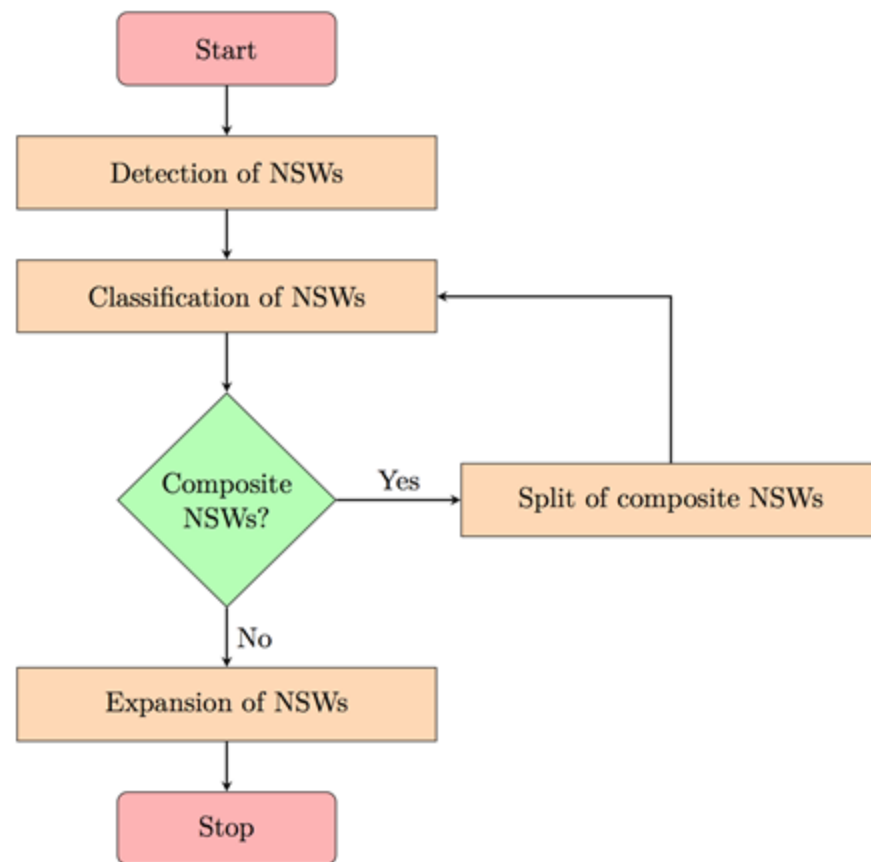


Text Analysis

- Transforms input text into linguistic features, including
 - **Text normalization:** *eg. Feb. 25th → "February twenty fifth"*
 - **Phrase/word/syllable segmentation:** *eg. "synthesis" → syn-the-sis*
 - **Part of speech (POS) tagging:** *eg. "I study at HUST" → [PRON, VERB, PREP, NOUN]*
 - **Grapheme-to-Phoneme conversion:** *eg. "speech" → s p i y ch*
 - ...
- Linguistic features can be:
 - Phoneme
 - Syllable, Word
 - Phrase and sentence-level features, ...

Text Normalization model

Group	Category	Description	Example
NUMBERS	NTIM	time	1:30
	NDAT	date	17/3/87, 1/3/2010
	NDAY	day and month	17/3, 03-05/3
	NMON	month and year	3/87, 3/2010, 3-5/87
	NNUM	number	2009, 70.000
	NTEL	telephone number	0915.334.577
	NDIG	number as digits	Mã số 999
	NSCR	score	Tỉ số là 3-5
	NRNG	range	Từ 3-5 ngày
	NPER	percentage	93%, 30-40%,
	NFRC	fraction	34/6, 45,6/145
	NADD	address	số 14/3/2 phố Huế
LETTERS	LWRD	read as a word	London, NATO
	LSEQ	letter sequence	ODA, GDP
	LABB	abbreviation	TS (tiền sĩ)
OTHERS	PUNC	speakable punctuation	... () [] ‘ ’ “ ” - /
	URLE	url, path name or email	http://soict.hut.vn
	MONY	Money	2\$, \$2, VNĐ 9.000
	CSEQ	read all characters	:), XXX
	DURA	duration (nghi)	“-” in scores (2-3)
	NONE	ignored	ascii art...



Vietnamese word segmentation

- Same delimiter (spaces) for Vietnamese syllables and words
 - E.g. Tôi | đang | nghiên cứu | về | tổng hợp | tiếng nói | Tiếng Việt | tại | phòng thí nghiệm | LIMSI (I am doing research on Vietnamese speech synthesis in the LIMSI laboratory).
- Ambiguity in Vietnamese word segmentation
 - E.g. Học sinh | học sinh | học OR Học sinh | học | sinh học
(Pupils, pupils study OR Pupils study biology)

Vietnamese Word Segmentation Tools

Tool	Provider	Programming Language	Technique	Strengths	Notes
VnCoreNLP (RDRSegmenter)	VNU-HCM (Vietnam National University)	Java	Rule-based (Ripple Down Rules)	High accuracy, robust, widely used	Core of most Vietnamese NLP pipelines
pyvi	Open-source community	Python	CRF-based	Lightweight, Pythonic	Good for quick prototyping
underthesea	UnderTheSea team (Vietnam)	Python	Hybrid (rules + statistical)	Easy to use, pip-installable	Lower accuracy than RDRSegmenter
VnTokenizer	Le Hong Phuong (HCMUS)	Java	Rule-based	Historical significance	Less used in modern applications

Building Vietnamese tonophone set

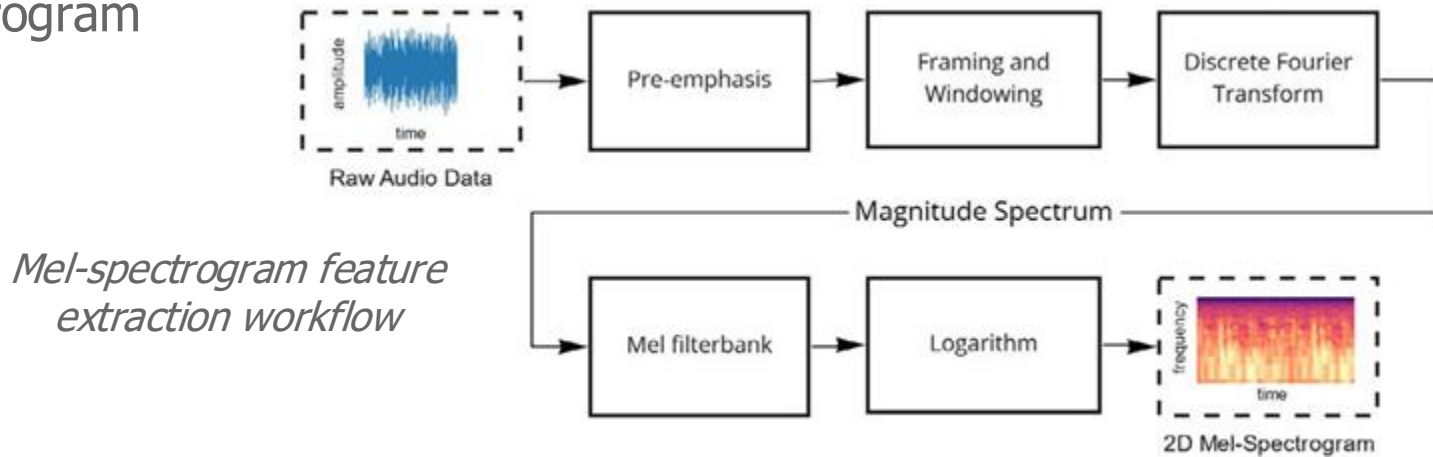
- **Tonophone**: An allophone with tone information
 - Adding tones to all elements of rhyme
 - E.g. chuyển [tɕwiən-3] (to move) => [tɕ], [w3], [iə3], [n3]

Element	Allophones	Lexical tones	TN #
Initial consonant	ɲ t ^h tɕ ɲ f X z ɣ d t k ɓ m n l s h v p	(NOT adding)	19x1
Medial	w	1-4, 5a, 5b, 6a, 6b	1x8
Nucleus	i e ɛ a ɤ ɯ u o ɔ ɛ̃ ǎ ǝ̃ ǝ̃ iə ɯə uə	1-4, 5a, 5b, 6a, 6b	16x8
Final consonant	m n ɲ ɲm ɲ. w j	1-4, 5a, 6a	7x6
	p t k kp kɿ	5b, 6b	5x2
TOTAL	48		207

- **Inventory** of all vietnamese pronounceable syllables
 - PRO-SYLDIC (PROnounceable SYLLable DICTIONary)
 - Maintaining non-existent but pronounceable syllables for loan words
 - For automatic transcription and text normalization

Acoustic model

- **Predicts** acoustics features from linguistic features.
- **Aligns** linguistic and acoustic features
 - Maps text-based sequences (eg. phonemes) to continuous speech representations
- **Acoustic features** can be:
 - Mel-scale Frequency Cepstral Coefficients (MFCC)
 - Mel-spectrogram



Acoustic Models (Text → Spectrogram)

Type	Description	Representative Models	Pros	Cons
Autoregressive	Predict spectrogram frame-by-frame with attention	Tacotron, Tacotron 2	Natural prosody	Slow, unstable attention
Non-Autoregressive	Parallel prediction using duration predictor	FastSpeech, FastPitch, FastSpeech 2	Fast, stable	Requires alignment
Variational / Flow-based	Latent modeling with VAE / Normalizing Flow	Glow-TTS, Flow-TTS	Expressive, flexible	Complex training
Diffusion-based	Generate spectrogram via denoising process	Grad-TTS, FastDiff, DiffGAN-TTS	High-quality, controllable	Slow inference

Acoustic Models

Mô hình	Kiến trúc	Autoregressive	Alignment	Tốc độ sinh (Inference)	Đặc điểm nổi bật
Tacotron 2	Seq2Seq + Attention	✅ Có	Attention (soft)	❌ Chậm (tuần tự)	Dễ lỗi lặp/thiếu từ; phổ biến
FastSpeech	Transformer + Duration	❌ Không	Dựa trên teacher-forced duration từ Tacotron	✅ Nhanh (song song)	Ổn định, nhanh hơn Tacotron
FastSpeech 2	Transformer + Duration	❌ Không	Dựa trên duration (tự học hoặc teacher)	✅ Rất nhanh	Thêm pitch & energy modeling
Glow-TTS	Flow-based (Glow)	❌ Không	Monotonic Alignment Search (MAS)	✅ Rất nhanh	Không cần attention; sinh mel theo flow
VITS	VAE + Flow + GAN	❌ Không	Variational + Duration latent align	✅ Nhanh + Real-time	End-to-end sinh waveform luôn

Acoustic model - Example

- **Glow-TTS**

- Log likelihood

$$\log P_X(x|c) = \log P_Z(z|c) + \log \left| \det \frac{\partial f_{dec}^{-1}(x)}{\partial x} \right|$$

- Prior distribution is learnt from phoneme text

$$\log P_Z(z|c; \theta, A) = \sum_{j=1}^{T_{mel}} \log \mathcal{N}(z_j; \mu_{A(j)}, \sigma_{A(j)}),$$

- Alignment matrix A is obtained by monotonic alignment search

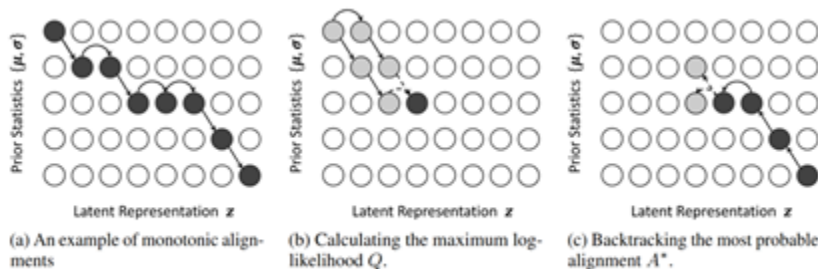
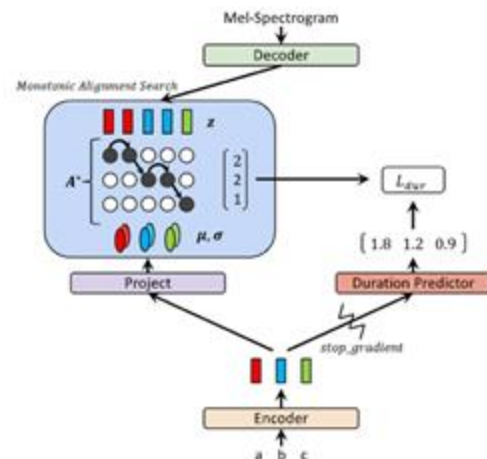
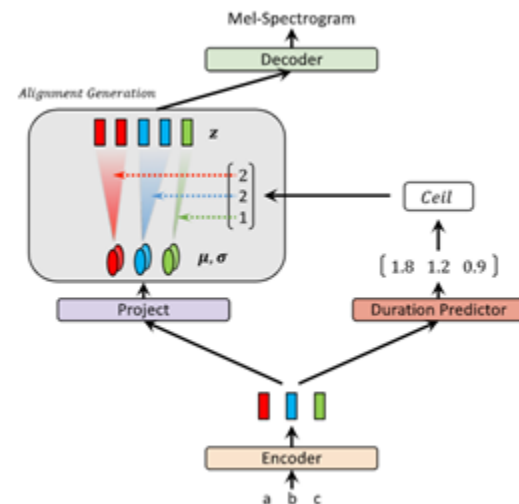


Figure 2: Illustrations of the monotonic alignment search.



(a) An abstract diagram of the training procedure.



(b) An abstract diagram of the inference procedure.

Vocoder

- **Voice encoder**
 - **Final stage in TTS pipelines, transforming** acoustics features **(typical spectrograms) into intelligible, natural-sounding speech.**
- Greatly impacts the **naturalness, quality, and intelligibility** of synthesized speech
 - Traditional vocoders like Griffin-Lim are simple but produce robotic speech, while neural vocoders such as HiFi-GAN offer high-quality, real-time synthesis.
 - Vocoder architectures have evolved from autoregressive models (WaveNet) to more efficient GAN- and diffusion-based approaches

Vocoder (Spectrogram → Waveform)

Type	Description	Representative Models	Pros	Cons
Autoregressive	Frame-by-frame waveform generation	WaveNet	Very high quality	Very slow
Flow-based	Invertible transformation from noise → waveform	WaveGlow, FloWaveNet	Faster than AR	Quality not as good as GANs
GAN-based	Adversarial learning to generate waveform	MelGAN, HiFi-GAN, UnivNet	Fast, high-quality	Needs careful training
Diffusion-based	Denoising process to generate waveform	DiffWave, WaveGrad	High fidelity	Slower inference

Vocoder - Key challenges

- **Trade-off** between quality and speed
 - Autoregressive models are slow but high-quality.
 - Non-autoregressive models are fast in inference but resource-intensive during training.
- **Generalization** Across Speakers & Datasets
 - Vocoder trained on one dataset may struggle with unseen voices

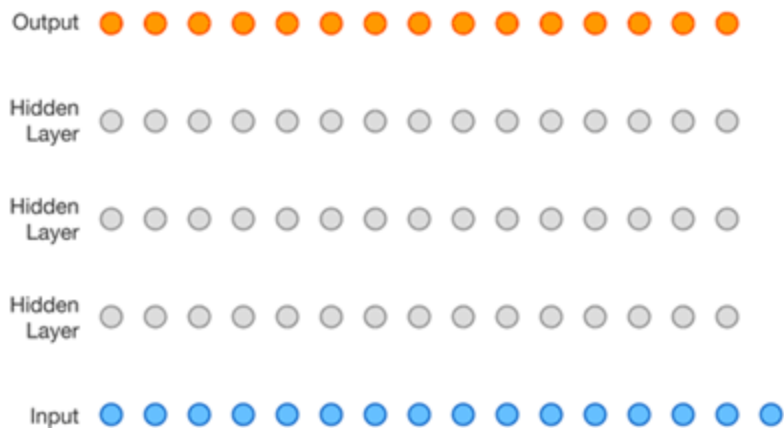
End-to-end TTS model

- **Advantages**

- Trained with text-speech pairs with minimum human annotation
- Do not require explicit alignment between text and speech
- Errors cannot accumulate and no error propagation since it is a single model

- **More end-to-end TTS models**

- Directly predict waveform instead of intermediate representation (eg. mel-spectrogram)



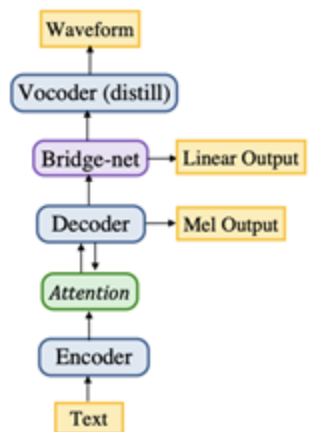
WaveNet: autoregressive model with dilated causal convolution

End-to-end TTS model

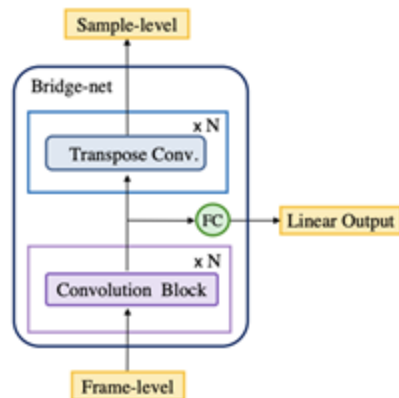
- **Advantages**

- Trained with text-speech pairs with minimum human annotation
- Do not require explicit alignment between text and speech
- Errors cannot accumulate and no error propagation since it is a single model

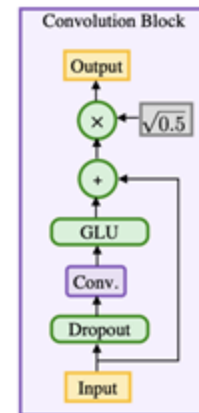
- **Fully en**



(a) Text-to-wave architecture



(b) Bridge-net



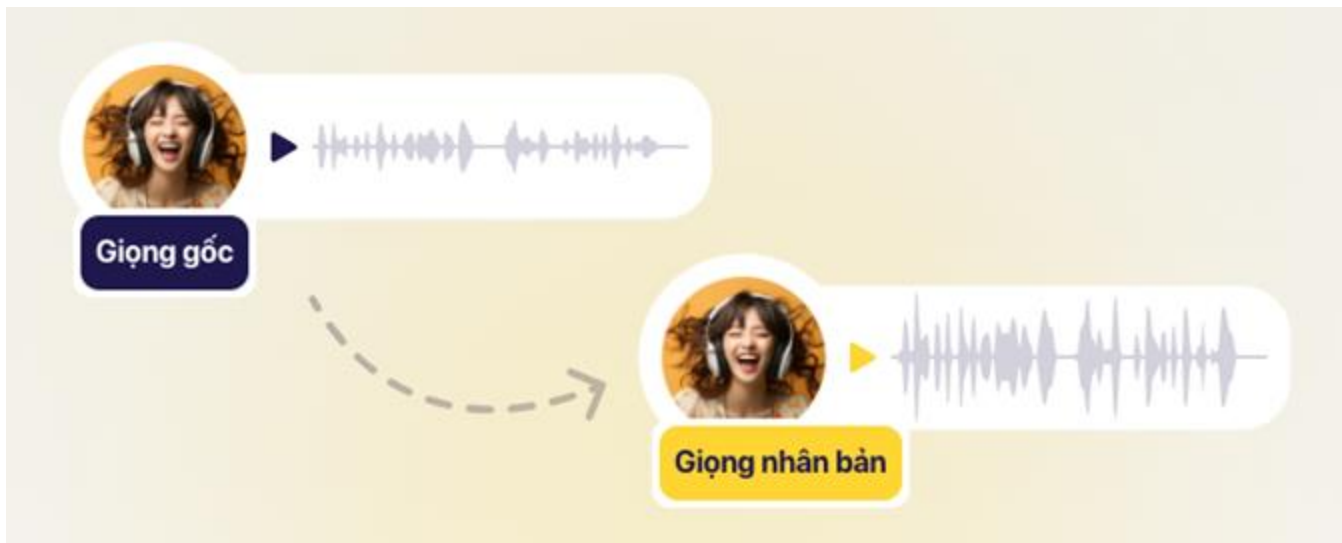
(c) Convolution block

End-to-End TTS Architecture (Text → Waveform)

Model Type	Description	Representative Models	Pros	Cons
VAE + Flow + GAN	Combines latent variables, flow-based decoder, and adversarial learning	VITS, VITS2	High-quality, compact, expressive	Complex training, harder to debug
Diffusion-based E2E	Text-to-waveform via diffusion	StyleTTS2, NaturalSpeech 2	Natural, controllable	Training/inference cost

Voice Cloning

Synthesizing speech that mimics the **voice identity** of a specific speaker — typically from limited data — while allowing arbitrary text as input.



TTS vs. Voice Cloning

Aspect	TTS (Text-to-Speech) 🗣️	Voice Cloning 🗣️👤
Definition	Converts text into synthetic speech.	Reproduces a specific person's voice.
Purpose	Generate natural, intelligible speech for general use.	Generate speech that sounds like a specific speaker .
Training Data	Large datasets from multiple speakers .	Requires data from one or a few target speakers .
Customization	Uses pre-trained voices , limited speaker selection.	Can replicate any speaker's voice.
Data Required	100+ hours (traditional), 1–10 hours (modern).	Few-shot learning (minutes to hours) for cloning.
Output Quality	Highly natural , but generic voice .	Highly realistic , captures unique speaker traits .
Speaker Consistency	Voice remains generic across all outputs.	Mimics speaker's tone, style, and pitch .

Voice Cloning Challenges

- Low-resource data:
 - Only a few minutes or samples of the target speaker's voice available
- Speaker identity loss:
 - Generated voice lacks vocal uniqueness.
- Flat prosody:
 - Robotic or emotionless delivery.
- Multilingual/tonal issues:
 - Difficulties with tone, pitch, or phoneme disambiguation (e.g., in Vietnamese).
- Evaluation:
 - Subjective and hard to automate.

Voice Cloning Models

- **Multi-speaker TTS + Embedding**

- Base: Tacotron 2, FastSpeech 2, VITS
- Speaker embedding is concatenated at encoder or decoder level.

- **End-to-End Voice Cloning**

- Text + reference audio → waveform
- No explicit mel-spectrogram stage; trained jointly (e.g. YourTTS, VITS2, StyleTTS2)

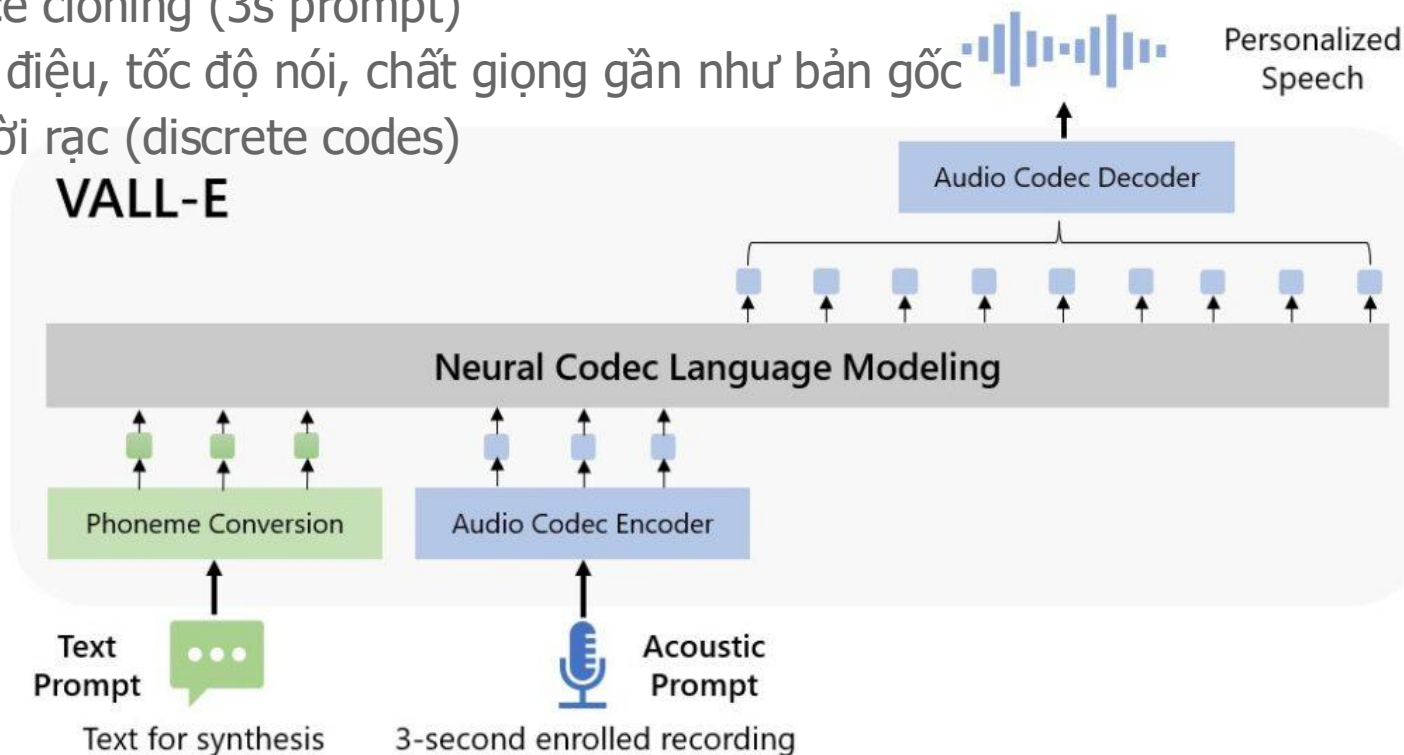
- **Zero-shot Cloning**

- Pretrained speaker encoder (e.g., GE2E) + shared TTS backbone
- Works with unseen speakers without fine-tuning (e.g. YourTTS, SV2TTS, hoặc iSTFT-VITS, VALL-E, StyleTTS2)

VALL-E: Neural Codec Language Model

- Microsoft Research (2023)
- Zero-shot voice cloning (3s prompt)
- Giữ được ngữ điệu, tốc độ nói, chất giọng gần như bản gốc
- Sử dụng mã rời rạc (discrete codes)

Không sinh spectrogram mà dự đoán mã RVQ rồi tái tạo



Các hướng nghiên cứu cải tiến VALL-E

- Prosody Modeling, Style Token, Emotion Modeling... để kiểm soát ngữ điệu, cảm xúc, tốc độ
- Giảm sự phụ thuộc vào prompt: Robust Speaker embedding, Speaker prompt từ text (zero-prompt), Speaker encoder
- Multilingual và cross-lingual voice cloning
- Mô hình hóa tốt hơn giữa text → code:
 - Semi-autoregressive/Diffusion model thay vì AR transformer → giảm lỗi lặp, mất đoạn
 - Tăng độ chính xác trong mapping giữa ngữ nghĩa câu và chuỗi code RVQ
- Tách biệt content / speaker / prosody: Speaker embedding (giọng), Content token (ngữ nghĩa), Style/prosody token
- Kết hợp với Large Language Models (LLMs):
 - LLM sinh ra prosody tokens, emotion hint, hoặc text chuẩn hóa trước
 - Dùng prompt text dạng ngữ cảnh: "nói với giọng nghiêm túc, trầm ấm..." → feed vào decoder

RVQ-VAE

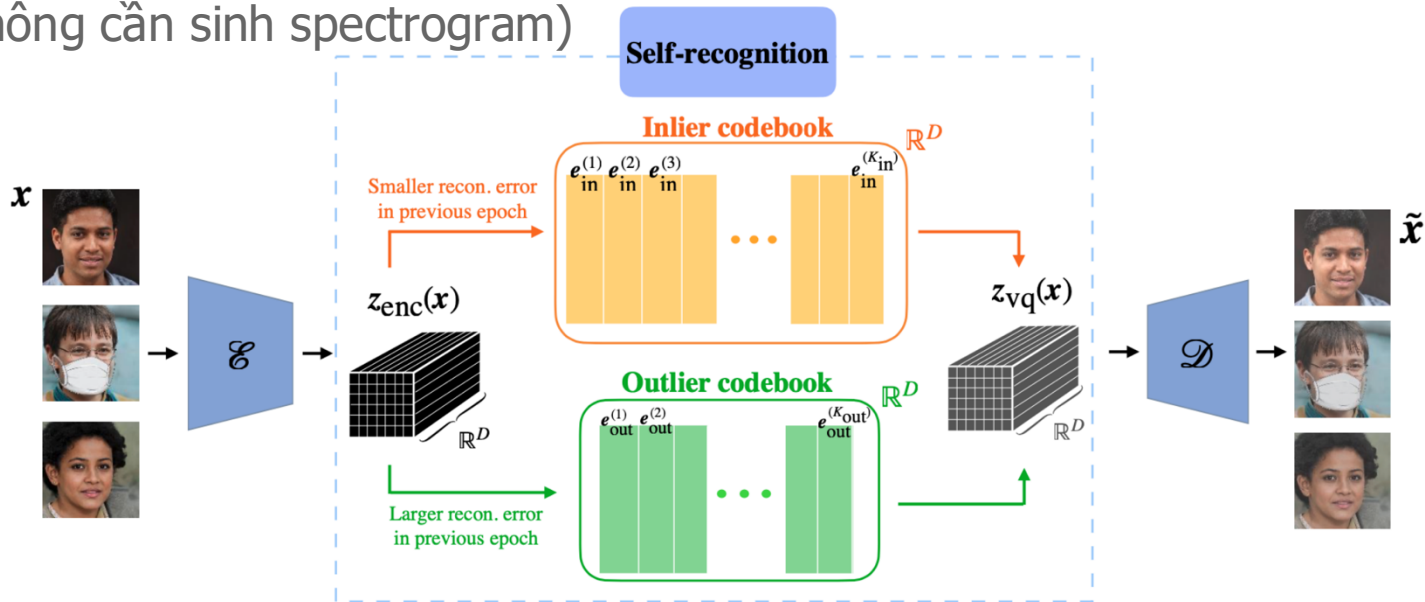
- **VAE** (Variational Autoencoder): học phân phối ẩn (latent distribution) để **nén và tái tạo dữ liệu**.
- **VQ-VAE** (Vector Quantized VAE): dùng **vector lượng tử hóa rời rạc (codebook)** để đại diện cho không gian ẩn.
- **RVQ (Residual Vector Quantization)**: cải tiến VQ bằng cách **sử dụng nhiều tầng lượng tử hóa kế tiếp**, mỗi tầng học phần dư (residual) còn lại từ tầng trước → Tăng khả năng biểu diễn mà không cần codebook quá lớn.

👉 **RVQ-VAE** = Autoencoder + multi-stage vector quantization

→ Học cách **mã hóa giọng nói thành chuỗi mã rời rạc giàu thông tin**, giúp mô hình dự đoán và tái tạo giọng nói linh hoạt và tự nhiên hơn, đồng thời hỗ trợ tốt cho **voice cloning, zero-shot TTS, và multilingual synthesis**.

Vai trò của RVQ-VAE với TTS

- Học latent speech representation: dùng làm mục tiêu huấn luyện
- Zero-shot speaker modeling: clone giọng từ mã speech ngắn
- Tách biệt giữa nội dung (content) và phong cách giọng nói (style)
- Giảm gánh nặng huấn luyện vocoder: chỉ cần học mapping từ text $\rightarrow$ mã RVQ (không cần sinh spectrogram)



AR Transformer trong TTS+RVQ

- Autoregressive decoding: Mỗi token dự đoán dựa trên toàn bộ các token trước đó
 - Cực kỳ chính xác về chất lượng giọng nói: Mô hình giữ được prosody, nhấn nhá, ngữ điệu tốt hơn
 - Chậm (không phù hợp với real-time): Vì phải sinh từng token một (không song song được)
- Phù hợp với offline high-quality TTS (giống như VALL-E), có thể dùng trong huấn luyện, sinh sample chất lượng cao làm "ground truth"
- Thường được dùng làm teacher model để tạo ground truth cho training NAR model (distillation)

NAR Transformer trong TTS+RVQ

- Sinh toàn bộ chuỗi mã cùng lúc hoặc song song theo nhóm Khả năng song song hóa cao
- Nhanh hơn nhiều lần so với AR: Rất phù hợp cho ứng dụng real-time TTS
- Khó giữ được nhịp điệu, intonation như AR: Dễ bị mất ngữ cảnh dài, đặc biệt nếu không có trợ lực (e.g. từ encoder)

→ Dùng để **deploy** online, hoặc dùng như student model trong distillation

→ Với sự kết hợp với RVQ, giúp giảm độ phức tạp của output (chỉ cần sinh mã discrete) nên NAR hoạt động hiệu quả hơn so với sinh mel trực tiếp.

Challenges in TTS developments

- **Inference Speedup**

- Reducing inference latency while maintaining high quality

- **Robust and Expressive TTS**

- Generating natural, expressive speech with accurate intonation, rhythm and emotion.

- **Low resource language**

- Achieving high-quality synthesis with minimal training data for low-resource languages.

Challenges and Solutions for Vietnamese TTS

- **Word Segmentation & Text Normalization**
 - Hybrid models
- **Lack of High-Quality Speech Data**
 - Semi-supervised or self-supervised learning
 - Data augmentation
- **Prosody Modeling**
 - Style, emotion tags
 - Style transfer or reference encoder

Summary

- TTS technology has **evolved** from **concatenative synthesis** and **statistical parametric synthesis** to **neural-based Text-to-Speech**, with modern approaches increasingly focusing on leveraging generative models.
- Mainstream TTS models typically use separate **acoustic models** and **vocoders**, but **fully end-to-end** models are emerging.
- The primary goal of TTS is to **improve quality** while **reducing costs**.

07. Automatic Speech Recognition



Nguyen Thi Thu Trang

Outline

7.1. Overview

7.2. System Architecture

7.3. ASR Techniques

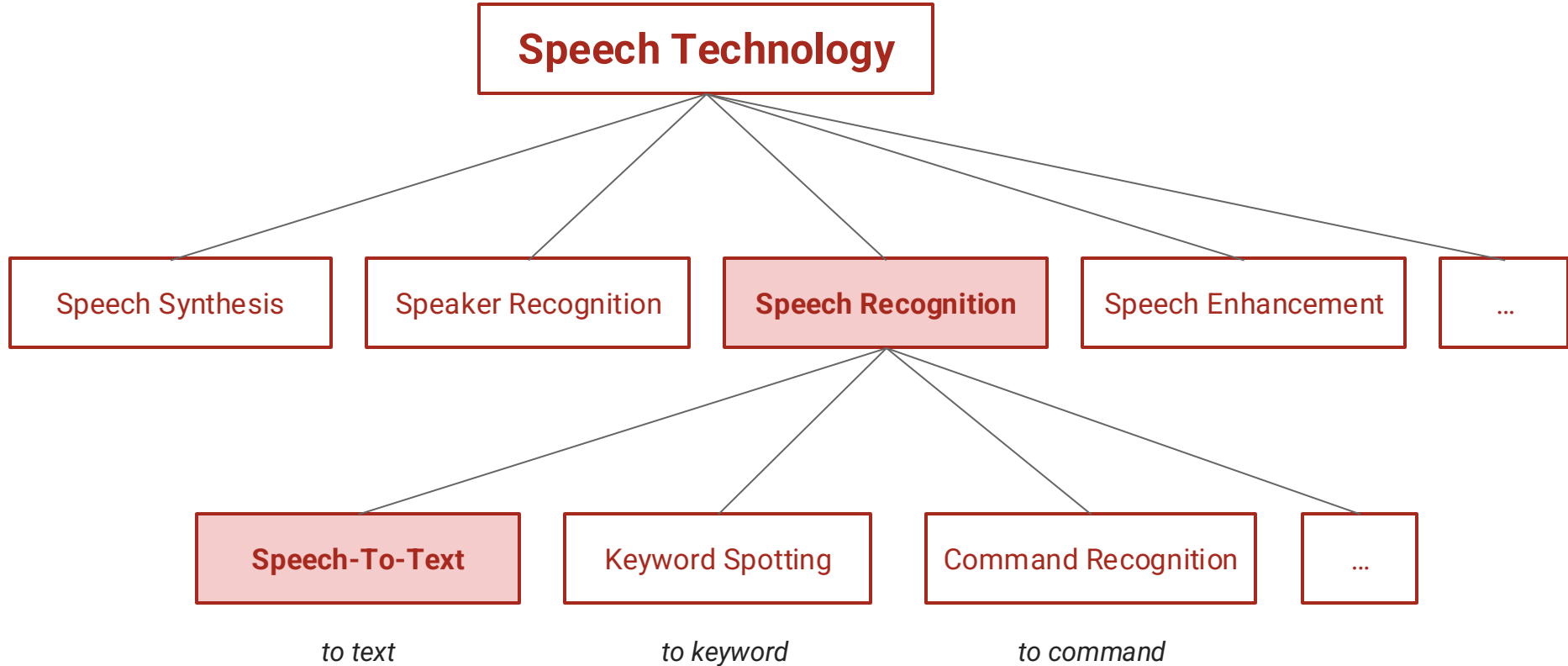
7.4. Challenges and Solutions

7.1 Overview

- ❖ Automatic Speech Recognition - Nhận dạng tiếng nói
 - Quá trình nhận diện và xử lý tiếng nói của con người thành những khía cạnh khác nhau
- ❖ Speech-To-Text (STT) - Chuyển từ tiếng nói thành văn bản
 - Một loại nhận dạng tiếng nói
 - Quá trình chuyển đổi từ đoạn âm thanh lời nói thành văn bản tương ứng



7.1. Overview



History of Automatic Speech Recognition

- 
- 1950s–60s: **Template Matching**. Audrey, limited **digits**, single speaker
 - 1970s : **Isolated Word**. DARPA funding, small vocabulary, feature engineering
 - 1980s : **HMM**. Modular pipeline, continuous speech, statistical modeling
 - 1990s : **GMM-HMM**. Dictation tools, speaker independence, large vocabularies
 - 2000s : **Advanced ML**. Context modeling, beam search, robust lexicons
 - 2010s : **Deep Learning**. End-to-end models (CTC, RNN-T), DNN-based AMs
 - 2020s : **SSL + Transformers**. Pretrained models (Whisper, Wav2Vec 2.0), multilingual, open-source.

ASR Application in Real-life?



<https://wall.sli.do/event/wPmQxU18pEZmnxHpv1Uo3m?section=c1a5f979-f9fd-41df-8ddd-d38a83ccd7e8>

Applications of Speech Recognition



**Customer
Service**



**Virtual
Assistant**



**Transcription
Services**



**Accessibility
Features**

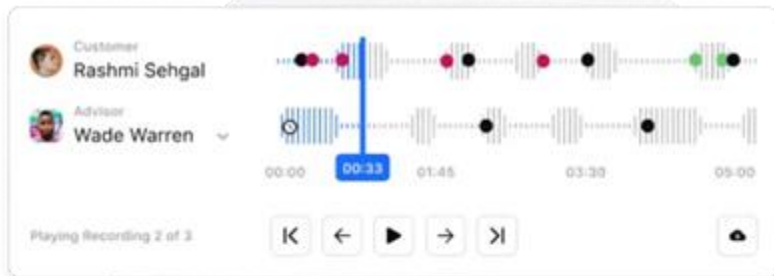


**Smart
Homes**

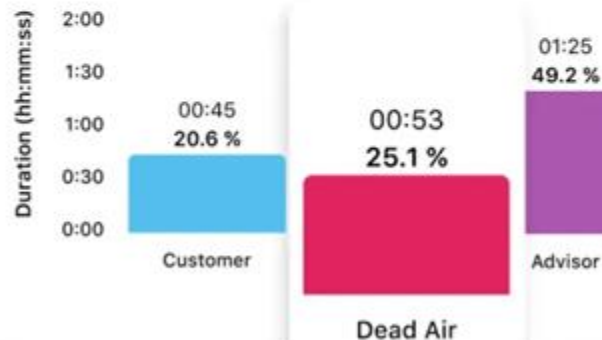
ASR Application in real-life

- ❖ Voice Assistants
- ❖ Call Center Automation
- ❖ Dictation / Voice Typing
- ❖ Subtitling
- ❖ Meeting Transcription
- ❖ Education
- ❖ In-car Voice Control
- ❖ Multilingual Interface

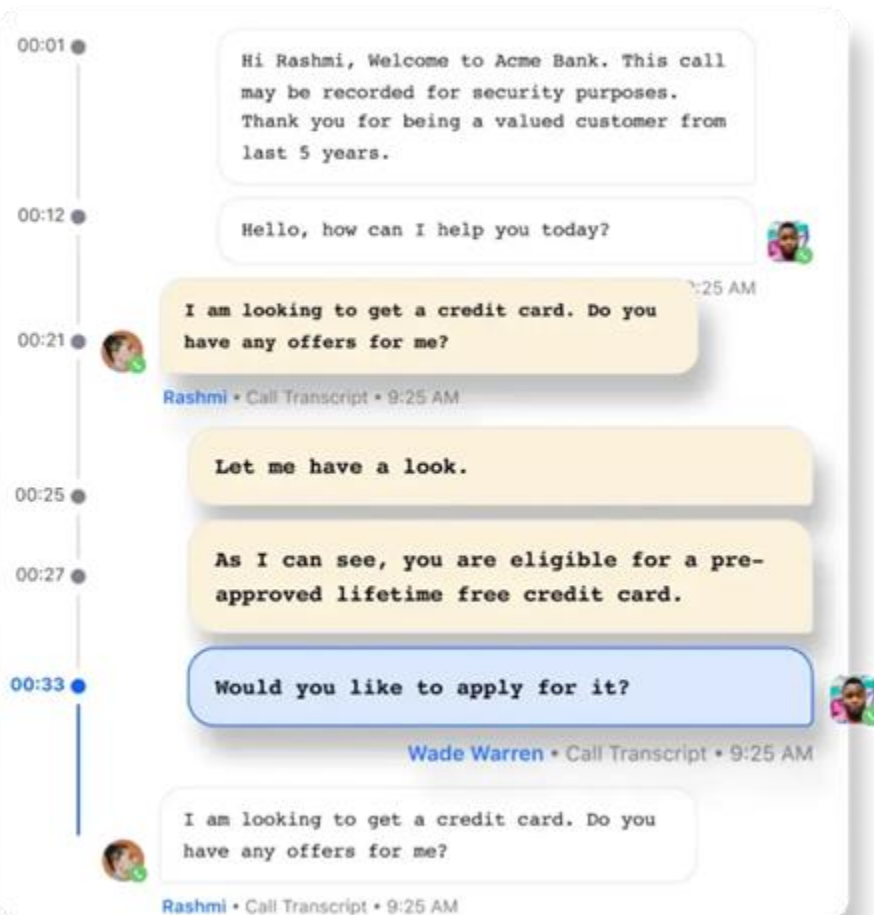




Time Utilization

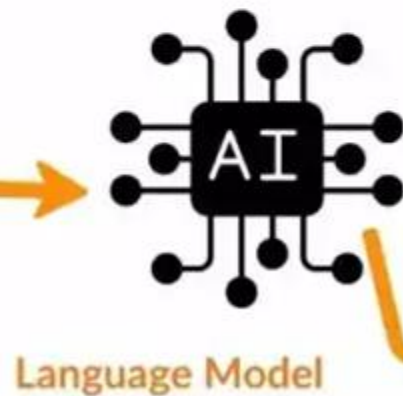
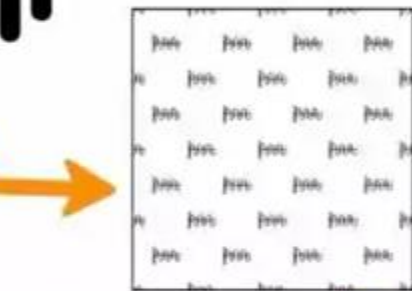


Call Transcript



Current State of ASR

English Tasks	WER%
LibriSpeech audiobooks 960hour clean	1.4
LibriSpeech audiobooks 960hour other	2.6
Switchboard telephone conversations between strangers	5.8
CALLHOME telephone conversations between family	11.0
Sociolinguistic interviews, CORAAL (AAL)	27.0
CHiMe5 dinner parties with body-worn microphones	47.9
CHiMe5 dinner parties with distant microphones	81.3
Chinese (Mandarin) Tasks	CER%
AISHELL-1 Mandarin read speech corpus	6.7
HKUST Mandarin Chinese telephone conversations	23.5

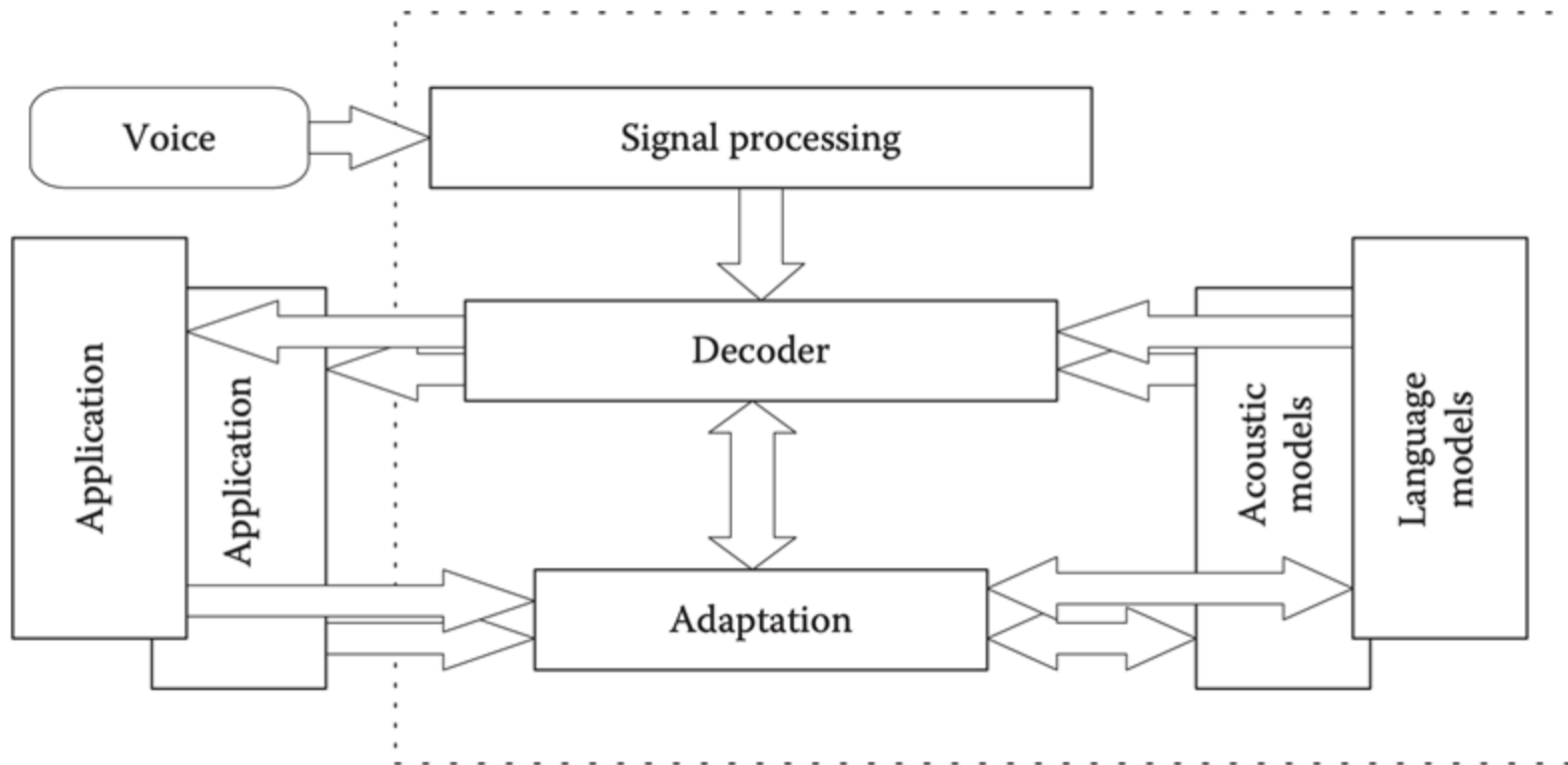


The Process of Speech Recognition

ASR: Acoustic Model & Language Model



7.2 Architecture of ASR System

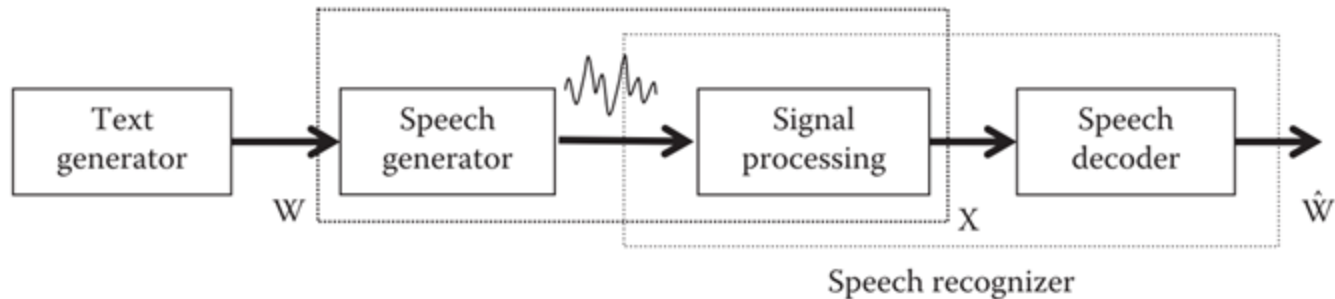


Basic components of an ASR system.

Main Components

- **Acoustic Model:** Representation of knowledge about **acoustics**, **phonetics**, **microphone** and **environment** variability, **gender** and **dialect** differences among **speakers**, etc.
- **Language Model:** System's knowledge of what constitutes a **possible word**, what words are likely to **co-occur**, and in what **sequence**.
- **Decoder:** Generate the **word sequence** that has the maximum posterior probability for the input feature vectors extracted from speech signal.
- **Adaptation:** Modify either the acoustic or language models to **improve performance**

ASR Problem



- **Input:** Acoustic observation or feature vector sequence $\mathbf{X} = X_1 X_2 \dots X_n$
- **Output:** Corresponding word sequence $\hat{\mathbf{W}} = w_1 w_2 \dots w_m$
having maximum posterior probability $P(W|X)$

$$\hat{\mathbf{W}} = \arg \max_{\mathbf{w}} P(\mathbf{W}|\mathbf{A}) = \arg \max_{\mathbf{w}} \frac{P(\mathbf{W})P(\mathbf{A}|\mathbf{W})}{P(\mathbf{A})} \stackrel{\text{Fixed } \mathbf{X}}{=} \arg \max_{\mathbf{w}} P(\mathbf{W})P(\mathbf{X}|\mathbf{W})$$

- $P(W)$: Language Model
~ Lexical and grammatical complexity and variations of spoken language
- $P(X|W)$: Acoustic Modeling ~ Phonetic Modeling
~ Account speaker variations, pronunciation variations, environmental variations, and context-dependent phonetic coarticulation variations

Best Word Sequence for Conventional Models

- λ_1 : Language model scaling factor (LMSF)
 - To recalibrate the trade-off between acoustics and common word sequences since the AM tends to underestimate $P(X|W)$ due to the independence assumption on phone probabilities

$$\hat{W} = \operatorname{argmax}_W P_{AM}(X|W) P_{LM}(W)^{\lambda_1}$$

$$\hat{W} = \operatorname{argmax}_W \log P_{AM}(X|W) + \lambda_1 \log P_{LM}(W)$$

Language Model

- Conventional models implicitly learn a language model for the output domain of characters from their training data
 - Training data (speech paired with text transcriptions) may not include sufficient text to train a good language model.
- Improve a model at least slightly by incorporating a very large language model

The roses in the vase by the door ?

Competing answers: is, are

$P(\text{The roses in the vase by the door are})$

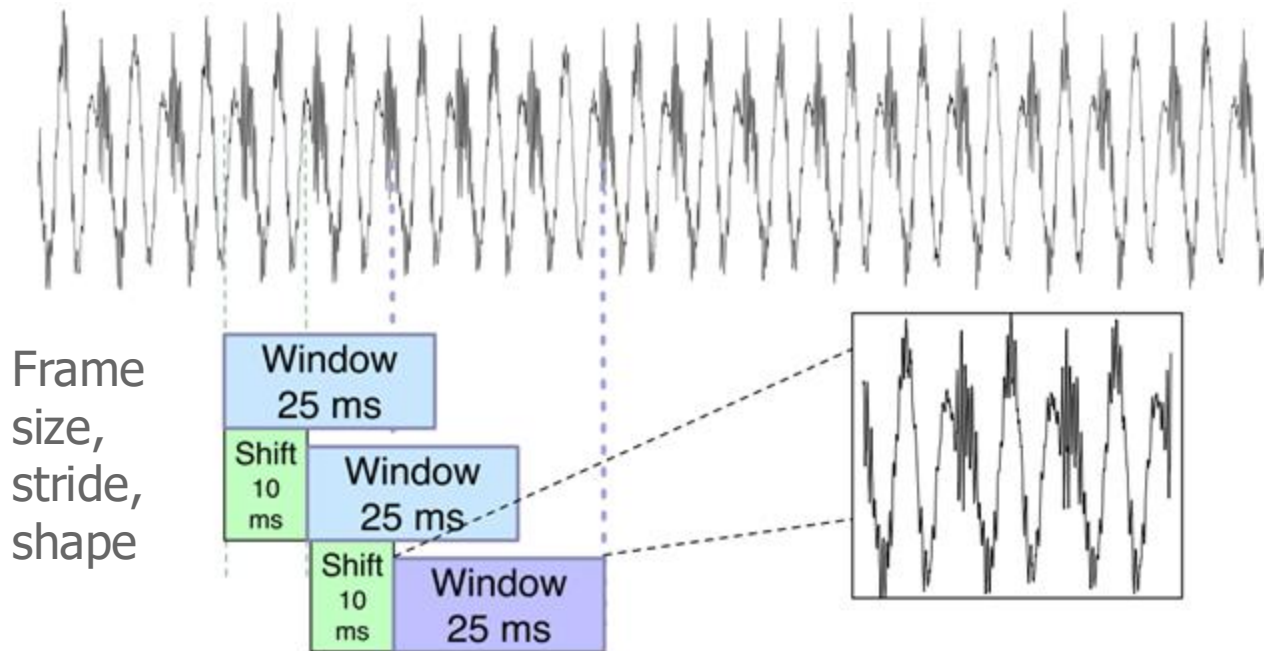
$P(\text{The roses in the vase by the door is})$

Is the correct answer
ranked higher?

$P(\dots \text{are}) > P(\dots \text{is})?$

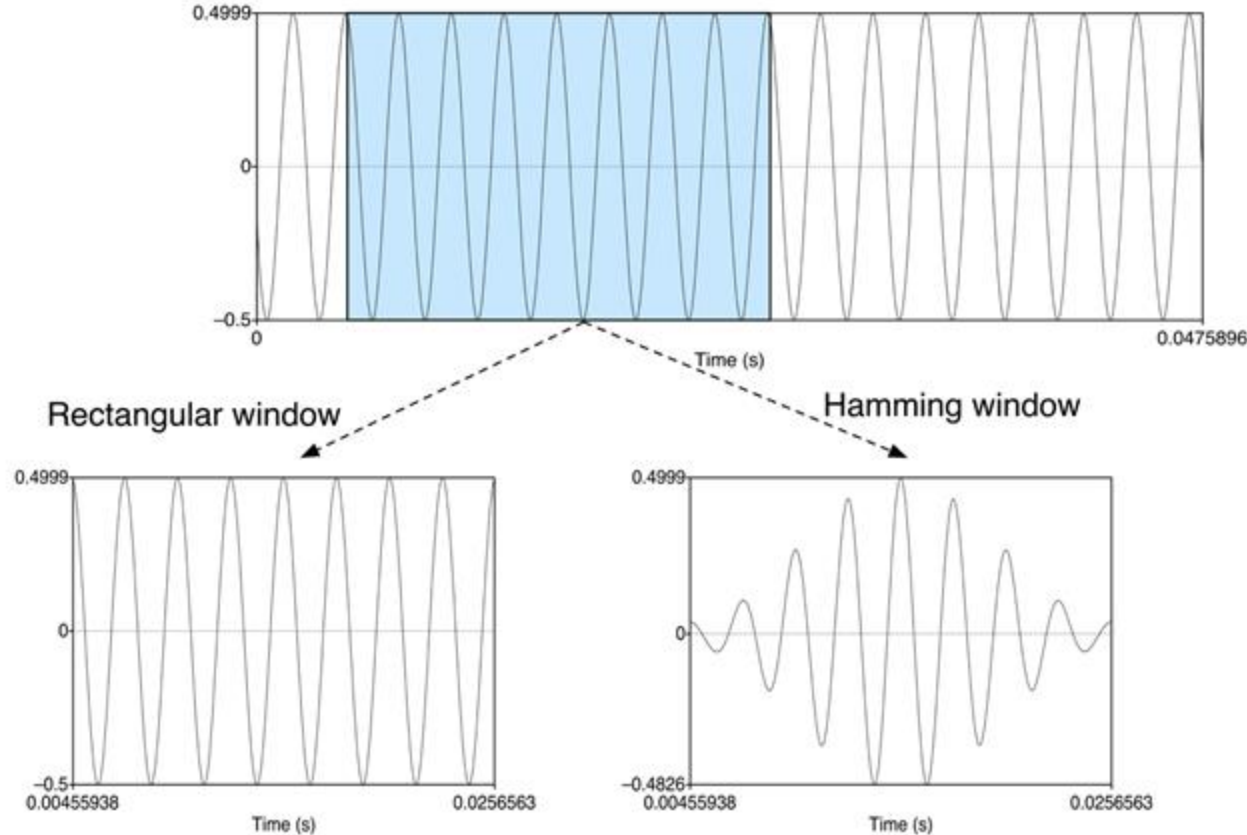
Windowing to reduce spectral leakage

Its statistical properties are constant within this region
→ *Extract acoustic feature vectors for each frame*



Windowing, showing a 25 ms rectangular window with a 10ms stride.

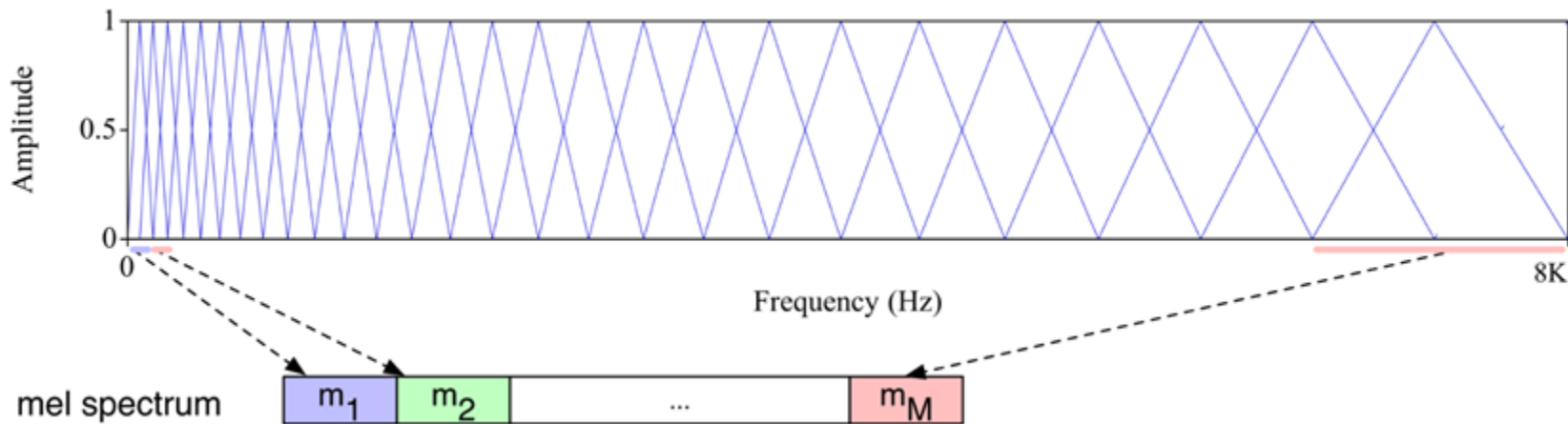
Hamming window



Hamming window **shrinks** the values of the signal **toward zero** at the window **boundaries**, **avoiding discontinuities** compared to Rectangular

Mel filter bank and Log

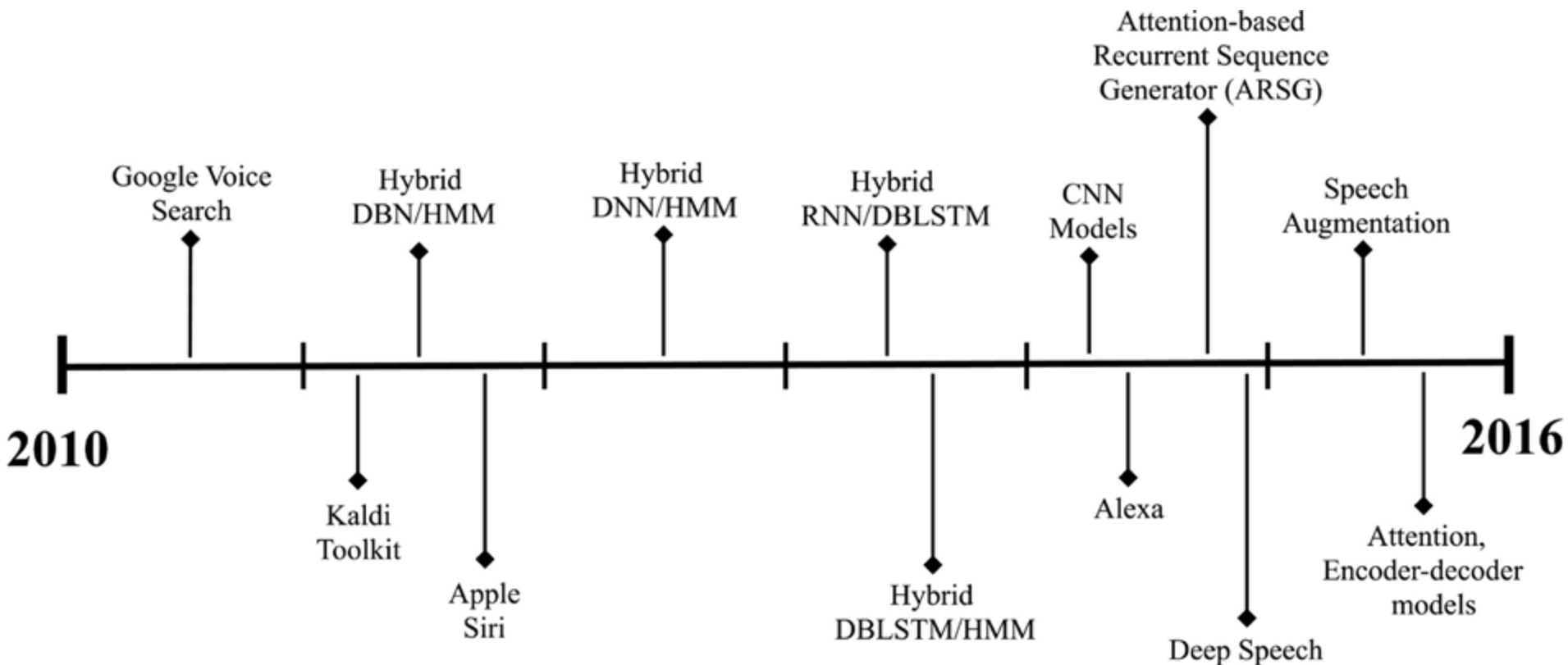
- **Mel scale:** Unit of pitch, auditory frequency scale for human hearing
- **Mel filter bank:** Bank of filters that collect energy from each frequency band, spread logarithmically: very fine resolution at low frequencies, and less resolution at high frequencies
 - Human hearing is not equally sensitive at all frequency bands, less sensitive at higher frequencies.



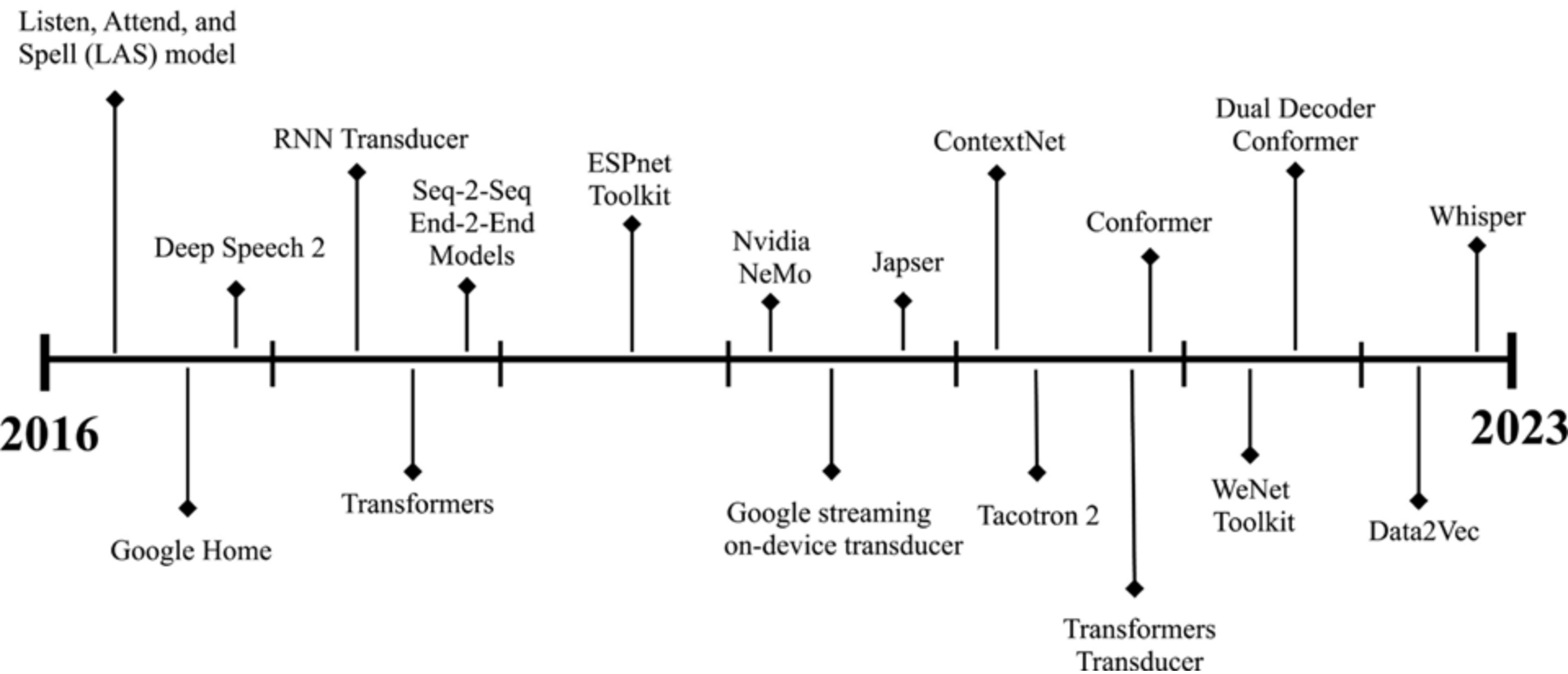
7.3. Speech Recognition Techniques

Technique	End-to-End	Streaming	Popular Models/Tools
HMM-based ASR	✗	✓	Kaldi, HTK, CMU Sphinx, Julius
CTC	✓	✓ (with tricks)	DeepSpeech, Jasper, Kaldi (tools)
RNN-T	✓	✓	Transducer-based models, TensorFlow ASR
LAS (Attention-based)	✓	✗	Listen, Attend, Spell
Transformer-based + SSL	✓	✗/✓	Wav2Vec 2.0, HuBERT, Whisper, SpeechBrain

From HMM to Attention Encoder-Decoder



From LAS to SSL & Transformer



Approaches / Techniques

1. Convolution Neural Network (CNN)

2. Recurrent Neural Network (RNN)

Long Short Term Memory (LSTM)

LSTM + RNN

3. Hybrid Architectures

CNN + RNN

LSTM + RNN + CNN

CNN + RNN + DNN

LSTM + CNN

DNN + RNN

ASR Techniques

4. Deep Neural Network (DNN)

5. Deep LSTM

11. Conformer

10. Transformer

Wav2Vec

Data2Vec

BERT

HuBERT

Whisper

9. Transfer Learning

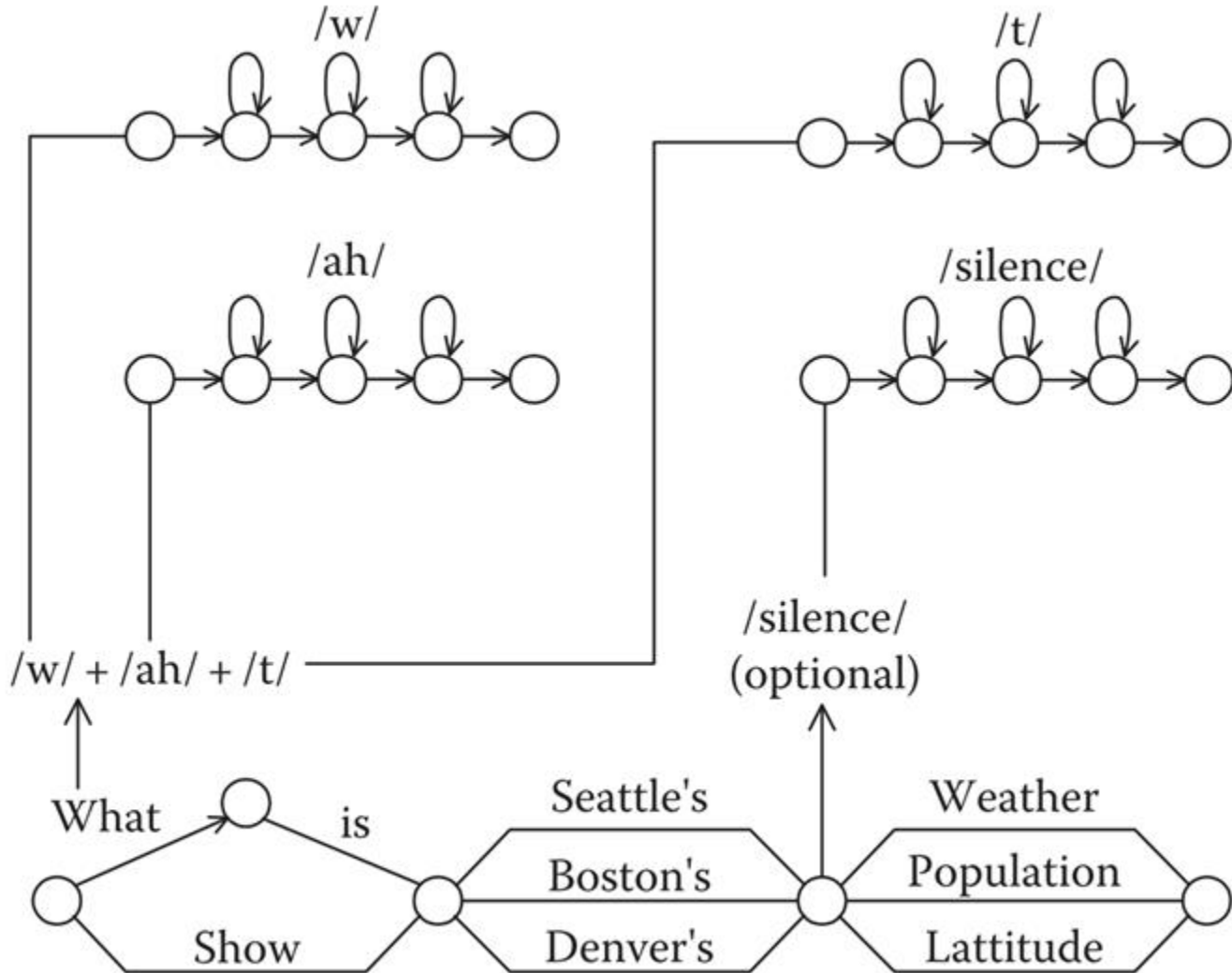
8. Time Delayed Neural Network (TDNN)

7. Deep Convolution Neural Network (DCNN)

6. Deep Belief Network (DBN)

7.3.1. HMM-based ASR

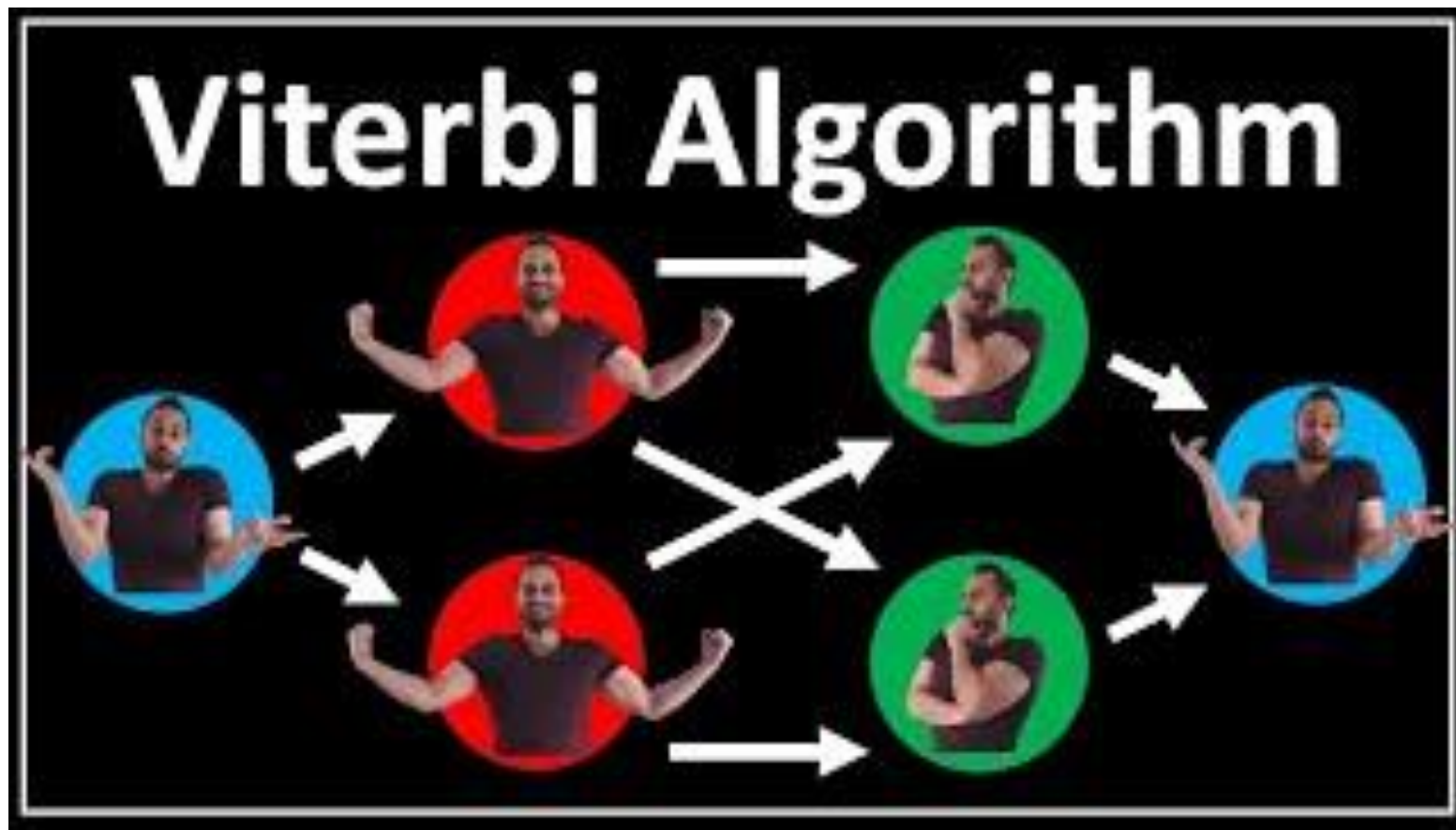
An illustration of how to compile a speech-recognition task with finite grammar into a composite HMM.



Hidden Markov Model

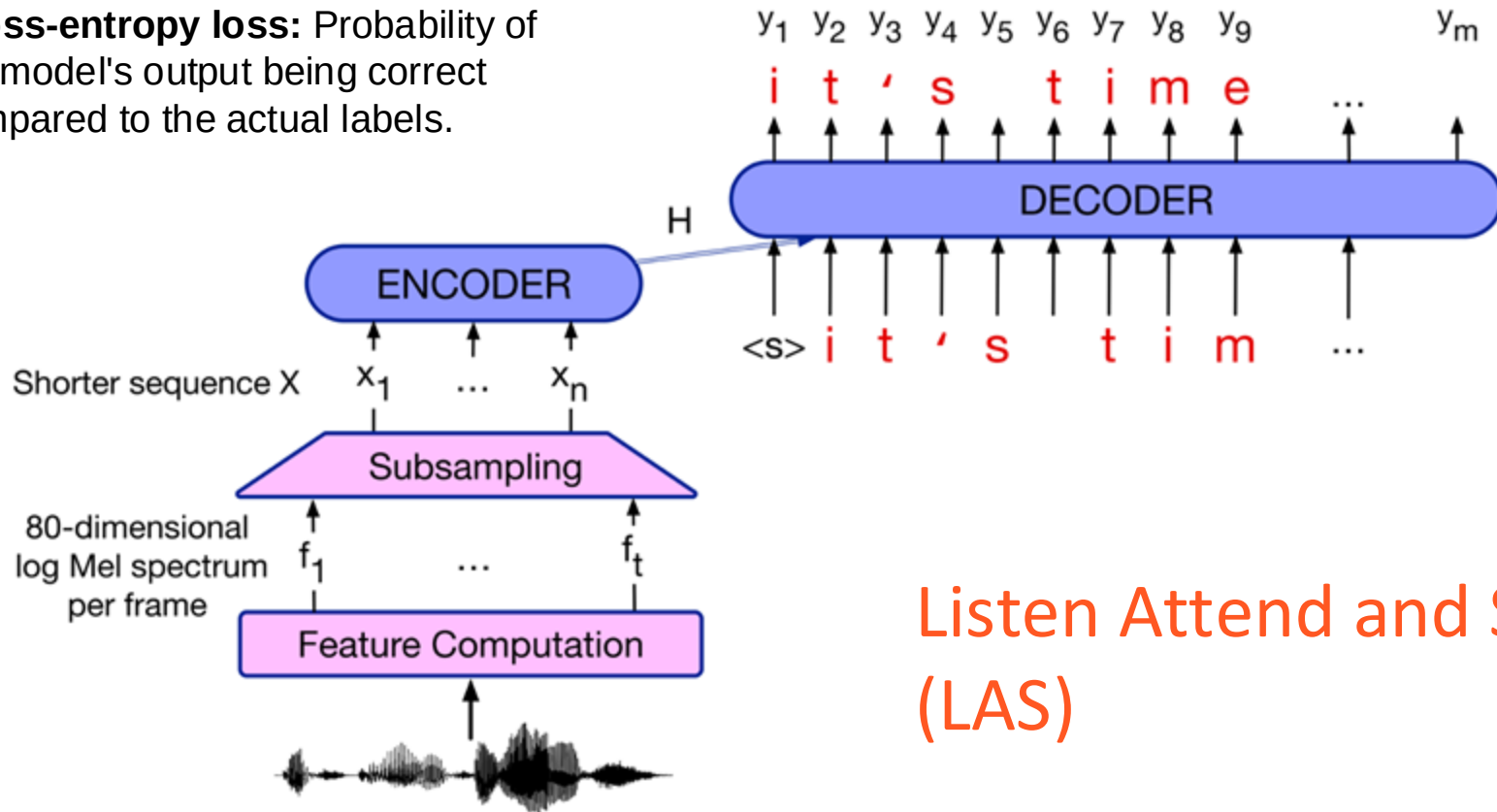


Viterbi algorithm



7.3.2. Attention-based Encoder-Decoder ASR

Cross-entropy loss: Probability of the model's output being correct compared to the actual labels.



Listen Attend and Spell
(LAS)

Two particular issues of ASR

- **Length differences**

- Very long acoustic feature sequences (200 acoustic 10ms-frames) mapping to much shorter sequences of letters (5 letters)

→ **AED** has a special **compression** stage that shortens the acoustic feature sequence before the encoder stage as well as an **attention** mechanism focusing on different parts of the input while predicting the output.

- **Input-output mapping:**

- Hard to know exactly which part of X maps to which part of Y

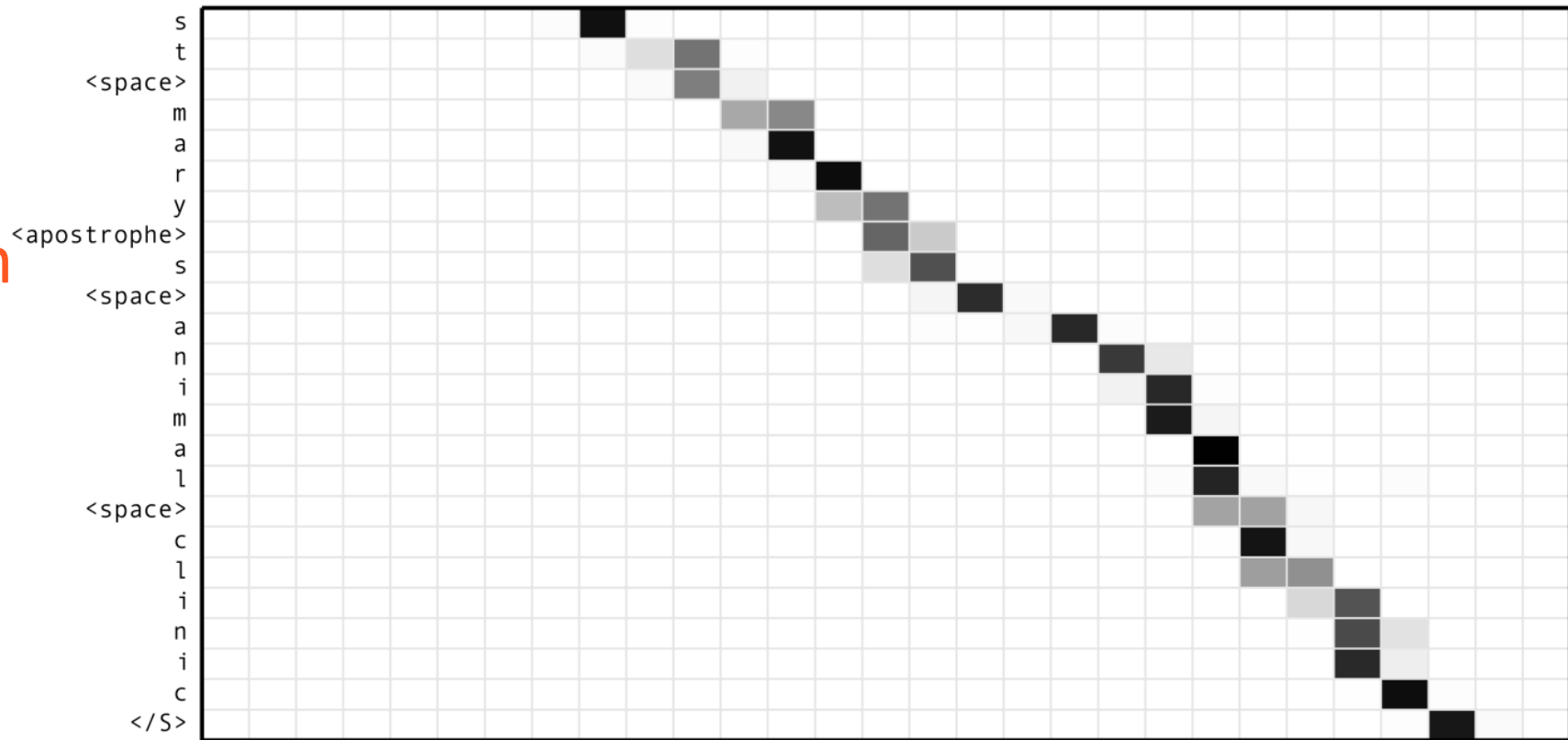
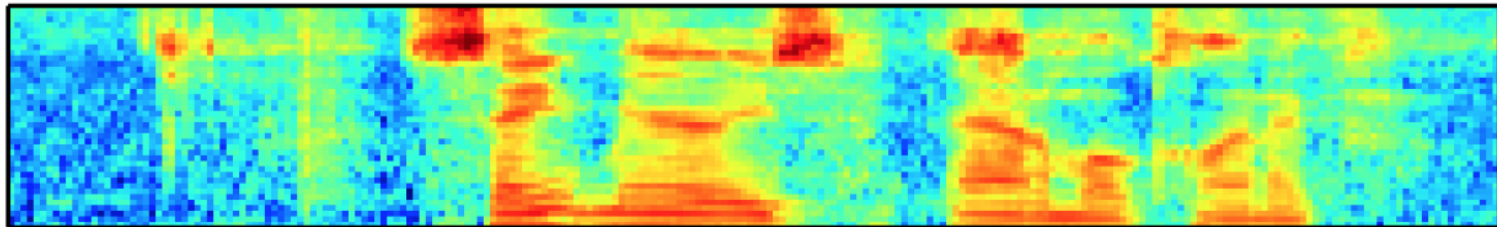
→ **AED** has an **attention** mechanism to align parts of the input with corresponding parts of the output, without requiring predefined structures.

Problems of AED in ASR

- The attention operation is expensive for long input sequences
 - The complexity of attending to the entire input for every output is $O(mn)$, and for audio, m and n are big.
- Attention models cannot be run *online* (in real time), since the entire input sequence needs to be available before the decoder can attend
- Attention models also don't take advantage of the fact that, for speech recognition, the alignment between inputs and outputs is **monotonic**
 - i.e. if word A comes after word B in the transcript, word A must come after word B in the audio signal.
 - The fact that attention models lack this inductive bias seems to make them harder to train for speech recognition; it's common to add auxiliary loss terms to stabilize training.

Monotonic Alignment

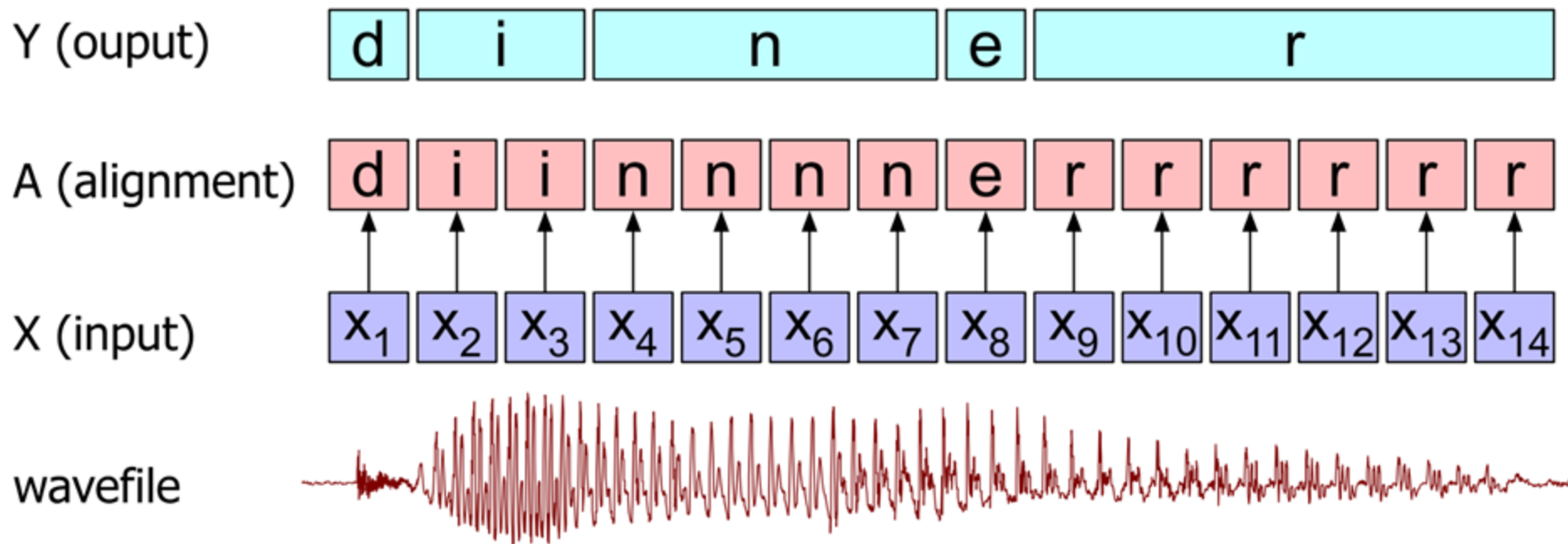
Audio



7.3.3. Connectionist Temporal Classification (CTC)

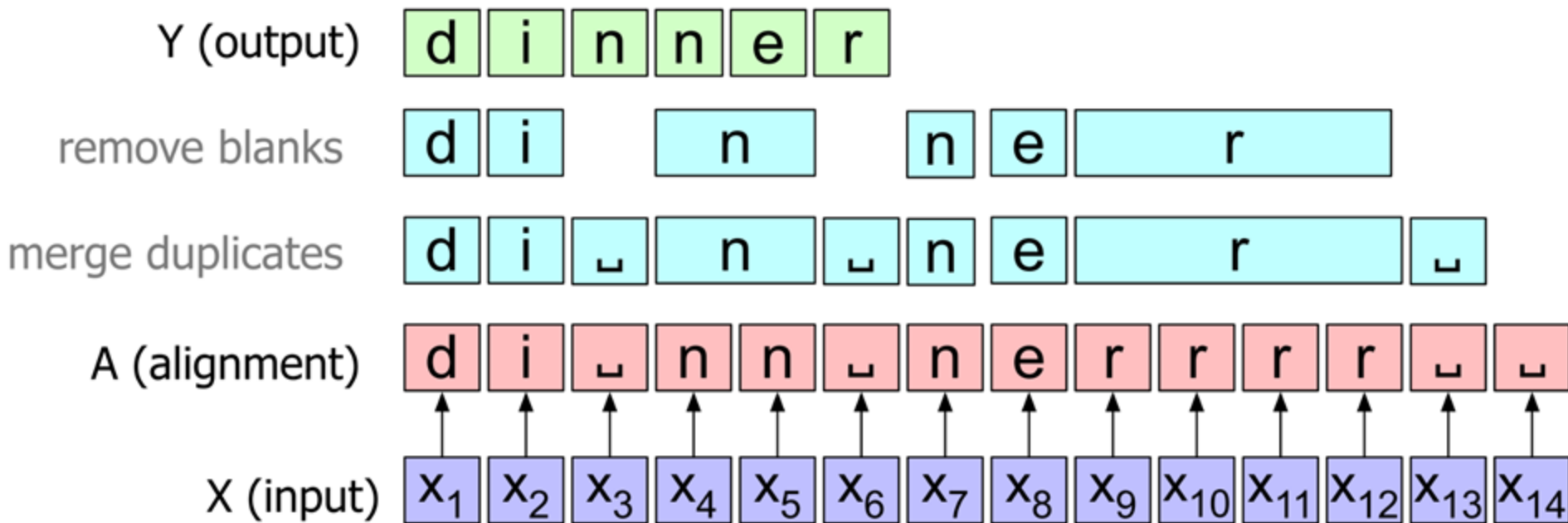
- Predict character sequences **without explicitly matching the input's length to the output**, enabling it to handle variable-length outputs and learn rare or uncommon words.
- **Algorithm:**
 - Output a single character for every frame of the input, so that the output is the same length as the input
 - Apply a collapsing function combining sequences of identical letters, resulting in a shorter sequence.
- **Loss function:**
 - Probability of all possible outputs to the actual outputs, including using a special symbol to represent gaps between words or parts of a word
 - Without requiring exact alignment between input and output sequences.

Naive collapsing an alignment



Naive algorithm for collapsing an alignment between input and letter

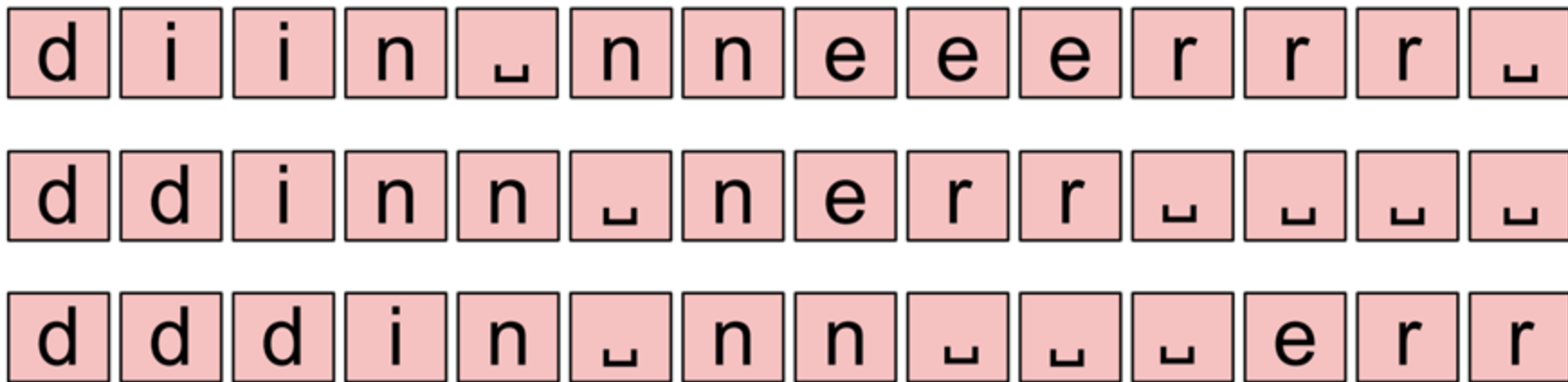
Blank token in CTC alignment



CTC adds to the transcription alphabet a special symbol for a blank

CTC Alignment

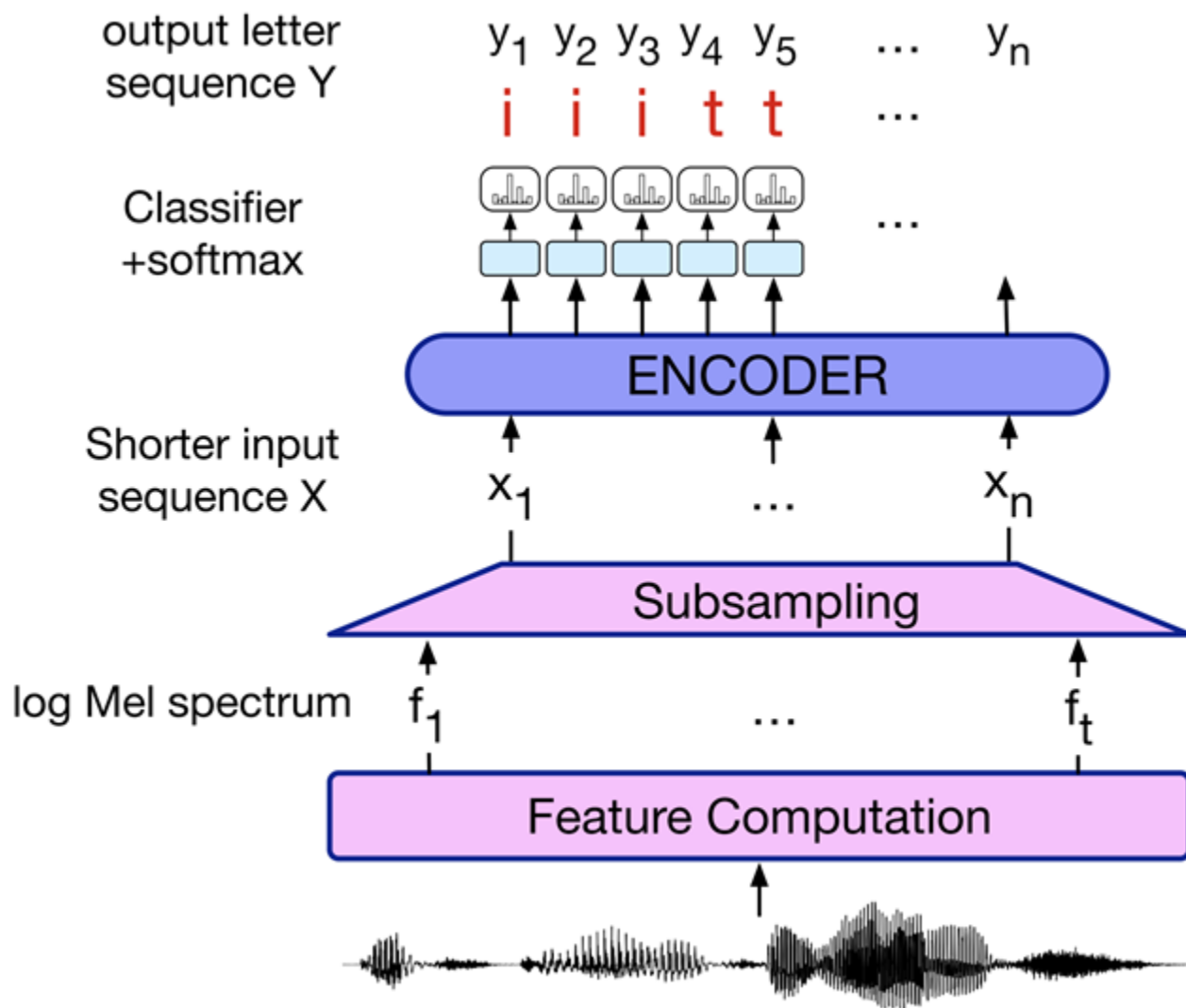
E.g. Different alignments that would produce the same output



CTC Inference

~ Sequence-modeling

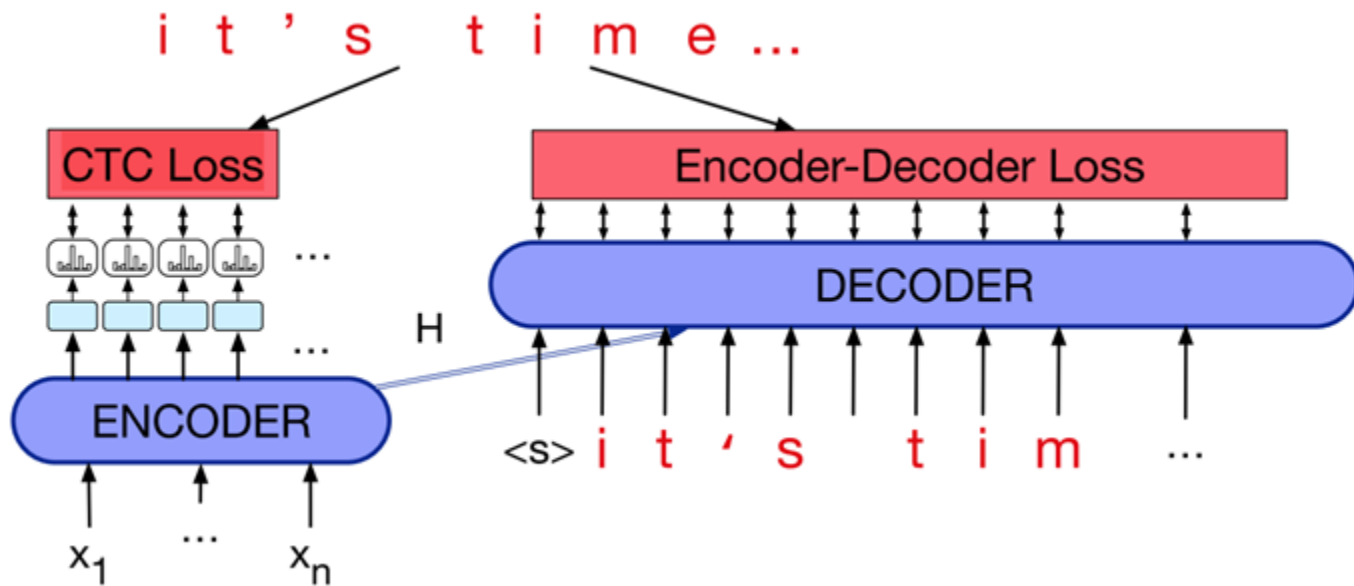
~ Encoder-only model,
with decoding done by
simple softmaxes over
the character
vocabulary at each
time step



Combining CTC and Encoder-Decoder

Combining cross-entropy loss from the encoder-decoder architecture, and the CTC loss

$$L = -\lambda \log P_{encdec}(Y|X) - (1 - \lambda) \log P_{ctc}(Y|X)$$

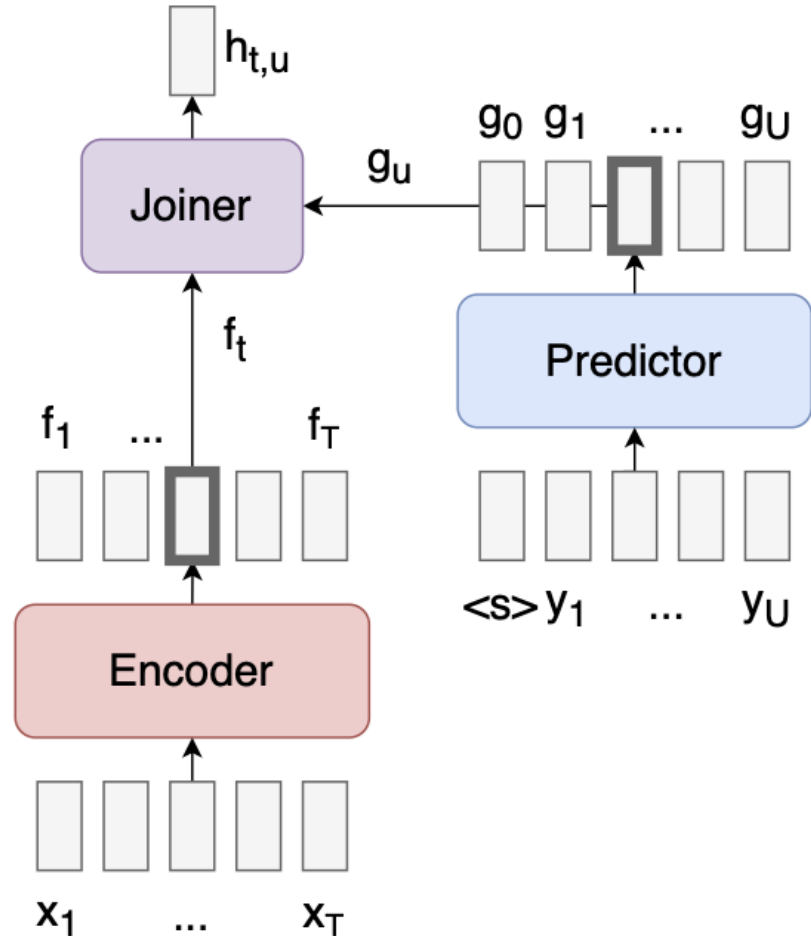


Problems of CTC

- *Problem 1: The output sequence length has to be smaller than the input sequence length*
 - This might not seem like a problem for speech recognition, where length of output T is much larger than length of input U —but it prevents us from using a model architecture that does a lot of pooling, which can make the model a lot faster.
- *Problem 2: The outputs are assumed to be independent of each other*
 - The result is that CTC models often produce outputs that are obviously wrong, like “I eight food” instead of “I ate food”. Getting good results with CTC usually requires a search algorithm that incorporates a secondary language model.

RNN-T

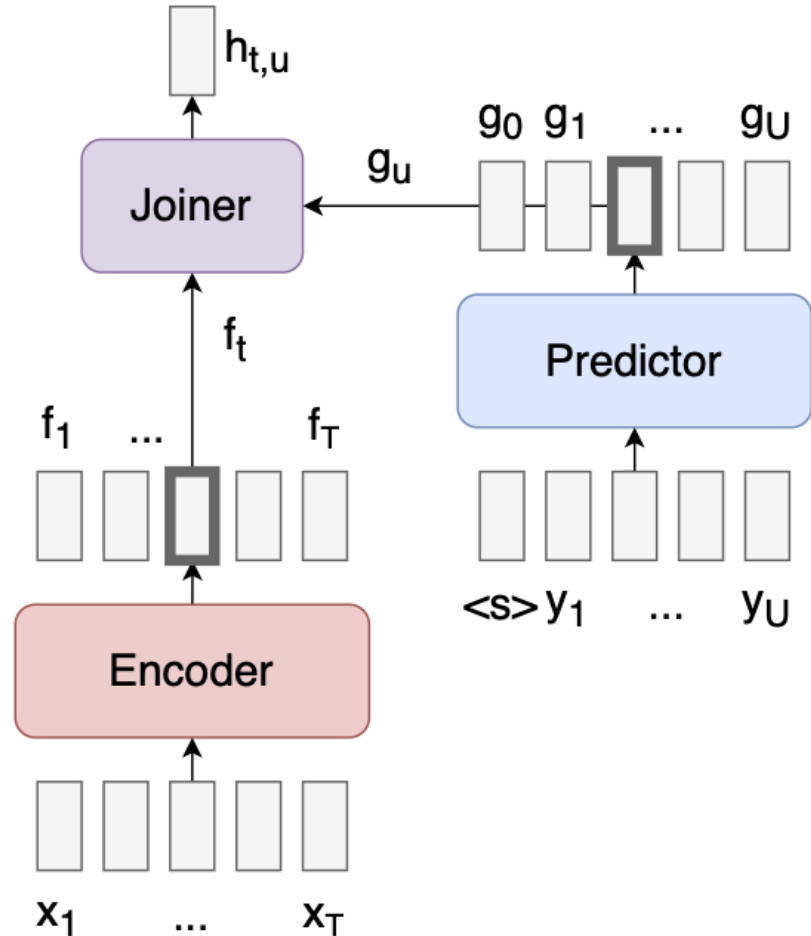
- RNN-T is an **end-to-end sequence-to-sequence model designed for ASR**. It **jointly learns the alignment between the input audio and the output text transcription** without requiring explicit frame-level alignment
 - Overcomes limitations of CTC which waits for a larger chunk or the full sentence, then collapses the output — introducing delay.
 - Allows for more flexible alignment between speech frames and text tokens
 - Supports streaming ASR, where transcriptions can be generated as the audio is being processed.



$$P_{AM}(X|W) = \frac{P_{RNNT}(W|X)P(X)}{P_{ILM}(W)}$$

7.3.4. Transducer model (RNN-T)

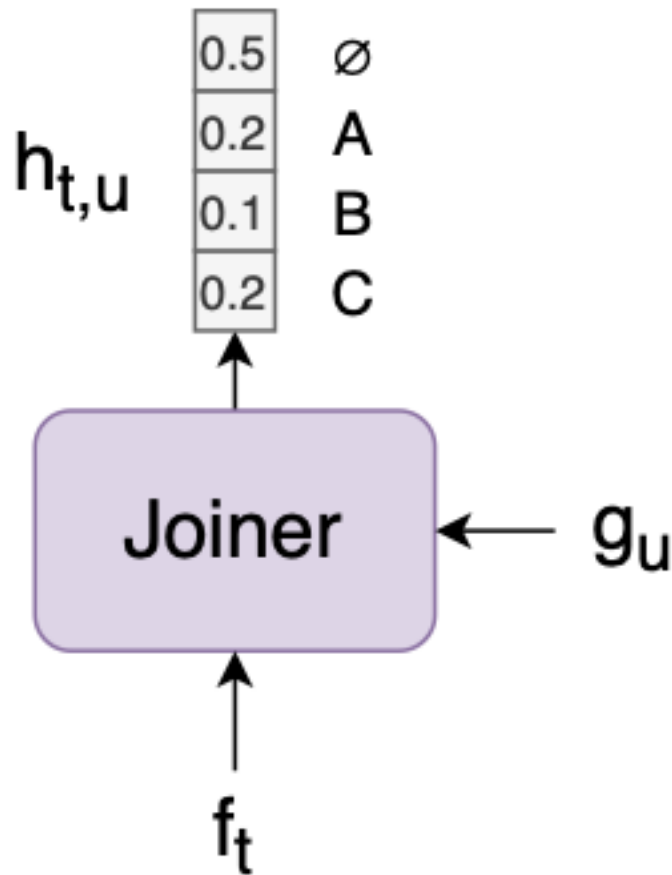
- The Transducer elegantly solves both problems associated with CTC, while retaining some of its advantages over attention models
 - It solves **Problem 1** by allowing multiple outputs for each input
 - It solves **Problem 2** by adding a predictor network and joiner network.



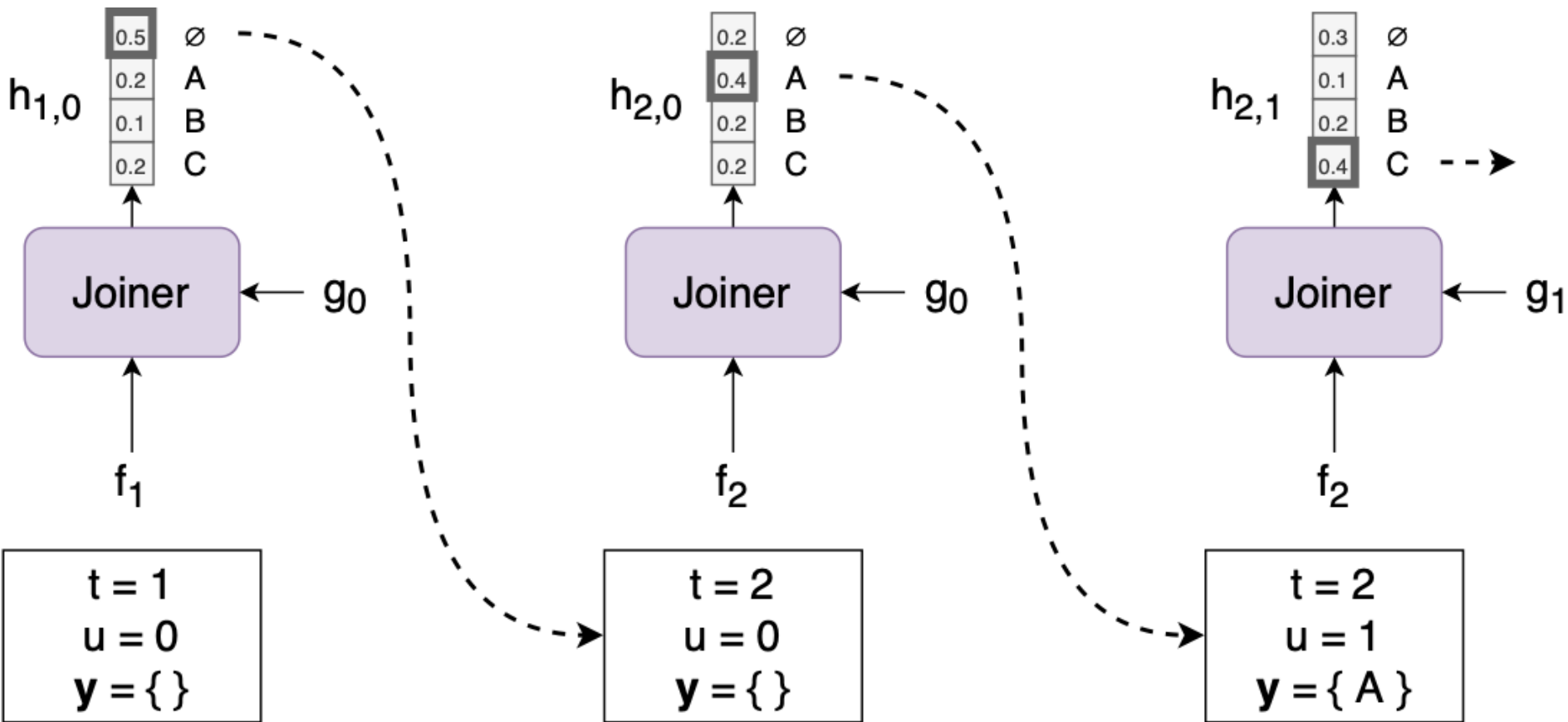
$$P_{AM}(X|W) = \frac{P_{RNNT}(W|X)P(X)}{P_{ILM}(W)}$$

RNN-T

- The predictor is autoregressive
 - it takes as input the previous outputs and produces features that can be used for predicting the next output, like a standard language model.
- The joiner is a simple feedforward network that combines the encoder vector f_t and predictor vector g_u and outputs a softmax $h_{t,u}$ over all the labels, as well as a “null” output $\emptyset$



Simple greedy search algorithm



Why RNN-T Is Suitable for Streaming ASR?

- Streaming-friendly
 - Processes audio frame-by-frame without needing the full sequence ahead of time
- No need for forced alignment
 - Learns alignment automatically (like CTC), no frame-level labels needed
- Supports incremental output
 - Can generate words while audio is still coming in — critical for low-latency ASR
- Monotonic decoding
 - Naturally maps well to the left-to-right nature of spoken language

Why RNN-T Is Suitable for Streaming ASR?

- Token-by-token prediction
 - RNN-T can emit tokens during audio input, frame-by-frame.
 - CTC often needs to wait until the full utterance or chunk is processed before collapsing repeated tokens and blanks.
- Context-aware decoding
 - RNN-T uses a Prediction Network (decoder) that retains previous outputs, giving it language modeling capabilities during decoding.
 - CTC has no such decoder, so it lacks contextual understanding during prediction.
- Joint network enables streaming
 - The Joint Network combines outputs from the encoder (acoustic info) and prediction network (linguistic context) to decide what to emit — ideal for real-time decoding.

CTC

vs.

RNN-T

✗ Not ideal for real-time streaming

Assumes monotonic alignment, no decoder context

✗ No decoder, output depends only on encoder

Outputs after processing entire input or a large chunk

High latency (especially on long utterances)

Offline transcription (e.g., post-processing voice recordings)

✓ Specifically designed for streaming

Learns alignment using joint acoustic + linguistic context

✓ Has a **prediction network** that models output history

Can emit tokens **incrementally** as audio frames arrive

Low latency; suitable for real-time interaction

Real-time ASR (e.g., voice assistants, live captioning)

Problems of RNN-T

- Latency
- Accuracy with long context
- Expensive training

How to improve RNN-T?

- Replacing RNNs with More Efficient Architectures (Transformer/Conformer)
- Replacing the Prediction Network (Decoder) with lightweight Transformers
- Enhancing the Loss Function with CTC Loss, LM Loss, Alignment-aware Loss...
- Integrating External Language Models (LM)
- Multilingual and Code-Switching ASR

External LM vs Internal LM (ILM) to RNN-T

- RNNT directly models the posterior $P(W|X)$ directly
 - Unlike the AM in conventional systems which models the likelihood $P(X|W)$
- λ_1, λ_2 : Language model scaling factor (LMSF)
 - To recalibrate the trade-off between acoustics and common word sequences since the AM tends to underestimate $P(X|W)$ due to the independence assumption on phone probabilities

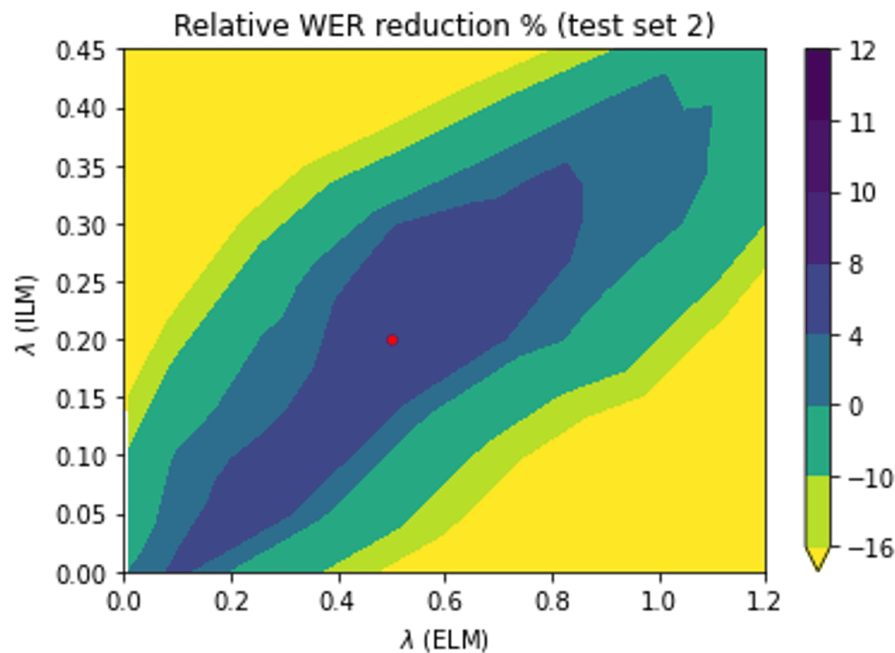
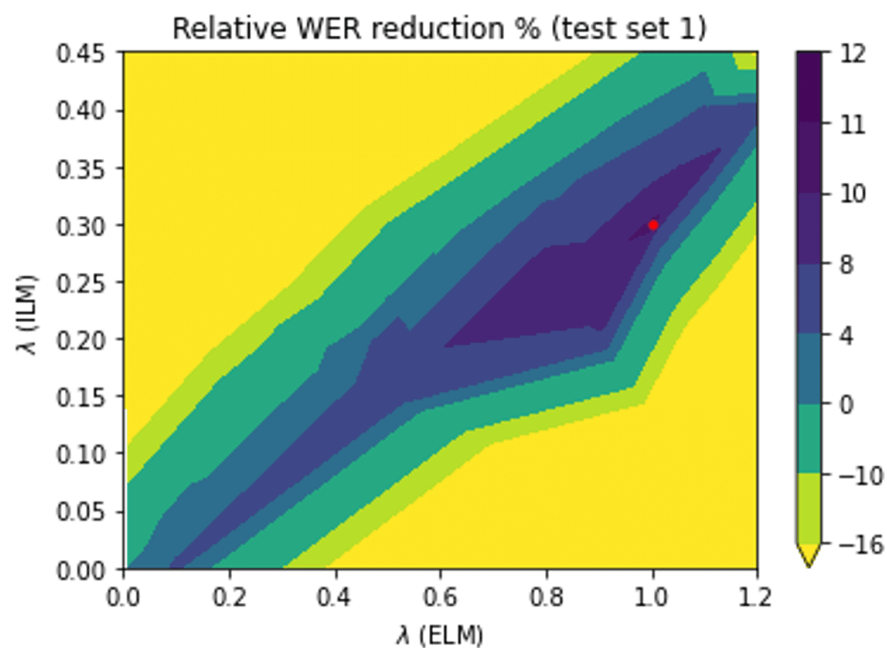
$$P_{AM}(X|W) = \frac{P_{RNNT}(W|X)P(X)}{P_{ILM}(W)}$$

$$\hat{W} = \operatorname{argmax}_W \frac{P_{RNNT}(W|X)}{P_{ILM}(W)^{\lambda_2}} P_{LM}(W)^{\lambda_1}$$

$$\hat{W} = \operatorname{argmax}_W \log P_{RNNT}(W|X) - \lambda_2 \log P_{ILM}(W) + \lambda_1 \log P_{LM}(W)$$

Tuning LMSF

- More ILM-subtraction introduces the need for more external LM addition to take up the slack



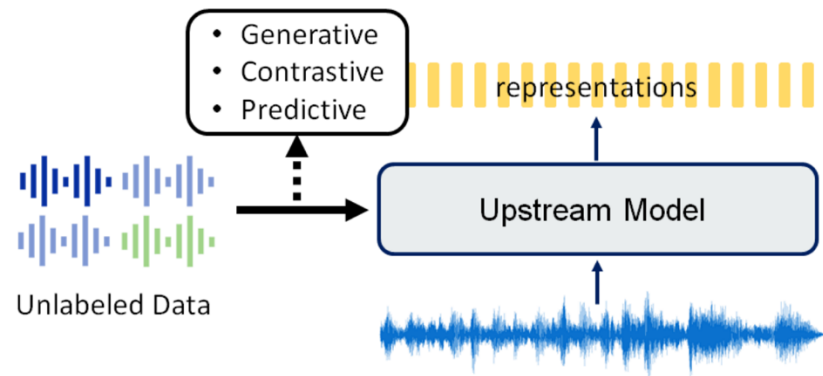
7.3.5. SSL + Transformer

- Self-Supervised Learning (SSL):
 - Pre-trains models on large amounts of **unlabeled data** to learn useful representations of speech.
 - Training the model to **predict masked parts** of the input or solve other self-defined tasks **without human annotations**
 - Pre-trained representations capture **general acoustic and phonetic** information
- Transformer
 - **Self-attention** mechanism to model relationships between different parts of a sequence
 - Can capture **long-range dependencies** in speech more effectively than traditional RNNs
 - Allows for **parallel processing** of the input sequence, leading to faster training times

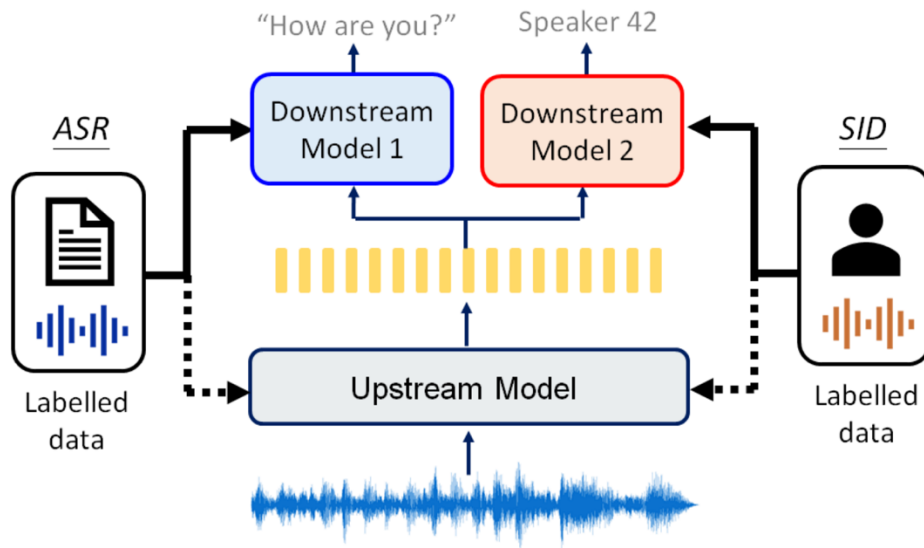
Why Combine SSL with Transformers in ASR?

- Learn from unlabeled audio
 - SSL enables pretraining on large-scale untranscribed speech
- Improved generalization
 - Robust across speakers, accents, and domains
- Low-resource friendly
 - Performs well even with limited fine-tuning data
- Multilingual & zero-shot
 - Especially effective in models like Whisper or HuBERT

Phase 1: Pre-train

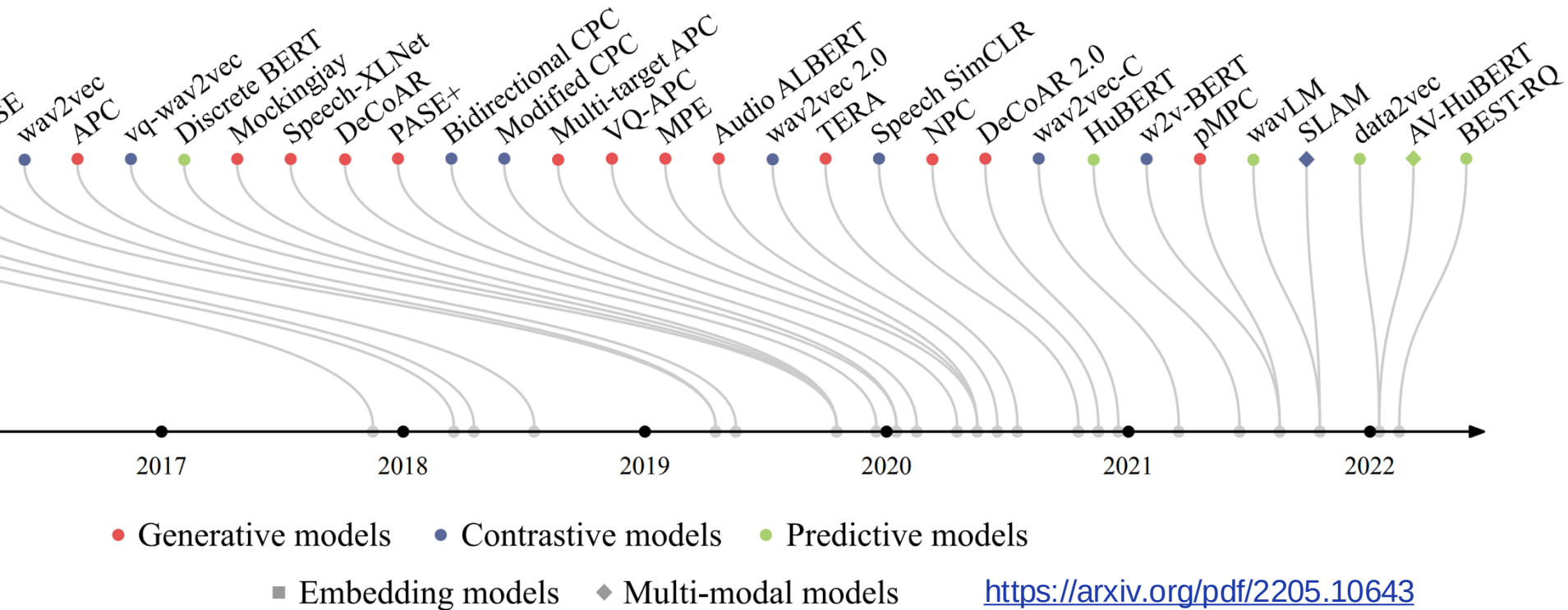


Phase 2: Downstream



SSL Model Categorization

A. Mohamed et al. (2022), "Self-Supervised Speech Representation Learning: A Review," in *IEEE Journal of Selected Topics in Signal Processing*, vol. 16, no. 6, pp. 1179-1210, Oct. 2022, doi: 10.1109/JSTSP.2022.3207050.



SSL Model Categorization

Model Type	Goal / Objective	Typical Use Cases	Examples / Models
Generative	Learn to generate raw or structured data	TTS, ASR decoding, data synthesis	GPT, Whisper, Tacotron, VALL-E, StableDiffusion
Contrastive	Learn to differentiate positive vs negative pairs	SSL pretraining, speaker ID, retrieval	Wav2vec 2.0 pretrain, SimCLR, CPC
Predictive	Learn to predict masked/missing parts	SSL, language modeling, audio modeling	HuBERT, WavLM, BERT, GPT, data2vec
Embedding	Learn to encode input into dense vector	Speaker embedding, clustering, retrieval	x-vector, ECAPA-TDNN, word2vec, BERT embeddings
Multimodal	Learn to integrate multiple modalities	ASR+translation, speech-text-image systems	Whisper, Flamingo, CLIP, AudioCLIP, SpeechT5

SSL Model Categorization

Model Type	Goal / Objective	Key Technique	Typical Use Cases
Generative	Learn to generate raw or structured data	Reconstruction, auto-regression, diffusion	TTS, ASR decoding, data synthesis
Contrastive	Learn to differentiate positive vs negative pairs	Contrastive loss (e.g., InfoNCE, SimCLR)	SSL pretraining, speaker ID, retrieval
Predictive	Learn to predict masked/missing parts	Masked modeling, next-step prediction	SSL, language modeling, audio modeling
Embedding	Learn to encode input into dense vector	Projection head, pooling, triplet/cosine loss	Speaker embedding, clustering, retrieval
Multimodal	Learn to integrate multiple modalities	Fusion (early/late), cross-attention	ASR+translation, speech-text-image systems

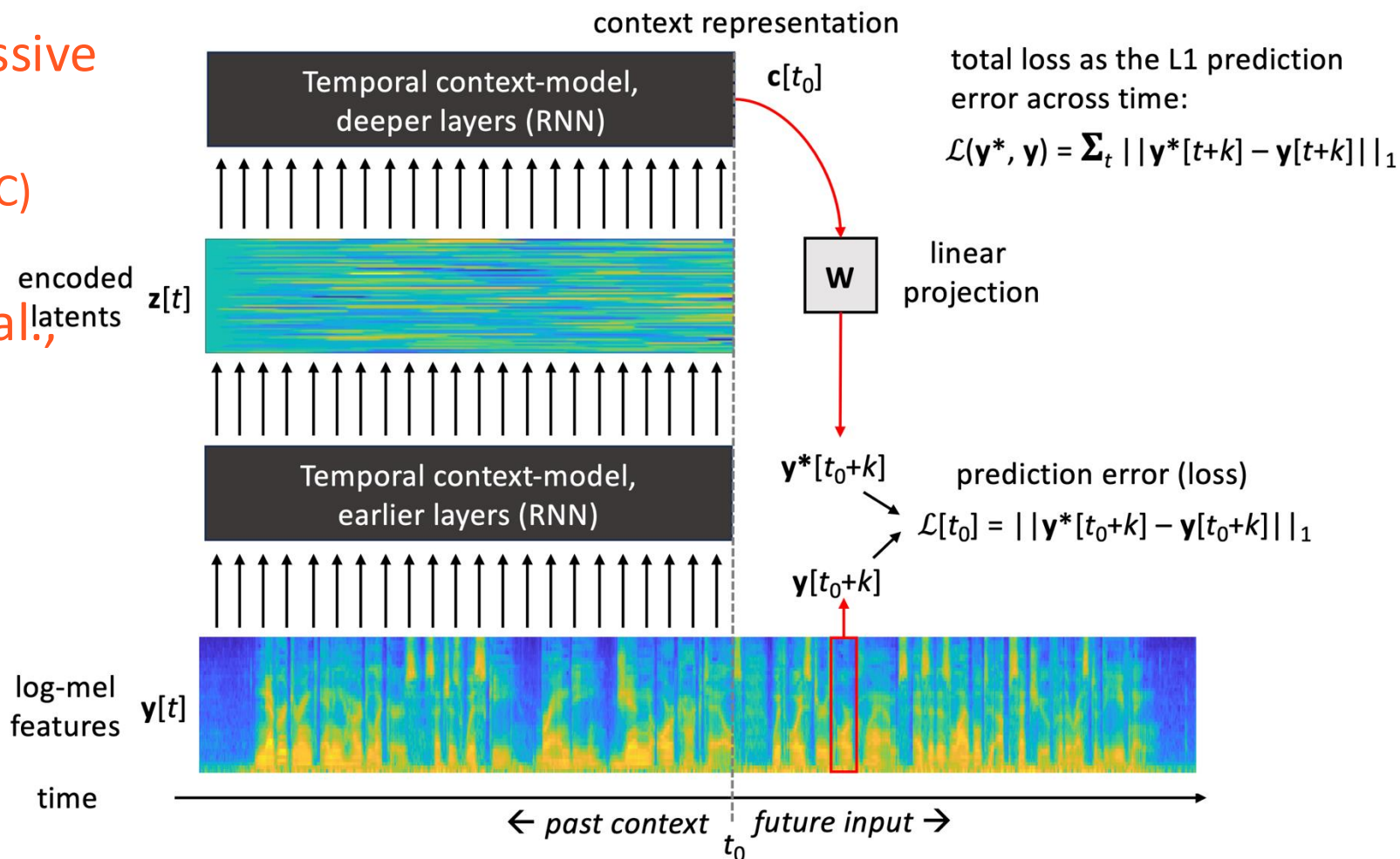
Popular SSL + Transformer Models

Model	Developer	Highlights
Wav2vec 2.0	Meta (Facebook AI)	SSL from raw audio + Transformer encoder, fine-tuned with CTC
HuBERT	Meta	Adds hidden unit clustering to improve SSL learning
WavLM	Microsoft	Advanced version of HuBERT, better for noisy/realistic speech
Whisper	OpenAI	Supervised Transformer encoder-decoder, multilingual ASR
Data2vec	Meta	Unified SSL model for speech, vision, and text

Popular SSL + Transformer Models

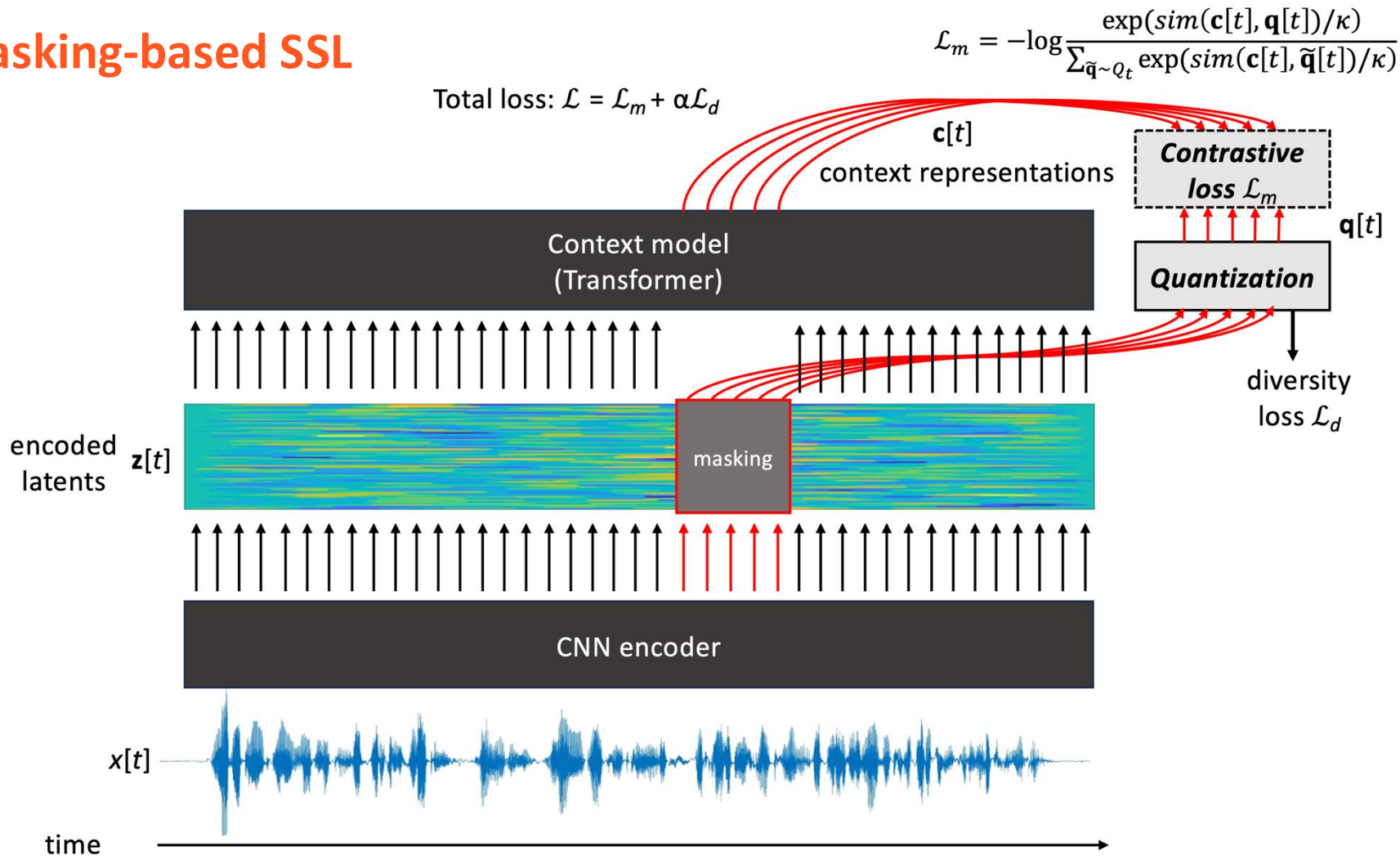
Model	Multilingual?	Details
Wav2vec 2.0	✗ (originally monolingual)	Trained on English (LibriSpeech), but can be fine-tuned on other languages
HuBERT	✗ (monolingual)	Pretrained mostly on English data; can be adapted to other languages
WavLM	✗ (mostly English)	Trained on large English corpora, but supports robust generalization to other languages when fine-tuned
Whisper	✓ Yes	Trained multilingually on 680k hours of speech across 96+ languages — supports multilingual ASR and translation out of the box
Data2vec	✗ (speech version is monolingual)	Cross-modality SSL, but multilingual support not default for speech
XLSR (Cross-lingual Wav2vec 2.0)	✓ Yes	Special version of Wav2vec 2.0 trained on 53+ languages (CommonVoice, BABEL, etc.)

Autoregressive
Predictive
Coding (APC)
algorithm
(Chung et al.,
2019)

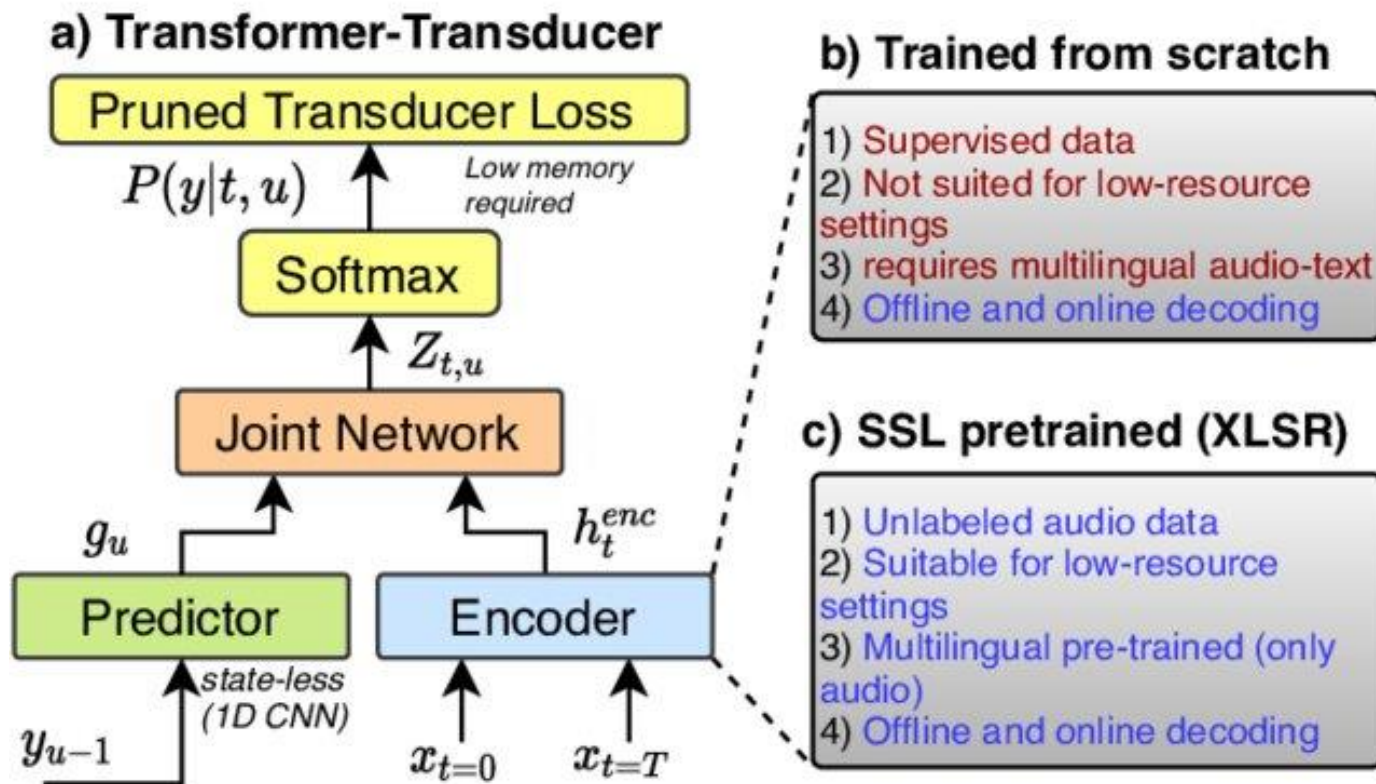


Predict spectral envelope features (e.g., log-Mel spectra or MFCCs) approximately 50 ms in the future, given access to the current and past observations of the input features

Masking-based SSL



XLSR-Transducer: Streaming ASR for Self-Supervised Pretrained Models (2024)



7.4. ASR Challenges & Solutions

- **Accuracy of Recognition:** ASR still struggles with accurately recognizing words or phrases, especially in complex contexts or unclear speech.
- **Language Complexity:** Natural language is diverse and nuanced, posing challenges for ASR in handling various language variations and accents.
- **Context Understanding:** ASR needs improvement in understanding context and deeper meanings of sentences, particularly in complex situations.

7.4. ASR Challenges & Solutions

- **Deep Learning Models:** Utilizing deep learning models such as Transformers and their variants has significantly improved ASR accuracy by learning from large datasets and self-adjusting.
- **Optimized Training Data:** Using representative and diverse datasets from multiple sources enhances ASR's recognition capabilities and confidence.
- **Integration with Other Technologies:** Combining ASR with Machine Learning and Natural Language Processing (NLP) helps in comprehensive language understanding and processing.
- **Context-awareness Integration:** To improve context understanding, ASR can integrate methods and models to recognize relationships between words and sentences in text or conversations.

Specific challenges on...

- **Limited Knowledge of Rare Words:** ASR often struggles with recognizing rare words, specialized terms, or uncommon vocabulary, leading to errors in recognition or misinterpretation of content.
- **Recognition of Proper Names:** ASR faces difficulties in accurately recognizing and pronouncing proper names such as personal names, place names, or company names, especially when these names are uncommon or have different phonetic variations.

Specific solutions...

- **Expand Training Data:** Incorporate diverse and extensive datasets that include rare words and proper names. This helps ASR learn and improve its ability to recognize these less common terms.
- **Utilize Transfer Learning Techniques:** Apply transfer learning techniques to train ASR on rich and diverse datasets, enabling it to learn how to recognize rare words and proper names effectively.
- **Integrate with NLP Technologies:** Use Natural Language Processing (NLP) methods such as Named Entity Recognition (NER) to enhance the recognition of proper names and specialized terms in various contexts.
- **Enhance Models and Algorithms:** Develop and optimize models and algorithms to ensure more accurate recognition of rare words and proper names, considering their linguistic complexities and variations.

ASR Trending

- **Self-supervised Learning:** Methods where ASR learns from audio data without detailed labels.
- **Multi-language ASR:** Models that handle multiple languages effectively.
- **Robustness and Adaptability:** Techniques for ASR to work well in diverse environmental conditions.
- **Context-aware ASR:** Models using context to improve accuracy.
- **Personalized ASR:** Customizing ASR for individual communication styles.
- **Integration with AI and NLP:** Combining ASR with AI and NLP to enhance understanding and responsiveness.

Multilingual datasets for ASR

Sr No.	Name	License Type	Languages	Hours/Sentence pairs (Approx)
	https://www.sciencedirect.com/science/article/pii/S2666307424000573			
1	MuST-C	CC BY-NC-ND 4.0	15 Languages	385-504 h
2	Europarl	Unknown	21 Languages	24M sentences
3	MediaSpeech	Open Source	4 Languages	10 h
4	CoVoST	CC0	16 Languages	2880 h
5	IWSLT TED	CC BY-NC-ND 4.0	109 Languages	Variable Dataset Size
6	VoxPopuli	CC0	23 Languages	400K h
7	Common Voice ¹	CC BY-SA 3.0	112 Languages	18.6K h

Multilingual datasets for ASR

Sr	Name	License Type	Languages	Hours/Sentence pairs
8	LibriVoxDeEn	CC BY-NC-SA 4.0	De-En	100 h
9	Microsoft Speech	C-UDA 1.0	Gujarati, Telugu, Tamil	Insufficient information
10	IIT Bombay	CC BY-NC 4.0	En-Hi	1.66M segments
11	MultiIndicMT	Open Source	11 Languages	11M sentence pair
12	IndicCorp	CC BY-NC 4.0	12 Languages	9B tokens
13	Dhwani	MIT License	40 Languages	17k h
14	MUCS 2021	Various licenses	6 Languages	145 h

Multilingual datasets for ASR

Sr No.	Name	License Type	Languages	Hours/Sentence pairs (Approx)
15	Shrutilipi	CC0	12 Languages	6457 h
16	Vistaar	MIT License	12 Languages	10736 h
17	NITK-IISc	Open Source	6 Languages	Insufficient information
18	Voxforge	GNU General Public License	6 Languages	Insufficient information
19	Tatoeba	Various CC licenses	Over 100 Languages	Variable Dataset Size
20	FLEURS	CC BY 4.0	102 Languages	1.4k h
21	Multilingual LibriSpeech	CC BY 4.0	8 Languages	44.5k h (Eng), 6k h (other)

Q/A



<https://app.sli.do/event/wPmQxU18pEZmnxHpv1Uo3m>

Final Test



<https://forms.office.com/r/UAbcquMvTa?origin=|prLink>